

Inverse q -Analogues and Their Series Expansions

A branch-aware, proof-complete synthesis

Human-readable research synthesis for
Vladimir Reshetnikov's ProveIt repository

Consolidated repository edition: 31 August 2026

Abstract

This volume replaces six overlapping reports on inverse q -analogues and their series expansions with one branch-aware account. Its organizing principle is that inversion is meaningful only after the forward observable, the parameter being recovered, a base point or asymptotic regime, and a branch have all been specified. Universal reversion is developed once and then specialized to finite and infinite q -Pochhammer products, Gaussian and multinomial coefficients, q -gamma and related special functions, cyclotomic limits, and the Fabius–Rvachev geometric-product setting.

Every nonconjectural result is accompanied by a complete human-readable proof. Claims backed by Lean are named explicitly; correct classical arguments that remain outside Lean are labelled as human-proved frontier results; numerical observations are not promoted to theorems. Statements that remain genuinely open appear only in theorem-style environments named *Conjecture*, with their outstanding obligations stated.

Contents

I	Foundations and conventions	5
1	What this volume consolidates	5
2	Status vocabulary	5
3	Basic normalizations	5
4	What an inverse symbol means	6
5	Corrections made during consolidation	6
II	Universal inversion calculus	7
6	Regular inverse germs	7
7	Formal triangular inversion	8
8	Formal and analytic Lagrange–Bürmann inversion	9
9	Bell-polynomial packaging	10
10	Passive-parameter sensitivities	11

11 Finite ramification and Puiseux inversion	12
12 Asymptotic inverse transfer and Lambert W	13
13 Multivariate triangular inversion	15
14 The product-form Lagrange–Good formula	16
15 Residual, Rouché, and interval-Newton certificates	17
III Finite and infinite q-Pochhammer inversion	19
16 The finite product as an inverse covering	19
16.1 Argument inversion at the unit level	19
16.2 All zero sheets and the complete real atlas	20
16.3 Cyclotomic collisions in the argument	21
17 Finite-product inversion in the base	22
17.1 The canonical real base branch	22
17.2 All-order inversion at $q = 0$	22
17.3 The complete logarithmic jet at $q = 1$	23
17.4 The exceptional slice $q = -1$	24
17.5 Reciprocal-base inversion at infinity	26
18 The infinite product: analytic domain and certified tails	26
19 Argument inversion and the complete real sheet atlas	28
19.1 Endpoint and central inverse jets	29
19.2 Remote real critical points	29
20 Large targets and the log-periodic principal inverse	30
21 Inversion in the base at $q = 0$ and on the real interval	31
22 Fixed-argument inversion as $q \rightarrow 1$	31
23 The coalescing parameter $a = q^x$	32
24 Fixed-target double scaling at $q = 1$	34
25 Raw root-of-unity endpoints and radial inverses	36
IV Gaussian and multinomial q-analogs	40
26 Polynomial structure and the positive real inverse	40
27 The inverse at $q = 0$	41
28 Complete expansion and inversion at $q = 1$	41
29 Centered palindromic inversion	43
30 Cyclotomic roots and the alternating base	44

31 The reciprocal and large-target inverse	45
32 Multinomial extension	46
V Other principal q-analogs	47
33 q -numbers	47
34 q -factorials and their exact ramification	49
35 The q -gamma function	50
36 q -digamma inversion	52
37 q -beta coordinate and base inverses	53
38 Euler q -exponentials	54
39 Basic hypergeometric argument inversion	55
40 Multiplicative theta products	55
VI Cyclotomic limits and the Fabius–Rvachev deformation	56
41 The boundary and probability regimes are different	56
42 The geometric uniform law and its sinc product	56
43 The exact Fabius and Rvachev member	58
44 Thue–Morse spline approximants	59
45 Cumulants as exact inverse observables	60
46 Scale-free recovery and an upgraded conjecture	62
47 Endpoint regimes and conditioning	62
48 Parameter scores, local CDF inversion, and the symmetry obstruction	63
49 The geometric comb as a q -Pochhammer interpolation basis	65
VII Certification and research frontiers	65
50 Branch-aware numerical certification	65
50.1 Certified tails for infinite products	66
50.2 Real branches: brackets, residuals, and interval Newton	68
50.3 Complex disks: contraction, Rouché, and total zero counts	69
50.4 Asymptotic truncation is part of the certificate	70

51 Two archived conjectures that are theorems	71
51.1 Total cumulants identify the geometric uniform law	72
51.2 A transcendental value determines the Gaussian rectangle	73
52 Precisely stated open frontiers	73
52.1 Generic monodromy of finite q -products	73
52.2 Continuous Gaussian monotonicity	74
52.3 A complete-Bernstein principal inverse	75
52.4 The minimum of the q -gamma function	75
52.5 Cyclotomic resurgence with fixed normalization	76
52.6 Editorial disposition of the remaining proposals	77
53 Lean source-status map	78
 VIII Reference appendices	 79
54 Source corpus and editorial provenance	79
55 The exhaustive result concordance	81
56 Notation and mandatory branch data	81
57 Source-to-canonical topical crosswalk	83
58 Formalization boundary	85
59 Repository and source note	85

Part I

Foundations and conventions

1 What this volume consolidates

The source corpus consisted of six independently delivered packages: *Inverse q -Analog Functions*, *Inverse q -Analog Jet Atlas*, *Inverse q -Analog: Extended Report*, two reports entitled *Inverse q -Analog*, and *q -Pochhammer and q -Binomial Expansions*. They treated largely the same inverse maps in different orders, repeatedly reproved the same reversion formulas, and occasionally assigned incompatible hypotheses or coefficients to the same claim. The present volume is an editorial and mathematical synthesis, not a mechanical concatenation.

The forward-expansion report supplies the strongest systematic engine for finite products, Gaussian coefficients, roots of unity, and the $q \rightarrow 1$ regimes. The jet atlas supplies the most useful all-order inverse, Bell-polynomial, mixed-parameter, and Lagrange–Good formulations. The function report supplies the clearest branchwise narrative. The extended and two earlier inverse reports contribute certification arguments, special regimes, and the Fabius–Rvachev applications. Repeated material appears once, with the strongest correct hypotheses and the most transparent proof.

2 Status vocabulary

Three statuses are used and are not interchangeable.

Lean-proved. The statement has a named, compiler-validated declaration in `Analysis/FabiusFunction`. Its exact module and declaration name are recorded in the reference appendix.

Human-proved frontier result. The statement is proved completely in this volume but does not yet have an exact Lean counterpart. It belongs in this research-frontier document rather than in the primary formalization-backed exposition.

Conjecture. The statement is plausible and mathematically meaningful, but the argument currently has a genuine missing step. Conjectures are isolated in explicitly named *Conjecture* environments. A numerical experiment, symbolic check, or manuscript label is never treated as a proof.

Definitions, notation, and descriptions of an algorithm are not themselves assertions of correctness. Whenever correctness, convergence, error, or complexity is claimed, it is stated as a theorem or proposition and proved.

3 Basic normalizations

For $n \in \mathbb{N}_0$, complex parameters a, q , and an empty product equal to one, the finite shifted factorial is

$$(a; q)_n := \prod_{j=0}^{n-1} (1 - aq^j).$$

When $|q| < 1$, the infinite shifted factorial is

$$(a; q)_\infty := \prod_{j=0}^{\infty} (1 - aq^j).$$

The convergence and holomorphy assertions implicit in the latter notation are proved in [section 18](#); outside $|q| < 1$, the symbol is used only after an explicit continuation or reciprocal normalization has been specified.

For integers $0 \leq k \leq n$, the Gaussian coefficient is normalized by

$$\begin{bmatrix} n \\ k \end{bmatrix}_q := \frac{(q; q)_n}{(q; q)_k (q; q)_{n-k}},$$

with its polynomial continuation at points where this quotient is formally indeterminate. This convention has degree $k(n - k)$, constant coefficient one, and satisfies the reciprocity law

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = q^{k(n-k)} \begin{bmatrix} n \\ k \end{bmatrix}_{q^{-1}}$$

for $q \neq 0$. A proof independent of quotient cancellation appears in [section 31](#).

We use

$$q = e^{-t} \quad (t \rightarrow 0)$$

for logarithmic expansions near $q = 1$, and $h = q - 1$ only when an ordinary Taylor coordinate is genuinely preferable. Conversion between the two is performed by formal composition, never by silently replacing t with $-h$. At a root of unity ζ , radial limits are written $q = \zeta e^{-t}$ with $t \downarrow 0$; this specifies a sectorial or radial asymptotic branch and does not assert the existence of an analytic germ across the unit circle.

4 What an inverse symbol means

An expression such as $a = a(y; q)$ or $q = q(y; a)$ is shorthand, not a globally defined function. Every inverse used below records the following data:

1. the forward map and which parameters are held fixed;
2. the parameter being recovered;
3. the base point, limiting regime, or real interval;
4. the local sheet or global monotone branch;
5. the logarithm or fractional-power determination when one is needed.

Thus a scalar equation with several free parameters defines a level set, not a simultaneous inverse. Several parameters can be recovered only after enough independent observables have been supplied and the corresponding Jacobian has been shown invertible. This elementary distinction removes a substantial fraction of the apparent disagreements among the source reports.

5 Corrections made during consolidation

The synthesis corrects, rather than propagates, the following recurrent defects from the source corpus:

- for $0 < q < 1$, the real zero lattice of $a \mapsto (a; q)_\infty$ is ordered $1 = q^0 < q^{-1} < q^{-2} < \dots$;
- a centered-palindromic inverse theorem requires nonzero equal endpoint coefficients, not palindromy alone;
- continuous lower-parameter Gaussian inversion must stay on the branch where the target lies strictly between the endpoint value and the central maximum;

- zeros introduced by a cyclotomic factor are simple unless a separate factor collision is proved; multiplicity cannot be inferred from the factor's mere presence;
- analytic-germ language at a natural boundary is replaced by radial or sectorial asymptotics with an explicit approach region;
- Newton and residual certificates include the differentiability, simple-root, derivative-separation, and domain-control hypotheses on which their conclusions depend;
- multivariate Lagrange–Good product formulas are used only for systems in the required product form; arbitrary locally invertible systems instead use the general total-degree recursion;
- conjectures contradicted by elementary counterexamples are removed rather than relabelled.

Later chapters give the corrected formulas and proofs at the point of use.

Part II

Universal inversion calculus

This part collects the inversion machinery used throughout the volume. Its purpose is not merely to quote familiar formulas: coefficient conventions, domains, branches, and certification hypotheses are made explicit so that later applications can invoke the results without changing their meaning. Unless otherwise stated, a power series is based at the origin and $[z^n]A(z)$ denotes its z^n -coefficient.

6 Regular inverse germs

Definition 6.1 (Partial inverse). Let $F(p, \lambda)$ be a scalar observable, where p is the variable to be recovered and $\lambda = (\lambda_1, \dots, \lambda_s)$ consists of passive parameters. A *partial inverse germ in p* at (p_0, λ_0) , with $y_0 = F(p_0, \lambda_0)$, is a germ $G(y, \lambda)$ at (y_0, λ_0) satisfying

$$F(G(y, \lambda), \lambda) = y, \quad G(y_0, \lambda_0) = p_0.$$

Thus the word *inverse* always includes the recovered variable, the base point, and the selected local branch.

Theorem 6.2 (Regular analytic partial inverse). *Let F be holomorphic in a neighbourhood of $(p_0, \lambda_0) \in \mathbb{C} \times \mathbb{C}^s$, put $y_0 = F(p_0, \lambda_0)$, and suppose*

$$F_p(p_0, \lambda_0) \neq 0.$$

Then there are neighbourhoods U of (y_0, λ_0) and V of p_0 , and a unique holomorphic map $G: U \rightarrow V$, after shrinking them if necessary, which is the partial inverse germ of [theorem 6.1](#). For each fixed nearby λ , the two maps $p \mapsto F(p, \lambda)$ and $y \mapsto G(y, \lambda)$ are inverse biholomorphic germs.

Proof. Apply the holomorphic implicit-function theorem to

$$\Psi(p, y, \lambda) := F(p, \lambda) - y.$$

At (p_0, y_0, λ_0) one has $\Psi = 0$ and $\Psi_p = F_p \neq 0$, so the theorem supplies a unique holomorphic solution $p = G(y, \lambda)$. Its defining identity is $F(G(y, \lambda), \lambda) = y$. Holding λ fixed and applying the same theorem with p as input shows that this identity holds between inverse germs in both directions. Uniqueness is the uniqueness clause of the implicit-function theorem. $\square$

Remark 6.3. The identical statement holds in the real C^r category, $1 \leq r \leq \infty$, with *holomorphic* replaced by C^r . This is the one-dimensional parameterized inverse-function theorem.

7 Formal triangular inversion

Theorem 7.1 (Triangular inverse recurrence). *Let R be a commutative ring and*

$$f(z) = \sum_{n \geq 1} c_n z^n \in R[[z]]$$

with c_1 a unit. There is a unique series $g(w) = \sum_{n \geq 1} b_n w^n$ satisfying both

$$f(g(w)) = w, \quad g(f(z)) = z.$$

Its coefficients are determined without division except by c_1 :

$$b_1 = c_1^{-1},$$

and, for $n \geq 2$,

$$b_n = -c_1^{-1} \sum_{k=2}^n c_k \sum_{\substack{j_1 + \dots + j_k = n \\ j_1, \dots, j_k \geq 1}} b_{j_1} \cdots b_{j_k}. \quad (1)$$

Proof. The coefficient of w in $f(g(w))$ is $c_1 b_1$, which forces $b_1 = c_1^{-1}$. For $n \geq 2$, the coefficient of w^n is

$$c_1 b_n + \sum_{k=2}^n c_k \sum_{\substack{j_1 + \dots + j_k = n \\ j_i \geq 1}} b_{j_1} \cdots b_{j_k}.$$

Every j_i in the second term is at most $n - 1$, so setting this coefficient equal to zero gives (1) and constructs a unique right compositional inverse g .

It remains to check the other composition. Put $h(z) = g(f(z))$. Then

$$f(h(z)) = (f \circ g)(f(z)) = f(z),$$

and the linear coefficient of h is $b_1 c_1 = 1$. If $h(z) \neq z$, let $r \geq 2$ be the first degree at which they differ and write $h(z) = z + dz^r + \mathcal{O}(z^{r+1})$ with $d \neq 0$. The first degree at which $f(h(z))$ and $f(z)$ differ is then r , with difference $c_1 d$. Since c_1 is a unit this is nonzero, contradicting $f \circ h = f$. Hence $g \circ f$ is also the identity. $\square$

Corollary 7.2 (The first five ordinary coefficients). *Over a field, the first coefficients in [theorem 7.1](#) are*

$$\begin{aligned} b_1 &= \frac{1}{c_1}, & b_2 &= -\frac{c_2}{c_1^3}, \\ b_3 &= \frac{2c_2^2 - c_1 c_3}{c_1^5}, & b_4 &= \frac{-5c_2^3 + 5c_1 c_2 c_3 - c_1^2 c_4}{c_1^7}, \\ b_5 &= \frac{14c_2^4 - 21c_1 c_2^2 c_3 + 6c_1^2 c_2 c_4 + 3c_1^2 c_3^2 - c_1^3 c_5}{c_1^9}. \end{aligned}$$

Proof. The recurrence specializes successively to

$$\begin{aligned} b_2 &= -c_1^{-1} c_2 b_1^2, \\ b_3 &= -c_1^{-1} (2c_2 b_1 b_2 + c_3 b_1^3), \\ b_4 &= -c_1^{-1} \{c_2(2b_1 b_3 + b_2^2) + 3c_3 b_1^2 b_2 + c_4 b_1^4\}, \\ b_5 &= -c_1^{-1} \{c_2(2b_1 b_4 + 2b_2 b_3) + c_3(3b_1^2 b_3 + 3b_1 b_2^2) + 4c_4 b_1^3 b_2 + c_5 b_1^5\}. \end{aligned}$$

Substituting $b_1 = c_1^{-1}$ from top to bottom and collecting terms gives the five displayed formulas. $\square$

Remark 7.3. These are ordinary-power-series coefficients. Factorial-normalized derivatives obey the Bell recurrence in [theorem 9.2](#).

8 Formal and analytic Lagrange–Bürmann inversion

Lemma 8.1 (Formal residue under a change of coordinate). *Let R be a commutative $\mathbb{Q}$ -algebra, let $f(z) = zU(z)$ with $U(0)$ a unit, and let $\text{res}_z A(z) := [z^{-1}]A(z)$. For every formal Laurent series $A(w) \in R((w))$,*

$$\text{res}_w A(w) = \text{res}_z (A(f(z))f'(z)).$$

Proof. Both sides are R -linear, and in any requested coefficient only finitely many terms of A contribute, so it suffices to take $A(w) = w^k$. If $k \neq -1$, then

$$f(z)^k f'(z) = \frac{1}{k+1} \frac{d}{dz} f(z)^{k+1},$$

whose residue is zero, as is the residue of w^k . Division by $k+1$ is legitimate because R is a $\mathbb{Q}$ -algebra. If $k = -1$, then

$$\frac{f'(z)}{f(z)} = \frac{1}{z} + \frac{U'(z)}{U(z)}.$$

The second summand is an ordinary power series and has zero residue, while the first has residue one. This is also $\text{res}_w w^{-1}$. $\square$

Theorem 8.2 (Formal Lagrange–Bürmann formula). *Let R be a commutative $\mathbb{Q}$ -algebra, let $\phi(z) \in R[[z]]$ have invertible constant term, and put*

$$f(z) = \frac{z}{\phi(z)}.$$

Let g be the compositional inverse of f . Then for every $H(z) \in R[[z]]$ and every $n \geq 1$,

$$[w^n]H(g(w)) = \frac{1}{n} [z^{n-1}]H'(z)\phi(z)^n. \quad (2)$$

In particular,

$$[w^n]g(w) = \frac{1}{n} [z^{n-1}]\phi(z)^n.$$

Proof. By [theorem 8.1](#), with $w = f(z)$,

$$\begin{aligned} [w^n]H(g(w)) &= \text{res}_w H(g(w))w^{-n-1} \\ &= \text{res}_z H(z)f'(z)f(z)^{-n-1}. \end{aligned}$$

The residue of a formal derivative is zero. Applying this to $H(z)f(z)^{-n}$ gives

$$0 = \text{res}_z (H'(z)f(z)^{-n} - nH(z)f'(z)f(z)^{-n-1}).$$

Consequently the preceding residue equals

$$\frac{1}{n} \text{res}_z H'(z)f(z)^{-n} = \frac{1}{n} \text{res}_z H'(z)z^{-n}\phi(z)^n,$$

which is (2). Taking $H(z) = z$ proves the special case. $\square$

Theorem 8.3 (Analytic Lagrange–Bürmann formula). *Under the hypotheses of [theorem 6.2](#) with the passive parameter fixed, write $y_0 = F(p_0)$. The selected inverse germ is*

$$G(y) = p_0 + \sum_{n \geq 1} \frac{(y - y_0)^n}{n!} \left[\frac{d^{n-1}}{dp^{n-1}} \left(\frac{p - p_0}{F(p) - F(p_0)} \right)^n \right]_{p=p_0}.$$

The quotient in parentheses is understood by its removable value $1/F'(p_0)$.

Proof. Set $z = p - p_0$, $w = y - y_0$, and

$$f(z) = F(p_0 + z) - F(p_0), \quad \phi(z) = \frac{z}{f(z)}.$$

Since $f'(0) = F'(p_0) \neq 0$, ϕ is holomorphic at zero with nonzero constant term. The analytic inverse-function theorem gives a convergent inverse germ $z = g(w)$, while [theorem 8.2](#) gives

$$[w^n]g(w) = \frac{1}{n} [z^{n-1}] \phi(z)^n = \frac{1}{n!} \left[\frac{d^{n-1}}{dz^{n-1}} \phi(z)^n \right]_{z=0}.$$

Translating back from z and w proves the formula. $\square$

Warning 8.4 (Radius is branchwise). Lagrange–Bürmann determines a germ, not its global sheet. Its radius of convergence is the distance from the expansion value to the nearest singularity of the *selected analytically continued inverse germ*. A critical value on another sheet, or a value reached only after crossing a cut, need not control that radius. Thus the phrase *the nearest critical value* is not a valid unqualified radius formula.

9 Bell-polynomial packaging

The complete and partial exponential Bell polynomials are fixed by

$$\exp \left(\sum_{j \geq 1} x_j \frac{t^j}{j!} \right) = \sum_{n \geq 0} B_n(x_1, \dots, x_n) \frac{t^n}{n!}, \quad (3)$$

$$\frac{1}{r!} \left(\sum_{j \geq 1} x_j \frac{t^j}{j!} \right)^r = \sum_{n \geq r} B_{n,r}(x_1, \dots, x_{n-r+1}) \frac{t^n}{n!}. \quad (4)$$

These identities are definitions and fix all factorial conventions.

Lemma 9.1 (Exponentiating logarithmic jets). *Let R be a commutative $\mathbb{Q}$ -algebra and let $A(z) \in R[[z]]$ have invertible constant term. If*

$$\log \frac{A(z)}{A(0)} = \sum_{j \geq 1} \ell_j \frac{z^j}{j!},$$

then

$$\frac{A(z)}{A(0)} = \sum_{n \geq 0} B_n(\ell_1, \dots, \ell_n) \frac{z^n}{n!}.$$

Proof. Exponentiate the displayed logarithmic identity. The right-hand side is exactly the defining generating function (3). $\square$

Theorem 9.2 (Bell recurrence for inverse derivatives). *Let R be a commutative $\mathbb{Q}$ -algebra and write*

$$f(z) = \sum_{r \geq 1} f_r \frac{z^r}{r!}, \quad g(w) = \sum_{n \geq 1} g_n \frac{w^n}{n!},$$

where f_1 is a unit and $f(g(w)) = w$. Then

$$g_1 = f_1^{-1}$$

and, for $n \geq 2$,

$$g_n = -f_1^{-1} \sum_{r=2}^n f_r B_{n,r}(g_1, \dots, g_{n-r+1}). \quad (5)$$

Proof. Substituting $g(w)$ into the r -th term of f , and using (4), shows that the coefficient multiplying $w^n/n!$ in $f(g(w))$ is

$$\sum_{r=1}^n f_r B_{n,r}(g_1, \dots, g_{n-r+1}).$$

Here $B_{n,1}(g_1, \dots, g_n) = g_n$. For $n = 1$, comparison with w gives $f_1 g_1 = 1$. For every $n \geq 2$, the displayed coefficient must vanish; isolating $f_1 g_n$ gives (5). Each right-hand side involves only g_j with $j < n$, so the recurrence is triangular. $\square$

10 Passive-parameter sensitivities

Theorem 10.1 (First and second inverse sensitivities). *Let F and its partial inverse G be C^2 (or holomorphic) and suppose $F_p \neq 0$ on the germ under consideration. All derivatives of F below are evaluated at $(G(y, \lambda), \lambda)$. Then*

$$G_y = \frac{1}{F_p}, \quad G_{\lambda_i} = -\frac{F_{\lambda_i}}{F_p},$$

$$G_{yy} = -\frac{F_{pp}}{F_p^3}, \quad G_{y\lambda_i} = \frac{F_{pp}F_{\lambda_i} - F_p F_{p\lambda_i}}{F_p^3},$$

and

$$G_{\lambda_i \lambda_j} = -\frac{F_{pp}G_{\lambda_i}G_{\lambda_j} + F_{p\lambda_i}G_{\lambda_j} + F_{p\lambda_j}G_{\lambda_i} + F_{\lambda_i \lambda_j}}{F_p}.$$

If F and G are C^r , the same differentiation gives all mixed derivatives through total order r recursively: at every new order the highest derivative of G occurs once, multiplied by F_p , and all remaining terms involve lower derivatives of G . In the holomorphic case this continues to every order.

Proof. Differentiate $F(G(y, \lambda), \lambda) = y$ with respect to y and to λ_i :

$$F_p G_y = 1, \quad F_p G_{\lambda_i} + F_{\lambda_i} = 0.$$

These give the first two formulas. Differentiating the first identity once more with respect to y gives

$$F_{pp}G_y^2 + F_p G_{yy} = 0,$$

and substituting $G_y = 1/F_p$ gives the formula for G_{yy} . Differentiating $F_p G_y = 1$ with respect to λ_i at fixed y gives

$$(F_{pp}G_{\lambda_i} + F_{p\lambda_i})G_y + F_p G_{y\lambda_i} = 0.$$

Substitution of the first derivatives yields the stated mixed formula. Finally, differentiate $F_p G_{\lambda_i} + F_{\lambda_i} = 0$ with respect to λ_j . The chain rule gives

$$F_p G_{\lambda_i \lambda_j} + F_{pp}G_{\lambda_i}G_{\lambda_j} + F_{p\lambda_i}G_{\lambda_j} + F_{p\lambda_j}G_{\lambda_i} + F_{\lambda_i \lambda_j} = 0,$$

which proves the last formula. Repeating the chain rule proves the triangular assertion at every order. $\square$

Remark 10.2 (Conditioning). For perturbations of the observed value with passive parameters fixed, the absolute local condition number is $|F_p|^{-1}$. When $p \neq 0$ and $y \neq 0$, the corresponding relative condition number is

$$\left| \frac{y}{pF_p} \right|.$$

The blow-up as $F_p \rightarrow 0$ is a conditioning statement; by itself it is not a proof of nonidentifiability.

11 Finite ramification and Puiseux inversion

Theorem 11.1 (Finite-ramification normal form). *Let F be holomorphic near p_0 , let $y_0 = F(p_0)$, and suppose, for some $m \geq 2$,*

$$F(p_0 + z) = y_0 + c_m z^m + c_{m+1} z^{m+1} + \mathcal{O}(z^{m+2}), \quad c_m \neq 0.$$

Choose a local value

$$\xi = \left(\frac{y - y_0}{c_m} \right)^{1/m}$$

on a sufficiently small simply connected punctured y -domain. Then the m inverse branches near p_0 , indexed by the m -th roots of unity, are

$$p - p_0 = \omega \xi - \frac{c_{m+1}}{m c_m} \omega^2 \xi^2 + \mathcal{O}(\xi^3), \quad \omega^m = 1. \quad (6)$$

For every sufficiently small $y \neq y_0$, these are precisely the m solutions near p_0 , counted without multiplicity; at $y = y_0$ they coalesce.

Proof. Factor

$$F(p_0 + z) - y_0 = c_m z^m H(z), \quad H(z) = 1 + \frac{c_{m+1}}{c_m} z + \mathcal{O}(z^2).$$

On a small disk, H is nonzero and has a holomorphic m -th root with value one at zero. Define $v = zH(z)^{1/m}$. Then $v(0) = 0$ and $v'(0) = 1$, so v has a regular holomorphic inverse $z = \psi(v)$. Moreover,

$$\frac{F(p_0 + z) - y_0}{c_m} = v^m.$$

Thus the solutions are $z = \psi(\omega \xi)$, $\omega^m = 1$. Since

$$v = z + \frac{c_{m+1}}{m c_m} z^2 + \mathcal{O}(z^3), \quad \psi(v) = v - \frac{c_{m+1}}{m c_m} v^2 + \mathcal{O}(v^3),$$

substitution gives (6). For nonzero ξ , the m values $\omega \xi$ are distinct, and the injectivity of ψ on a smaller disk makes their images distinct. Conversely every nearby solution yields one of these m roots of $v^m = \xi^m$, proving exhaustion. $\square$

Corollary 11.2 (Universal fold jet through order two). *Let a holomorphic germ, in the notation of theorem 11.1 with $m = 2$, have the expansion*

$$y - y_* = Az^2 + Bz^3 + Cz^4 + Dz^5 + \mathcal{O}(z^6), \quad A \neq 0,$$

and choose a square root $\sqrt{A}$. The two inverse branches, indexed by $\sigma \in \{1, -1\}$, are

$$\begin{aligned} z = & \sigma \frac{(y - y_*)^{1/2}}{\sqrt{A}} - \frac{B}{2A^2} (y - y_*) \\ & + \sigma \frac{5B^2 - 4AC}{8A^{7/2}} (y - y_*)^{3/2} - \frac{A^2 D - 3ABC + 2B^3}{2A^5} (y - y_*)^2 + \mathcal{O}((y - y_*)^{5/2}). \end{aligned} \quad (7)$$

The powers $A^{7/2}$ use the same chosen square root as the leading term.

Proof. Put $t = (y - y_*)^{1/2}$ on the chosen ramified coordinate and seek

$$z = at + bt^2 + ct^3 + dt^4 + \mathcal{O}(t^5).$$

Matching the coefficient of t^2 gives $Aa^2 = 1$, hence $a = \sigma/\sqrt{A}$. Successively setting the coefficients of t^3, t^4, t^5 equal to zero gives

$$\begin{aligned} 2Aab + Ba^3 &= 0, \\ A(b^2 + 2ac) + 3Ba^2b + Ca^4 &= 0, \\ A(2ad + 2bc) + B(3a^2c + 3ab^2) + 4Ca^3b + Da^5 &= 0. \end{aligned}$$

Solving these triangular equations yields

$$b = -\frac{B}{2A^2}, \quad c = \sigma \frac{5B^2 - 4AC}{8A^{7/2}}, \quad d = -\frac{A^2D - 3ABC + 2B^3}{2A^5}.$$

Substitution into the ansatz proves (7). $\square$

Warning 11.3 (What a singular Jacobian does not say). The condition $F_p(p_0, \lambda_0) = 0$ says only that the regular inverse-function theorem is unavailable. A finite first nonzero derivative can produce finitely ramified branches as above; an injective real map such as $p \mapsto p^3$ remains identifiable despite its zero derivative; and more degenerate behaviour is also possible. Neither nonidentifiability nor Puiseux behaviour follows from a singular Jacobian alone.

12 Asymptotic inverse transfer and Lambert W

Proposition 12.1 (Residual transfer along a monotone inverse branch). *Let F be real C^1 and injective on an interval I , with inverse G on $F(I)$. Let $\tilde{G}(y) \in I$. If $|F'(p)| \geq m(y) > 0$ on the segment joining $\tilde{G}(y)$ to $G(y)$, then*

$$|\tilde{G}(y) - G(y)| \leq \frac{|F(\tilde{G}(y)) - y|}{m(y)}.$$

In particular, a forward residual estimate transfers to an inverse asymptotic estimate once the derivative scale is controlled on the selected branch.

Proof. The mean-value theorem supplies a point θ between $\tilde{G}(y)$ and $G(y)$ such that

$$F(\tilde{G}(y)) - F(G(y)) = F'(\theta)(\tilde{G}(y) - G(y)).$$

Use $F(G(y)) = y$ and the assumed lower bound for $|F'(\theta)|$. $\square$

Theorem 12.2 (Reciprocal-scale reversion). *Let $A \neq 0$ and suppose, on a real or complex sector as $t \rightarrow 0$,*

$$x = X(t) = \frac{A}{t} + B + Ct + Dt^2 + \mathcal{O}(t^3).$$

For any inverse branch $t = T(x) \rightarrow 0$, put $S = x - B$. Then $t \sim A/S$ and, in the corresponding sector as $S \rightarrow \infty$,

$$T(x) = \frac{A}{S} + \frac{CA^2}{S^3} + \frac{DA^3}{S^4} + \mathcal{O}(S^{-5}). \quad (8)$$

In particular, the coefficient of S^{-2} is zero.

Proof. After subtracting B and multiplying by t , the inverse equation is

$$St = A + Ct^2 + Dt^3 + \mathcal{O}(t^4).$$

Since $t \rightarrow 0$, this first gives $St \rightarrow A$, hence $t \sim A/S$. Dividing the same identity by S and using $t = \mathcal{O}(S^{-1})$ gives

$$t = \frac{A}{S} + \mathcal{O}(S^{-3}).$$

It follows that

$$t^2 = \frac{A^2}{S^2} + \mathcal{O}(S^{-4}), \quad t^3 = \frac{A^3}{S^3} + \mathcal{O}(S^{-5}).$$

Substituting these two estimates once more into

$$t = \frac{A}{S} + \frac{C}{S}t^2 + \frac{D}{S}t^3 + \mathcal{O}(t^4/S)$$

gives (8). This purely algebraic bootstrap does not differentiate the asymptotic remainder. $\square$

Remark 12.3. The theorem deliberately excludes logarithmic corrections. If the forward expansion contains $t \log t$, substitution of $t \sim A/S$ generally produces a $(\log S)/S^3$ term in the inverse. Such a term must not be hidden inside an ordinary power-series ansatz.

Theorem 12.4 (Lambert- W preconditioner with branch compatibility). *Fix a simply connected domain $D \subset \mathbb{C} \setminus \{0\}$ and a holomorphic branch of $\log t$ on D . Consider the truncated equation*

$$\Lambda = \frac{A}{t} + B \log t + C, \quad A \neq 0, \quad t \in D. \quad (9)$$

If $B = 0$, its solution is $t = A/(\Lambda - C)$, provided the denominator is nonzero and this value lies in D . If $B \neq 0$, put

$$Z = -\frac{A}{B} \exp\left(-\frac{\Lambda - C}{B}\right).$$

For every branch W_k defined at Z with $W_k(Z) \neq 0$, set

$$t_k = -\frac{A}{B W_k(Z)}.$$

Then t_k solves (9) if and only if $t_k \in D$ and the branch-compatibility identity

$$\log t_k = \frac{\Lambda - C}{B} + W_k(Z) \quad (10)$$

holds for the chosen logarithm. Conversely, every solution of (9) has this form for some value of the multivalued relation $ue^u = Z$, and hence for a local branch of W whenever that relation is locally regular.

Proof. The case $B = 0$ is immediate. Suppose $B \neq 0$ and set

$$u = -\frac{A}{Bt}.$$

If t solves (9), then

$$\log t = \frac{\Lambda - C}{B} + u.$$

Exponentiating and using $t = -A/(Bu)$ gives

$$ue^u = -\frac{A}{B} \exp\left(-\frac{\Lambda - C}{B}\right) = Z.$$

Hence u is a value of $W(Z)$; wherever $1 + u \neq 0$, it belongs to a locally holomorphic branch W_k . The displayed formula for t_k follows, and (10) is the original, unexponentiated equation. Conversely, if $u = W_k(Z)$, $t_k = -A/(Bu)$, and (10) holds, then

$$\frac{A}{t_k} + B \log t_k + C = -Bu + B \left(\frac{\Lambda - C}{B} + u \right) + C = \Lambda.$$

$\square$

Warning 12.5. The compatibility test in (10) is essential over $\mathbb{C}$. Exponentiation forgets integer multiples of $2\pi i$; choosing a W -branch without also fixing and checking the logarithm branch can therefore create spurious solutions or omit genuine ones.

Corollary 12.6 (Accuracy of the Lambert preconditioner). *On a fixed real or complex branch, suppose the full equation is*

$$\Lambda = \frac{A}{t} + B \log t + C + R(t), \quad A \neq 0, \quad R(t) = \mathcal{O}(t)$$

uniformly in the chosen branch domain. Let $t_0 \rightarrow 0$ be a branch-compatible solution of the truncated equation (9), and let t solve the full equation with $t/t_0 \rightarrow 1$. In the complex case assume the segment joining t_0 and t stays in the chosen logarithm domain. Then

$$t = t_0 + \mathcal{O}(t_0^3).$$

Proof. Write $L(t) = A/t + B \log t + C$. The two equations give

$$L(t) - L(t_0) = -R(t) = \mathcal{O}(t_0).$$

On the joining segment, the assumption $t/t_0 \rightarrow 1$ gives uniformly

$$L'(t) = -\frac{A}{t^2} + \frac{B}{t} = -\frac{A}{t_0^2}(1 + o(1)).$$

The real mean-value theorem, or integration of L' along the complex segment, therefore gives

$$L(t) - L(t_0) = (t - t_0) \left(-\frac{A}{t_0^2} \right) (1 + o(1)).$$

Since its left-hand side is $\mathcal{O}(t_0)$, division proves $t - t_0 = \mathcal{O}(t_0^3)$. □

13 Multivariate triangular inversion

Theorem 13.1 (Total-degree inverse recurrence). *Let R be a commutative ring and let*

$$\mathbf{F}(\mathbf{p}) = A\mathbf{p} + \sum_{r \geq 2} \mathbf{F}_r(\mathbf{p})$$

be a d -component formal map, where each $\mathbf{F}_r$ is homogeneous of total degree r and $A \in R^{d \times d}$ is invertible. There is a unique two-sided formal inverse $\mathbf{G}(\mathbf{y}) = \sum_{n \geq 1} \mathbf{G}_n(\mathbf{y})$, with each $\mathbf{G}_n$ homogeneous of degree n . It is determined by

$$\mathbf{G}_1 = A^{-1}\mathbf{y}$$

and, for $n \geq 2$,

$$\mathbf{G}_n = -A^{-1}[\text{total degree } n] \sum_{r=2}^n \mathbf{F}_r \left(\sum_{j=1}^{n-1} \mathbf{G}_j \right). \quad (11)$$

If $\mathbf{F}$ is analytic over $\mathbb{R}$ or $\mathbb{C}$, then after translating a base point $\mathbf{p}_0$ and its value to the origin, the hypothesis $\det D\mathbf{F}(\mathbf{p}_0) \neq 0$ makes the formal series the Taylor series of the local analytic inverse, and

$$D\mathbf{G}(\mathbf{y}) = D\mathbf{F}(\mathbf{G}(\mathbf{y}))^{-1}.$$

Proof. In total degree one, the equation $F(G(y)) = y$ is $AG_1 = y$. At degree $n \geq 2$, the only occurrence of the unknown G_n is the linear term AG_n : every nonlinear F_r uses at least two positive-degree inputs, so its degree- n part involves only $G_1, \dots, G_{n-1}$. Isolating G_n gives (11) and constructs a unique right inverse.

For two-sidedness, let $H = G \circ F$. Then $F \circ H = F$ and $DH(0) = I$. If $H - \text{id}$ first has a nonzero homogeneous part K_n , then $F \circ H - F$ first has homogeneous part AK_n , which cannot vanish because A is invertible. Thus $H = \text{id}$.

The analytic inverse-function theorem proves convergence in the analytic case. Differentiating $F(G(y)) = y$ gives $DF(G(y))DG(y) = I$, and hence the derivative formula. $\square$

Remark 13.2. When $\det DF = 0$, the regular multivariate theorem is simply inapplicable. Rank loss alone does not identify a ramification type; a separate normal-form or elimination argument is required.

14 The product-form Lagrange–Good formula

Lemma 14.1 (Multivariate formal residue substitution). *Let R be a commutative $\mathbb{Q}$ -algebra and*

$$T_i(\mathbf{x}) = x_i U_i(\mathbf{x}), \quad U_i(\mathbf{0}) \in R^\times, \quad 1 \leq i \leq d.$$

For a formal Laurent series $A(\mathbf{T})$, whose support is bounded below in each coordinate, define

$$\text{Res}_{\mathbf{T}} A(\mathbf{T}) \, dT_1 \cdots dT_d := [T_1^{-1} \cdots T_d^{-1}] A(\mathbf{T}).$$

Then

$$\text{Res}_{\mathbf{T}} A(\mathbf{T}) \, dT_1 \cdots dT_d = \text{Res}_{\mathbf{x}} A(\mathbf{T}(\mathbf{x})) \det \left(\frac{\partial T_i}{\partial x_j} \right) dx_1 \cdots dx_d.$$

Proof. By linearity it is enough to consider a Laurent monomial $\prod_i T_i^{k_i}$. If some $k_i \neq -1$, its associated top form is exact:

$$\left(\prod_j T_j^{k_j} \right) dT_1 \wedge \cdots \wedge dT_d = d \left(\frac{(-1)^{i-1} T_i^{k_i+1}}{k_i + 1} \prod_{j \neq i} T_j^{k_j} \wedge_{j \neq i} dT_j \right).$$

Formal differentiation has zero residue, and pullback commutes with d , so both residues vanish. If every $k_i = -1$, the form is

$$\bigwedge_{i=1}^d \frac{dT_i}{T_i} = \bigwedge_{i=1}^d \left(\frac{dx_i}{x_i} + \frac{dU_i}{U_i} \right).$$

The product of all dx_i/x_i has residue one. Every other wedge term replaces at least one logarithmic pole by a regular power-series differential, so its full $x_1^{-1} \cdots x_d^{-1}$ residue is zero. Thus the pullback also has residue one, proving the assertion. $\square$

Theorem 14.2 (Lagrange–Good inversion). *Let R be a commutative $\mathbb{Q}$ -algebra and let $\phi_i(\mathbf{x}) \in R[[x_1, \dots, x_d]]$ have invertible constant terms. The product-form system*

$$w_i = t_i \phi_i(\mathbf{w}), \quad 1 \leq i \leq d,$$

has a unique formal solution $\mathbf{w}(t)$ with zero constant term. For every $H(\mathbf{x}) \in R[[x_1, \dots, x_d]]$,

$$H(\mathbf{w}(t)) = \sum_{\mathbf{n} \in \mathbb{Z}_{\geq 0}^d} t^{\mathbf{n}} [\mathbf{x}^{\mathbf{n}}] H(\mathbf{x}) \prod_{i=1}^d \phi_i(\mathbf{x})^{n_i} \det \left(\delta_{ij} - \frac{x_j}{\phi_i(\mathbf{x})} \frac{\partial \phi_i}{\partial x_j}(\mathbf{x}) \right)_{i,j=1}^d. \quad (12)$$

Proof. Existence and uniqueness follow from [theorem 13.1](#) applied to $t_i = w_i/\phi_i(\mathbf{w})$. Fix $\mathbf{n} \in \mathbb{Z}_{\geq 0}^d$. Coefficient extraction as a formal residue gives

$$[t^n]H(\mathbf{w}(t)) = \text{Res}_{\mathbf{t}} H(\mathbf{w}(t)) \prod_i t_i^{-n_i} \frac{dt_i}{t_i}.$$

Use the formal coordinate change

$$t_i = \frac{x_i}{\phi_i(\mathbf{x})}.$$

Since $\mathbf{w}(t(\mathbf{x})) = \mathbf{x}$, one has

$$t_i^{-n_i} = x_i^{-n_i} \phi_i(\mathbf{x})^{n_i}$$

and

$$\frac{dt_i}{t_i} = \sum_{j=1}^d \left(\delta_{ij} - \frac{x_j}{\phi_i(\mathbf{x})} \frac{\partial \phi_i}{\partial x_j}(\mathbf{x}) \right) \frac{dx_j}{x_j}.$$

Taking the wedge product produces the determinant in (12); taking the remaining residue is exactly $[\mathbf{x}^{\mathbf{n}}]$. Summing over $\mathbf{n}$ proves the formula. $\square$

Remark 14.3 (Equivalent determinant conventions). If $D = \text{diag}(x_1, \dots, x_d)$ and $Q_{ij} = \phi_i^{-1} \partial \phi_i / \partial x_j$, the determinant in (12) is $\det(I - QD)$. Some sources instead print $\det(I - DQ)$, whose entries contain x_i rather than x_j . The two are equal because $\det(I - AB) = \det(I - BA)$ for square matrices over a commutative ring. For completeness, this identity follows by taking the determinant after reducing

$$\begin{pmatrix} I & A \\ B & I \end{pmatrix}$$

in two different ways: the operation $\text{row}_2 \leftarrow \text{row}_2 - B\text{row}_1$ gives diagonal blocks $(I, I - BA)$, while $\text{row}_1 \leftarrow \text{row}_1 - A\text{row}_2$ gives diagonal blocks $(I - AB, I)$. Both block row operations have determinant one.

Remark 14.4 (Scope of Good's determinant). [Theorem 14.2](#) applies to systems already written in the componentwise product form $w_i = t_i \phi_i(\mathbf{w})$. It is not a coefficient formula for an arbitrary locally invertible square system. For such a system one should first use the invariant total-degree recurrence (11); a reduction to product form requires additional structure and must be justified.

15 Residual, Rouché, and interval-Newton certificates

Proposition 15.1 (Real residual certificate). *Let F be differentiable on an interval I , suppose $|F'(x)| \geq m > 0$ throughout I , and let $x_* \in I$ satisfy $F(x_*) = y$. Then x_* is the unique solution of $F(x) = y$ in I , and every $\tilde{x} \in I$ satisfies*

$$|\tilde{x} - x_*| \leq \frac{|F(\tilde{x}) - y|}{m}.$$

Proof. Derivatives have the Darboux property, so a derivative bounded away from zero cannot change sign; hence F is strictly monotone on I and the root is unique. Apply [theorem 12.1](#) with $G(y) = x_*$, $\tilde{G}(y) = \tilde{x}$, and $m(y) = m$; its segment hypothesis follows from the bound on all of I . $\square$

Theorem 15.2 (Rouché disk certificate). *Let $r > 0$, let F be holomorphic on a neighbourhood of the closed disk $\overline{D}(z_0, r)$, let $y \in \mathbb{C}$, and assume*

$$F'(z_0) \neq 0, \quad \sup_{\overline{D}(z_0, r)} |F''(z)| \leq M.$$

If

$$\frac{Mr^2}{2} < |F'(z_0)|r - |F(z_0) - y|, \quad (13)$$

then $F(z) = y$ has exactly one solution in $D(z_0, r)$, and that solution is simple.

Proof. Put $H(z) = F(z) - y$ and

$$L(z) = H(z_0) + F'(z_0)(z - z_0).$$

Taylor's formula along the segment from z_0 to z , valid because a disk is convex, gives

$$|H(z) - L(z)| \leq \frac{M}{2}|z - z_0|^2.$$

On the boundary $|z - z_0| = r$,

$$|L(z)| \geq |F'(z_0)|r - |H(z_0)| > \frac{Mr^2}{2} \geq |H(z) - L(z)|.$$

Rouché's theorem says that H and L have the same number of zeros in the disk, counted with multiplicity. Condition (13) also implies $|H(z_0)/F'(z_0)| < r$, so the affine function L has exactly one zero there. Hence H has total zero multiplicity one, which means its unique zero is simple. $\square$

Theorem 15.3 (Scalar interval-Newton certificate). *Let $I = [a, b] \subset \mathbb{R}$, let $F \in C^1(I)$, choose $x_0 \in I$, and let $D = [d_-, d_+]$ be a closed real interval containing $F'(I)$, with $0 \notin D$. Interpret $F(x_0)/D$ as its interval image and define*

$$N = x_0 - \frac{F(x_0)}{D}.$$

If $N \subseteq I$, then F has exactly one zero x_ in I , and $x_* \in N$. If $N \subseteq \text{int } I$, the certified zero lies in $\text{int } I$.*

Proof. Replacing F by $-F$ and D by $-D$, if necessary, leaves N unchanged, so assume $0 < d_- \leq F'(x) \leq d_+$ on I . Thus F is strictly increasing. If $F(x_0) = 0$, the conclusion is immediate. If $F(x_0) > 0$, then

$$N = \left[x_0 - \frac{F(x_0)}{d_-}, x_0 - \frac{F(x_0)}{d_+} \right] \subseteq [a, b].$$

The left-end inclusion implies $F(x_0) \leq d_-(x_0 - a)$. Since

$$F(x_0) - F(a) = \int_a^{x_0} F'(s) \, ds \geq d_-(x_0 - a),$$

we obtain $F(a) \leq 0 < F(x_0)$, so the intermediate-value theorem gives a zero between a and x_0 . If $F(x_0) < 0$, the right-end inclusion similarly gives $-F(x_0) \leq d_-(b - x_0)$, hence $F(b) \geq 0$, and again a zero exists. Strict monotonicity makes it unique.

For that zero x_* , the mean-value theorem gives some θ between x_0 and x_* such that

$$F(x_0) = F'(\theta)(x_0 - x_*).$$

Because $F'(\theta) \in D$,

$$x_* = x_0 - \frac{F(x_0)}{F'(\theta)} \in N.$$

The final interior assertion follows from $x_* \in N$. $\square$

These certificates answer different logical questions. A residual estimate controls the error once existence and the intended branch are already known. The Rouché test proves complex existence, uniqueness, and simplicity in a disk. The scalar interval-Newton test proves real existence, uniqueness, and localization in an interval. A small numerical residual alone must never be reported as an existence or branch certificate.

Part III

Finite and infinite q -Pochhammer inversion

16 The finite product as an inverse covering

Fix $n \geq 1$ and write

$$P_n(a, q) := (a; q)_n = \prod_{j=0}^{n-1} (1 - aq^j).$$

Throughout, $[n]_q$ and $\begin{bmatrix} n \\ r \end{bmatrix}_q$ denote their polynomial continuations at $q = 0, 1$ and at roots of unity; no quotient with a vanishing denominator is being used there. This section consolidates the finite-product results that were repeated in the six source reports. The results are human-proved frontier results: no Lean formalization is claimed here. We first invert in a , then in q , and keep the two branch geometries separate.

Whenever no factor vanishes, logarithmic differentiation gives

$$\partial_a \log P_n(a, q) = - \sum_{j=0}^{n-1} \frac{q^j}{1 - aq^j}, \quad (14)$$

$$\partial_q \log P_n(a, q) = -a \sum_{j=1}^{n-1} \frac jq^{j-1}}{1 - aq^j}. \quad (15)$$

Both identities also hold after clearing denominators at a zero. They will be used only on zero-free sets, where the logarithms are unambiguous locally.

16.1 Argument inversion at the unit level

Euler's finite q -binomial identity is

$$P_n(a, q) = \sum_{r=0}^n (-1)^r q^{\binom{r}{2}} \begin{bmatrix} n \\ r \end{bmatrix}_q a^r. \quad (16)$$

Set $s = 1 - y$ and

$$b_r = (-1)^{r+1} q^{\binom{r}{2}} \begin{bmatrix} n \\ r \end{bmatrix}_q, \quad f(a) = 1 - P_n(a, q) = \sum_{r \geq 1} b_r a^r.$$

Theorem 16.1 (All-order argument inverse at $y = 1$). *If $[n]_q = b_1 \neq 0$, there is a unique analytic branch $a = a(s)$ with $a(0) = 0$ and $P_n(a(s), q) = 1 - s$. Its coefficients are*

$$[s^m]a(s) = \frac{1}{m} [z^{m-1}] \left(\frac{z}{1 - P_n(z, q)} \right)^m, \quad m \geq 1. \quad (17)$$

In particular,

$$a = \frac{s}{[n]_q} + \frac{q \begin{bmatrix} n \\ 2 \end{bmatrix}_q}{[n]_q^3} s^2 + \left(\frac{2q^2 \begin{bmatrix} n \\ 2 \end{bmatrix}_q^2}{[n]_q^5} - \frac{q^3 \begin{bmatrix} n \\ 3 \end{bmatrix}_q}{[n]_q^4} \right) s^3 + \mathcal{O}(s^4). \quad (18)$$

Proof. The linear coefficient of f is $f'(0) = [n]_q \neq 0$, so the analytic inverse-function theorem supplies one and only one local branch. Rewrite $s = f(a)$ as $a = s\phi(a)$, where $\phi(a) = a/f(a)$. Lagrange inversion gives $[s^m]a = m^{-1}[z^{m-1}]\phi(z)^m$, which is [equation \(17\)](#). Finally

$$s = [n]_q a - q \begin{bmatrix} n \\ 2 \end{bmatrix}_q a^2 + q^3 \begin{bmatrix} n \\ 3 \end{bmatrix}_q a^3 + \mathcal{O}(a^4).$$

Substitute $a = A_1 s + A_2 s^2 + A_3 s^3 + \dots$ and compare coefficients. This gives

$$A_1 = [n]_q^{-1}, \quad A_2 = \frac{q \begin{bmatrix} n \\ 2 \end{bmatrix}_q}{[n]_q^3}, \quad A_3 = \frac{2q^2 \begin{bmatrix} n \\ 2 \end{bmatrix}_q^2}{[n]_q^5} - \frac{q^3 \begin{bmatrix} n \\ 3 \end{bmatrix}_q}{[n]_q^4},$$

proving both displayed claims. $\square$

16.2 All zero sheets and the complete real atlas

For $q \neq 0$, the argument zeros are $a_j = q^{-j}$, $0 \leq j < n$. They are distinct precisely when $q^r \neq 1$ for $1 \leq r < n$.

Proposition 16.2 (Exact normalization at every simple zero). *Assume $q \neq 0$ and $q^r \neq 1$ for $1 \leq r < n$. Then*

$$\partial_a P_n(q^{-j}, q) = (-1)^{j+1} q^{-j(j-1)/2} (q; q)_j (q; q)_{n-1-j}. \quad (19)$$

Consequently the inverse sheet through $(y, a) = (0, q^{-j})$ begins

$$a(y) = q^{-j} + \frac{(-1)^{j+1} q^{j(j-1)/2}}{(q; q)_j (q; q)_{n-1-j}} y + \mathcal{O}(y^2). \quad (20)$$

Proof. Differentiate the unique vanishing factor:

$$\partial_a P_n(q^{-j}, q) = -q^j \prod_{\ell \neq j} (1 - q^{\ell-j}).$$

For $\ell < j$, write $1 - q^{\ell-j} = -q^{-(j-\ell)}(1 - q^{j-\ell})$. Their product is $(-1)^j q^{-j(j+1)/2} (q; q)_j$, while the factors with $\ell > j$ give $(q; q)_{n-1-j}$. Multiplication by $-q^j$ proves [equation \(19\)](#). Its nonvanishing permits ordinary analytic inversion, and division by the derivative gives [equation \(20\)](#). $\square$

Theorem 16.3 (Complete real argument atlas). *Let $q > 0$, $q \neq 1$, and order the zeros increasingly as $r_0 < \dots < r_{n-1}$. Then:*

1. $P_n(\cdot, q)$ is positive and strictly decreasing on $(-\infty, r_0)$, and maps that interval bijectively onto $(0, \infty)$.
2. Every gap (r_j, r_{j+1}) contains exactly one critical point c_j . It is simple, so the two inverse sheets meeting at $P_n(c_j, q)$ have square-root ramification.
3. There is no critical point on (r_{n-1}, ∞) .

For $n \geq 2$, the maximal real intervals on which $P_n(\cdot, q)$ is one-to-one are

$$(-\infty, c_0], \quad [c_{j-1}, c_j] \quad (1 \leq j \leq n-2), \quad [c_{n-2}, \infty).$$

Each contains exactly one simple zero r_j , and together they exhaust the maximal real injective branches; their interiors are the maximal open domains of regular real inversion. Equivalently, each zero-free gap (r_j, r_{j+1}) consists of two half-branches meeting at the fold c_j . When $n = 1$, the sole maximal branch is all of $\mathbb{R}$.

Proof. Since $1 - aq^j = -q^j(a - q^{-j})$, away from the roots

$$\frac{\partial_a P_n(a, q)}{P_n(a, q)} = \sum_{j=0}^{n-1} \frac{1}{a - r_j}, \quad \frac{d}{da} \frac{\partial_a P_n}{P_n} = - \sum_{j=0}^{n-1} \frac{1}{(a - r_j)^2} < 0. \quad (21)$$

On the leftmost interval every summand in the first expression is negative. All product factors are positive there, the product tends to $+\infty$ at $-\infty$, and it tends to zero at r_0 ; this proves the first item. On (r_j, r_{j+1}) , the logarithmic derivative decreases continuously from $+\infty$ to $-\infty$, so it has exactly one zero. At that zero, P_n''/P_n equals the strictly negative second expression in equation (21); hence $P_n'' \neq 0$. Taylor's formula and reversion of its first nonzero quadratic term give two square-root sheets. To the right of the largest root every summand in equation (21) is positive. The preceding argument, together with the nonvanishing derivatives at the simple zeros, proves that $c_0, \dots, c_{n-2}$ are all the real zeros of $\partial_a P_n$. Its sign is therefore fixed and nonzero on each component of their complement. The function is strictly monotone on the closure of each such component. Because every c_j is a strict local extremum, no closure can be extended past one of its finite endpoints while preserving injectivity. These closures are therefore precisely the maximal injective intervals, while their interiors are the maximal regular inverse domains. The interlacing $c_{j-1} < r_j < c_j$, with the evident one-sided versions at the extremes, shows that each contains exactly one simple zero; analytic inversion continues through that zero. For $n = 1$, the polynomial is linear and the same conclusion is immediate. $\square$

Put $N = \binom{n}{2}$, $A_n = (-1)^n q^N$, and $e_1 = \sum_{j=0}^{n-1} q^{-j}$. Factoring the leading power of a gives

$$P_n(a, q) = A_n a^n \left(1 - \frac{e_1}{a} + \mathcal{O}(a^{-2}) \right).$$

Thus, on each choice $T = (y/A_n)^{1/n}$,

$$a = T + \frac{e_1}{n} + \mathcal{O}(T^{-1}). \quad (22)$$

Indeed, substitution of $a = T + c + \mathcal{O}(T^{-1})$ makes the coefficient of T^{-1} equal to $nc - e_1$, which forces $c = e_1/n$.

16.3 Cyclotomic collisions in the argument

Theorem 16.4 (Exact root-of-unity factorization and ramification). *Let ζ be primitive of order d , and write $n = Md + r$ with $0 \leq r < d$. Then*

$$(a; \zeta)_n = (1 - a^d)^M (a; \zeta)_r. \quad (23)$$

At $a_s = \zeta^{-s}$, $0 \leq s < d$, the zero multiplicity is

$$\nu_s = M + \mathbf{1}_{\{s < r\}}. \quad (24)$$

For every s with $\nu_s \geq 1$, the point a_s is a zero of multiplicity ν_s . On the ramified cover of a sufficiently small punctured target neighborhood, the inverse at target zero then has exactly ν_s local Puiseux sheets

coalescing at a_s . Equivalently, after choosing a ν_s -th root on a simply connected punctured target domain, these are the ν_s determinations obtained by multiplying that root by the ν_s -th roots of unity. When $\nu_s = 0$, a_s is not a zero and contributes no inverse germ at target zero. At the unit target $y = 1$, the branch through $a = 0$ is regular when $r > 0$. When $r = 0$, it has the d local sheets

$$a = \omega \left(\frac{1-y}{M} \right)^{1/d} + \mathcal{O}((1-y)^{2/d}), \quad \omega^d = 1. \quad (25)$$

Proof. Partition the indices into M complete residue blocks modulo d and one residual block. Each complete block contributes $\prod_{j=0}^{d-1} (1 - a\zeta^j) = 1 - a^d$, proving equation (23). A factor vanishes at a_s exactly when its index is congruent to $s \pmod{d}$. That residue occurs M times in the complete blocks and once more precisely when $s < r$, proving equation (24). When $\nu_s \geq 1$, removing the nonvanishing factors leaves a nonzero constant times $(a - a_s)^{\nu_s}$, whose equation with y has ν_s Puiseux determinations. When $\nu_s = 0$, no factor vanishes at a_s , so there is no inverse germ over target zero there.

At $a = 0$, the derivative is $-\sum_{j=0}^{n-1} \zeta^j = -\sum_{j=0}^{r-1} \zeta^j$, which is nonzero for $r > 0$. If $r = 0$, equation (23) gives $(a; \zeta)_n = (1 - a^d)^M = 1 - Ma^d + \mathcal{O}(a^{2d})$; solving its leading equation gives equation (25). $\square$

At $q = -1$, this includes

$$(a; -1)_{2M} = (1 - a^2)^M, \quad (a; -1)_{2M+1} = (1 - a)(1 - a^2)^M.$$

For even order every argument sheet is obtained from $a = \pm \sqrt{1 - \eta y^{1/M}}$, where $\eta^M = 1$.

17 Finite-product inversion in the base

17.1 The canonical real base branch

Theorem 17.1 (Global finite base inverse on $0 \leq q \leq 1$). *Let $n \geq 2$.*

1. *If $0 < a < 1$, then $q \mapsto P_n(a, q)$ is strictly decreasing from $1 - a$ to $(1 - a)^n$. Every $(1 - a)^n < y < 1 - a$ has exactly one inverse $q \in (0, 1)$.*
2. *If $a < 0$, the map is strictly increasing from $1 - a$ to $(1 - a)^n$. Every $1 - a < y < (1 - a)^n$ has exactly one inverse $q \in (0, 1)$.*

For $a = 0$, $a = 1$, or $n = 1$, the map is constant in q .

Proof. For $0 < q < 1$, every denominator in equation (15) is positive. Its sign is therefore $-\text{sgn}(a)$, and the nonzero derivative proves strict monotonicity. Direct substitution gives $P_n(a, 0) = 1 - a$ and $P_n(a, 1) = (1 - a)^n$. Continuity and strict monotonicity now give the stated ranges and uniqueness. The three constant cases follow immediately. $\square$

17.2 All-order inversion at $q = 0$

Assume $n \geq 2$ and $a \notin \{0, 1\}$, and normalize the target by

$$\eta = \frac{(1 - a) - y}{a(1 - a)}. \quad (26)$$

Define $F_{n,a}(q) = (1 - P_n(a, q)/(1 - a))/a$. Then $F_{n,a}(q) = q + \mathcal{O}(q^2)$, and $P_n(a, q) = y$ is precisely $F_{n,a}(q) = \eta$.

Theorem 17.2 (Exact coefficient engine at $q = 0$). *The inverse branch $q = Q_{n,a}(\eta)$ is analytic at zero and satisfies*

$$[\eta^m]Q_{n,a}(\eta) = \frac{1}{m}[z^{m-1}] \left(\frac{z}{F_{n,a}(z)} \right)^m. \quad (27)$$

Moreover,

$$\log \frac{P_n(a, q)}{1-a} = \sum_{r \geq 1} \ell_r^{(n)}(a) q^r, \quad \ell_r^{(n)}(a) = - \sum_{\substack{d|r \\ 1 \leq d \leq n-1}} \frac{a^{r/d}}{r/d}. \quad (28)$$

For $n \geq 6$,

$$q = \eta - \eta^2 + (1+a)\eta^3 - (1+4a)\eta^4 + (1+11a+3a^2)\eta^5 + \mathcal{O}(\eta^6). \quad (29)$$

For $n = 2$, $q = \eta$ exactly; for $n = 3$, the cubic coefficient is $2 + a$; and for every $n \geq 4$ it is $1 + a$.

Proof. The coefficient of q in $F_{n,a}$ is one, so analytic inversion and Lagrange's formula give [equation \(27\)](#). For the divisor formula, expand

$$\log \frac{P_n(a, q)}{1-a} = - \sum_{j=1}^{n-1} \sum_{m \geq 1} \frac{a^m q^{jm}}{m}.$$

The coefficient of q^r comes from $jm = r$, proving [equation \(28\)](#).

When $n \geq 6$, subset expansion gives

$$F_{n,a}(q) = q + q^2 + (1-a)q^3 + (1-a)q^4 + (1-2a)q^5 + \mathcal{O}(q^6).$$

Substitute $q = \eta + A_2\eta^2 + \dots + A_5\eta^5$ and compare powers. Successively, $A_2 = -1$, $A_3 = 1 + a$, $A_4 = -(1 + 4a)$, and $A_5 = 1 + 11a + 3a^2$. The same calculation with the available positive indices gives the stated low-order exceptions. $\square$

17.3 The complete logarithmic jet at $q = 1$

Put $q = e^{-t}$, $S_m(n) = \sum_{j=0}^{n-1} j^m$, and choose the local logarithm of $y/(1-a)^n$.

Theorem 17.3 (All-order regular base inverse at $q = 1$). *Let $n \geq 2$ and $a \in \mathbb{C} \setminus \{0, 1\}$. With*

$$u = \log \frac{y}{(1-a)^n}, \quad c_m = \frac{(-1)^{m+1}}{m!} \text{Li}_{1-m}(a) S_m(n),$$

the forward germ and its inverse are

$$u = \sum_{m \geq 1} c_m t^m, \quad (30)$$

$$t = \frac{u}{c_1} - \frac{c_2}{c_1^3} u^2 + \left(\frac{2c_2^2}{c_1^5} - \frac{c_3}{c_1^4} \right) u^3 + \mathcal{O}(u^4), \quad q = e^{-t}. \quad (31)$$

Here $c_1 = a \binom{n}{2} / (1-a) \neq 0$, and all subsequent inverse coefficients are obtained uniquely by triangular coefficient comparison.

Proof. For $|a| < 1$,

$$\log(1 - ae^{-x}) = - \sum_{r \geq 1} \frac{a^r e^{-rx}}{r} = \log(1-a) + \sum_{m \geq 1} \frac{(-1)^{m+1}}{m!} \text{Li}_{1-m}(a) x^m.$$

Substitute $x = jt$ and sum over $0 \leq j < n$. Both sides are analytic in a away from $a = 1$, and the nonpositive-index polylogarithms are rational, so analytic continuation proves the formula for every stated a . Since $S_1(n) = \binom{n}{2}$, the linear coefficient is nonzero. Substitution of $t = A_1 u + A_2 u^2 + A_3 u^3 + \dots$ gives $A_1 = c_1^{-1}$, $A_2 = -c_2 c_1^{-3}$, and $A_3 = 2c_2^2 c_1^{-5} - c_3 c_1^{-4}$. Every later equation is linear in the new coefficient. $\square$

For $n \geq 2$, holding $y \neq 0$ fixed while $q \rightarrow 1$ gives a useful mixed inverse. Choose $r = y^{1/n}$, put $a_0 = 1 - r$, and keep that root determination fixed. Substitution in the logarithm of the level equation gives

$$a(t) = a_0 + \frac{n-1}{2}a_0t + \frac{a_0(n-1)(2n-4-3a_0(n-1))}{24(1-a_0)}t^2 + \mathcal{O}(t^3). \quad (32)$$

Indeed, writing $a = a_0 + At + Bt^2$ makes the constant term $n \log(1 - a_0) = \log y$; the linear and quadratic equations give $A = (n-1)a_0/2$ and the displayed B .

17.4 The exceptional slice $q = -1$

Theorem 17.4 (Complete zero-free criticality at $q = -1$). *Let $n \geq 3$ and $a \notin \{0, \pm 1\}$. Then $\partial_q P_n(a, -1) = 0$ if and only if*

$$a = a_*(n) = \begin{cases} 1/(n-1), & n \text{ even}, \\ -1/n, & n \text{ odd}. \end{cases} \quad (33)$$

For $n = 2s$, $s \geq 2$,

$$y_* = \left(\frac{4s(s-1)}{(2s-1)^2} \right)^s, \quad (34)$$

$$L_2 := \partial_q^2 \log P_{2s}(a_*, -1) = -\frac{(s+1)(2s-1)^2}{12s(s-1)}. \quad (35)$$

For $n = 2s+1$, $s \geq 2$,

$$y_* = \left(\frac{2s+2}{2s+1} \right)^{s+1} \left(\frac{2s}{2s+1} \right)^s, \quad (36)$$

$$L_2 = -\frac{(s-1)(2s+1)^2}{12s(s+1)}. \quad (37)$$

All these slices are quadratic folds, with local inverse

$$q = -1 \pm \sqrt{\frac{2(y-y_*)}{y_* L_2}} + \mathcal{O}(y-y_*). \quad (38)$$

Within this zero-free family, the sole higher degeneracy is

$$P_3(-1/3, q) = \frac{32}{27} + \frac{4}{27}(q+1)^3. \quad (39)$$

Proof. Let $L = \log P_n$. At $q = -1$, separation of even and odd indices in [equation \(15\)](#) gives

$$L_q(a, -1) = a \left(\frac{E}{1-a} - \frac{O}{1+a} \right),$$

where E and O are the sums of the positive even and odd indices below n . For $n = 2s$, $E = s(s-1)$, $O = s^2$, giving $a = 1/(2s-1)$. For $n = 2s+1$, $E = s(s+1)$, $O = s^2$, giving $a = -1/(2s+1)$. Substitution in $P_n(a, -1) = (1-a)^{\lceil n/2 \rceil} (1+a)^{\lfloor n/2 \rfloor}$ gives the two critical values.

Differentiating [equation \(15\)](#) once more gives

$$L_{qq} = -a \sum_{j=1}^{n-1} \left(\frac{j(j-1)q^{j-2}}{1-aq^j} + \frac{aj^2q^{2j-2}}{(1-aq^j)^2} \right).$$

For completeness, the four parity sums needed here are

	$\sum j(j-1)$	$\sum j^2$
$2, 4, \dots, 2(s-1)$	$\frac{s(s-1)(4s-5)}{3}$	$\frac{2s(s-1)(2s-1)}{3}$
$1, 3, \dots, 2s-1$	$\frac{s(s-1)(4s+1)}{3}$	$\frac{s(4s^2-1)}{3}$
$2, 4, \dots, 2s$	$\frac{s(s+1)(4s-1)}{3}$	$\frac{2s(s+1)(2s+1)}{3}$

At $q = -1$, the first summand in L_{qq} uses the even row with a plus sign and the odd row with a minus sign, whereas its j^2 summand uses both rows with a plus sign. For $n = 2s$, insert the first two rows and $a = 1/(2s-1)$; for $n = 2s+1$, insert the last two rows and $a = -1/(2s+1)$. Cancelling the common factors gives exactly [equations \(35\) and \(37\)](#). The even expression never vanishes for $s \geq 2$; the odd expression vanishes only at $s = 1$, where direct multiplication proves [equation \(39\)](#). At a critical point $P_{qq} = P_n L_{qq} = y_* L_2 \neq 0$. Taylor expansion and reversion of the quadratic term give [equation \(38\)](#). $\square$

The exclusions in [theorem 17.4](#) are structural, not hidden exceptions. At $a = 0$ the product is identically one, and at $a = 1$ it is identically zero. At $a = -1$,

$$P_n(-1, q) = \prod_{j=0}^{n-1} (1 + q^j)$$

has at $q = -1$ a zero of exact order $r = \lfloor n/2 \rfloor$, because exactly the odd-indexed factors vanish and each does so simply. More precisely, with $e = \lceil n/2 \rceil$ and $J_n = \prod_{\substack{1 \leq j < n \\ j \text{ odd}}} j$,

$$P_n(-1, q) = 2^e J_n (q+1)^r + \mathcal{O}((q+1)^{r+1}). \quad (40)$$

Indeed, every even-indexed factor tends to two and $1 + q^j = j(q+1) + \mathcal{O}((q+1)^2)$ for odd j . Thus, on the ramified target cover,

$$q = -1 + \omega \left(\frac{y}{2^e J_n} \right)^{1/r} + \mathcal{O}(y^{2/r}), \quad \omega^r = 1.$$

This zero-target cyclotomic ramification is the repeated-factor phenomenon seen in [theorem 16.4](#); it is different from the nonzero-level critical points classified above.

Corollary 17.5 (The cubic slice lies on a Morse saddle). *For every odd $n = 2s + 1$, $s \geq 1$, the point $(a, q) = (-1/(2s+1), -1)$ is a nondegenerate saddle of $(a, q) \mapsto P_n(a, q)$. Thus [equation \(39\)](#) is a tangent one-dimensional slice, not a cusp of the full parameter surface.*

Proof. At this point both L_a and L_q vanish. Direct differentiation of $L = \sum_{j=0}^{2s} \log(1 - aq^j)$, followed by the parity sums, gives

$$L_{aa} = -\frac{(2s+1)^3}{4s(s+1)}, \quad L_{aq} = -\frac{(2s+1)^2}{4(s+1)}, \quad L_{qq} = -\frac{(s-1)(2s+1)^2}{12s(s+1)}.$$

For instance,

$$L_{aa} = -\frac{s+1}{(1-a)^2} - \frac{s}{(1+a)^2}, \quad L_{aq} = \frac{s(s+1)}{(1-a)^2} - \frac{s^2}{(1+a)^2},$$

which yield the first two entries after substituting $a = -1/(2s+1)$; the third is [equation \(37\)](#). Hence

$$\det \text{Hess } L = -\frac{(2s+1)^4(s^2+s+1)}{48s^2(s+1)^2} < 0. \quad (41)$$

Since $P_n \neq 0$ and its gradient vanishes, its Hessian is P_n times that of L . It is therefore also nonsingular and indefinite. $\square$

17.5 Reciprocal-base inversion at infinity

Let $n \geq 2$, $a \notin \{0, 1\}$, $N = \binom{n}{2}$, $C = (1 - a)(-a)^{n-1}$, and

$$H(z) = \prod_{j=1}^{n-1} (1 - a^{-1}z^j).$$

The exact factorization $P_n(a, q) = Cq^N H(q^{-1})$ converts infinity into an ordinary point.

Theorem 17.6 (All-order finite base inverse at infinity). *Choose an N -th root $x = (C/y)^{1/N}$. The corresponding branch $s = q^{-1}$ is the convergent solution of*

$$s = xH(s)^{1/N}, \quad [x^m]s(x) = \frac{1}{m}[z^{m-1}]H(z)^{m/N}. \quad (42)$$

In particular,

$$q = x^{-1} + \frac{1}{aN} + \mathcal{O}(x). \quad (43)$$

All N complex sheets arise from the N choices of x .

Proof. The level equation is $y = Cs^{-N}H(s)$, hence $x = sH(s)^{-1/N}$, equivalently the first equation in [equation \(42\)](#). Since $H(0) = 1$, a consistent analytic N -th root exists near zero and the analytic inverse-function theorem gives a unique convergent branch. Lagrange inversion proves the coefficient formula. Finally $H(s) = 1 - a^{-1}s + \mathcal{O}(s^2)$, so $s = x - a^{-1}x^2/N + \mathcal{O}(x^3)$. Taking the reciprocal gives [equation \(43\)](#). $\square$

Warning 17.7 (Claims deliberately not promoted). The source reports contain exact-looking secondary-discriminant factorizations for a proposed first Maxwell collision at $n = 5$. No independent exact certificate for the large factorization is currently part of the canonical package, so that special assertion is not stated here as a theorem. The proved branch atlas remains valid, and a certified resultant/Sturm computation would suffice to add the collision theorem later.

18 The infinite product: analytic domain and certified tails

For $|q| < 1$, put

$$P(a, q) := (a; q)_\infty = \prod_{j=0}^{\infty} (1 - aq^j).$$

The domain condition is part of the definition. Values on $|q| = 1$, radial limits toward that boundary, and exterior reciprocal normalizations are different objects and will be treated separately.

Theorem 18.1 (Normal convergence, zeros, and derivative kernels). *Let $0 < \rho < 1$ and $R > 0$. Uniformly for $|q| \leq \rho$, $|a| \leq R$, and $N \geq 0$,*

$$\left| \prod_{j=N}^{\infty} (1 - aq^j) - 1 \right| \leq \exp\left(\frac{R\rho^N}{1 - \rho}\right) - 1. \quad (44)$$

If $R\rho^N < 1$, the logarithm obtained by continuation from $a = 0$ satisfies

$$\left| \sum_{j=N}^{\infty} \log(1 - aq^j) \right| \leq \frac{R\rho^N}{(1 - \rho)(1 - R\rho^N)}. \quad (45)$$

Consequently P is holomorphic on $\{(a, q) \in \mathbb{C}^2 : |q| < 1\}$, and is entire in a for fixed q . For $0 < |q| < 1$, its zeros as a function of a are exactly $a = q^{-m}$, $m \geq 0$, and they are simple. Away from those zeros,

$$\partial_a \log P(a, q) = - \sum_{j \geq 0} \frac{q^j}{1 - aq^j}, \quad (46)$$

$$\partial_a^2 \log P(a, q) = - \sum_{j \geq 0} \frac{q^{2j}}{(1 - aq^j)^2}, \quad (47)$$

$$\partial_q \log P(a, q) = -a \sum_{j \geq 1} \frac{j q^{j-1}}{1 - aq^j}. \quad (48)$$

All three series converge locally uniformly on the indicated zero-free set.

Proof. Put $u_j = -aq^j$. Expanding finite products and applying the triangle inequality gives

$$\left| \prod_{j=N}^M (1 + u_j) - 1 \right| \leq \prod_{j=N}^M (1 + |u_j|) - 1 \leq \exp \left(\sum_{j=N}^M |u_j| \right) - 1.$$

Since $\sum_{j \geq N} |u_j| \leq R\rho^N/(1 - \rho)$, the finite tails form a uniformly Cauchy family and their limit obeys [equation \(44\)](#). If $R\rho^N < 1$, then

$$|\log(1 - z)| \leq \sum_{k \geq 1} \frac{|z|^k}{k} \leq \frac{|z|}{1 - |z|}$$

for $|z| < 1$. With $|aq^j| \leq R\rho^N \rho^{j-N}$, summation proves [equation \(45\)](#).

Locally uniform limits of the finite holomorphic products are holomorphic. If no factor vanishes, choose N so that $|aq^j| < 1/2$ for $j \geq N$. The tail is the exponential of an absolutely convergent logarithmic sum and is therefore nonzero; hence the only zeros are the factor zeros. At $a = q^{-m}$, exactly one factor vanishes, so the zero is simple. On a compact zero-free set, the denominators are separated from zero for the finite head, while the tails in [equations \(46\) to \(48\)](#) are bounded by convergent geometric series with at most a polynomial factor in j . Normal convergence therefore permits termwise differentiation and proves all three identities. $\square$

Lemma 18.2 (Lambert logarithm). For $|a| < 1$ and $|q| < 1$,

$$\log(a; q)_\infty = - \sum_{m \geq 1} \frac{a^m}{m(1 - q^m)}, \quad (49)$$

where the logarithm is the branch equal to zero at $a = 0$. The double series used in its derivation is absolutely and locally uniformly convergent.

Proof. For each factor, $\log(1 - aq^j) = - \sum_{m \geq 1} a^m q^{jm}/m$. On compact subdiscs $|a| \leq \alpha < 1$, $|q| \leq \rho < 1$,

$$\sum_{j \geq 0} \sum_{m \geq 1} \frac{|a|^m |q|^{jm}}{m} \leq \sum_{m \geq 1} \frac{\alpha^m}{m(1 - \rho^m)} \leq \frac{-\log(1 - \alpha)}{1 - \rho} < \infty.$$

Thus Fubini's theorem applies, and summing $\sum_{j \geq 0} q^{jm} = (1 - q^m)^{-1}$ gives [equation \(49\)](#). $\square$

19 Argument inversion and the complete real sheet atlas

Theorem 19.1 (Principal inverse, all gaps, and exact zero normalization). Fix $0 < q < 1$, write $P_q(a) = (a; q)_\infty$, and put $r_m = q^{-m}$.

1. P_q is positive and strictly decreasing from $+\infty$ to 0 on $(-\infty, 1)$. It therefore has a unique real-analytic inverse $\mathcal{A}_q : (0, \infty) \rightarrow (-\infty, 1)$.
2. Every r_m is a simple zero, with

$$d_m := P'_q(r_m) = (-1)^{m+1} q^{-m(m-1)/2} (q; q)_m (q; q)_\infty. \quad (50)$$

Hence its inverse germ is $a_m(y) = r_m + y/d_m + \mathcal{O}(y^2)$.

3. On the gap $I_m = (r_m, r_{m+1})$, the sign of P_q is $\sigma_m = (-1)^{m+1}$. There is exactly one critical point $c_m \in I_m$, and $P''_q(c_m) \neq 0$. If $M_m = |P_q(c_m)|$, then I_m contains two roots of $P_q(a) = y$ when $0 < \sigma_m y < M_m$, one double root when $\sigma_m y = M_m$, and no root when $\sigma_m y < 0$ or $\sigma_m y > M_m$. At $y = 0$, its two endpoint roots are r_m, r_{m+1} .

The principal inverse is strictly decreasing and convex, and, with

$$S_k(a) = \sum_{j \geq 0} \frac{q^{kj}}{(1 - aq^j)^k},$$

it satisfies

$$\mathcal{A}'_q(y) = -\frac{1}{y S_1(\mathcal{A}_q(y))}, \quad (51)$$

$$\mathcal{A}''_q(y) = \frac{S_1(\mathcal{A}_q(y))^2 - S_2(\mathcal{A}_q(y))}{y^2 S_1(\mathcal{A}_q(y))^3} > 0. \quad (52)$$

Proof. For $a < 1$, all factors are positive and $\partial_a \log P_q < 0$ by [equation \(46\)](#). As $a \uparrow 1$,

$$P_q(a) = (1 - a)(aq; q)_\infty \sim (q; q)_\infty (1 - a) \rightarrow 0,$$

where normal convergence gives the limit of the second factor. If $a = -x$, then $P_q(-x) = \prod_{j \geq 0} (1 + xq^j) \geq 1 + x \rightarrow \infty$. This proves the first bijection and, since its derivative never vanishes, real analyticity of its inverse.

To compute the derivative at r_m , remove the vanishing factor:

$$P'_q(r_m) = -q^m \prod_{j \neq m} (1 - q^{j-m}).$$

The indices $j < m$ contribute $(-1)^m q^{-m(m+1)/2} (q; q)_m$; the indices $j > m$ contribute $(q; q)_\infty$. This proves [equation \(50\)](#) and the inverse germ.

On every compact subset avoiding the zeros, [equation \(46\)](#) may be written

$$\frac{P'_q(a)}{P_q(a)} = \sum_{j \geq 0} \frac{1}{a - r_j}, \quad \left(\frac{P'_q}{P_q} \right)'(a) = - \sum_{j \geq 0} \frac{1}{(a - r_j)^2} < 0.$$

In I_m , the first sum tends to $+\infty$ at the left endpoint and to $-\infty$ at the right endpoint. It therefore has one and only one zero. At that point P''_q/P_q equals the strictly negative second sum, so the critical point is nondegenerate. There are $m + 1$ negative factors in the gap, giving the sign σ_m . The logarithmic derivative has opposite signs on the two sides of c_m ; hence the absolute value rises strictly from zero to M_m and falls strictly to zero. The root count follows.

Implicit differentiation gives [equation \(51\)](#). Since

$$S_1(a)^2 - S_2(a) = 2 \sum_{0 \leq i < j} \frac{q^{i+j}}{(1 - aq^i)(1 - aq^j)} > 0$$

for $a < 1$, a second differentiation gives [equation \(52\)](#) and proves convexity. $\square$

19.1 Endpoint and central inverse jets

Euler's expansion

$$(a; q)_\infty = \sum_{m \geq 0} \frac{(-1)^m q^{\binom{m}{2}}}{(q; q)_m} a^m$$

and Lagrange inversion give, with $s = 1 - y$,

$$\begin{aligned} \mathcal{A}_q(y) &= (1 - q)s + \frac{q(1 - q)}{1 + q} s^2 \\ &\quad + \frac{q^2(1 - q)(q^2 + q + 2)}{(1 + q)^2(q^2 + q + 1)} s^3 + \mathcal{O}(s^4), \end{aligned} \quad (53)$$

$$[s^m] \mathcal{A}_q(1 - s) = \frac{1}{m} [z^{m-1}] \left(\frac{z}{1 - (z; q)_\infty} \right)^m. \quad (54)$$

This follows exactly as in [theorem 16.1](#): the forward linear coefficient is $1/(1 - q)$, and direct coefficient comparison gives the three displayed terms.

At the other endpoint, let

$$C_q = (q; q)_\infty, \quad L_1(q) = \sum_{j \geq 1} \frac{q^j}{1 - q^j}, \quad v = \frac{y}{C_q}.$$

Writing $a = 1 - \delta$ gives

$$(a; q)_\infty = C_q \delta (1 + L_1(q) \delta + \mathcal{O}(\delta^2)).$$

Reversion therefore yields

$$\mathcal{A}_q(y) = 1 - v + L_1(q) v^2 + \mathcal{O}(v^3). \quad (55)$$

The radius of either inverse series is bounded by the nearest singular value encountered by its chosen sheet; such a singular value may be critical or asymptotic, and need not be the numerically nearest critical value belonging to some other sheet.

19.2 Remote real critical points

Theorem 19.2 (Remote-gap asymptotics). *Let $m = k + 1$. As $k \rightarrow \infty$,*

$$c_k(q) = q^{-m} \left(1 - \frac{1}{m+1} + \mathcal{O}(m^{-2}) \right), \quad (56)$$

$$\begin{aligned} |P_q(c_k)| &= (q; q)_\infty^2 q^{-m(m+1)/2} \frac{m^m}{(m+1)^{m+1}} (1 + \mathcal{O}(m^{-1})) \\ &= \frac{(q; q)_\infty^2}{e(m+1)} q^{-m(m+1)/2} (1 + \mathcal{O}(m^{-1})). \end{aligned} \quad (57)$$

The error constants are locally uniform for q in compact subsets of $(0, 1)$.

Proof. Parameterize the gap by $a = q^{-m}(1 - \varepsilon)$, $0 < \varepsilon < 1 - q$. Separating the m factors before the right endpoint gives the exact identity

$$\begin{aligned} P_q(a) &= (-1)^m q^{-m(m+1)/2} \varepsilon (1 - \varepsilon)^m G_m(\varepsilon), \\ G_m(\varepsilon) &= \left(\frac{q}{1 - \varepsilon}; q \right)_m ((1 - \varepsilon)q; q)_\infty. \end{aligned} \quad (58)$$

If $\varepsilon = \mathcal{O}(m^{-1})$, logarithmic differentiation of the two products shows uniformly that

$$\partial_\varepsilon \log G_m(\varepsilon) = \mathcal{O}(1).$$

Indeed, the resulting summands are bounded by constant multiples of $q^j/(1 - q^j)$, whose sum converges. The same logarithmic estimate, together with the omitted tail $\sum_{j>m} q^j = \mathcal{O}(q^m)$, gives

$$G_m(\varepsilon) = (q; q)_\infty^2 (1 + \mathcal{O}(m^{-1})).$$

It remains to verify that the unknown extremum lies in this range. The logarithmic derivative of [equation \(58\)](#) is positive as $\varepsilon \downarrow 0$. At $\varepsilon = 2/(m+1)$, which lies inside the gap for all sufficiently large m , the first two terms below have size $-m/2 + \mathcal{O}(1)$, while the already established $G'_m/G_m = \mathcal{O}(1)$ estimate applies there. The derivative is therefore negative. Uniqueness of the gap extremum gives $\varepsilon_* < 2/(m+1)$, so $\varepsilon_* = \mathcal{O}(m^{-1})$ without circular reasoning.

At that unique extremum, logarithmic differentiation of [equation \(58\)](#) now gives

$$0 = \frac{1}{\varepsilon} - \frac{m}{1 - \varepsilon} + \partial_\varepsilon \log G_m(\varepsilon).$$

The bounded last term perturbs the zero of the first two terms by $\mathcal{O}(m^{-2})$, so $\varepsilon = (m+1)^{-1} + \mathcal{O}(m^{-2})$. Substitution proves [equations \(56\)](#) and [\(57\)](#); finally $(1 - 1/(m+1))^m = e^{-1}(1 + \mathcal{O}(m^{-1}))$. $\square$

20 Large targets and the log-periodic principal inverse

Theorem 20.1 (Theta-modulated growth and its inverse). *Let $q = e^{-\tau}$, $\tau > 0$. For $x = e^L \geq 1$, write uniquely $x = q^{-m}r$, $m \in \mathbb{N}_0$, $1 \leq r < q^{-1}$, and $\rho = \log r \in [0, \tau)$. Define on that interval*

$$\Psi_q(\rho) = \log \left((-e^\rho; q)_\infty (-qe^{-\rho}; q)_\infty \right) - \frac{\rho}{2} - \frac{\rho^2}{2\tau}, \quad (59)$$

and extend it τ -periodically. Then Ψ_q is smooth and

$$\log(-e^L; q)_\infty = \frac{L^2}{2\tau} + \frac{L}{2} + \Psi_q(L \bmod \tau) + \mathcal{O}(e^{-L}). \quad (60)$$

If $Y = (-x; q)_\infty \rightarrow \infty$, set

$$R = \sqrt{2\tau \log Y}, \quad \rho_0 = (R - \tau/2) \bmod \tau.$$

Then

$$\log x = R - \frac{\tau}{2} + \frac{\tau(\tau/8 - \Psi_q(\rho_0))}{R} + \mathcal{O}((\log Y)^{-1}). \quad (61)$$

Proof. The finite shift is exact:

$$(-q^{-m}r; q)_\infty = r^m q^{-m(m+1)/2} (-q/r; q)_m (-r; q)_\infty.$$

Uniformly for $1 \leq r \leq q^{-1}$, replacing the finite product by $(-q/r; q)_\infty$ has relative error $1 + \mathcal{O}(q^m)$, by [equation \(44\)](#). Since $L = m\tau + \rho$,

$$m\rho + \frac{\tau m(m+1)}{2} = \frac{L^2}{2\tau} + \frac{L}{2} - \frac{\rho^2}{2\tau} - \frac{\rho}{2}.$$

Taking logarithms proves [equation \(60\)](#). Normal convergence permits all ρ -derivatives on the compact fundamental interval. Under $\rho \mapsto \rho + \tau$, the product inside the logarithm is multiplied by $e^{\rho+\tau}$, exactly the change of the two subtracted terms, so the extension is smooth and periodic.

Completing the square rewrites the forward equation as

$$2\tau \log Y = (L + \tau/2)^2 + 2\tau(\Psi_q(\rho) - \tau/8) + \mathcal{O}(e^{-L}).$$

Taking the positive square root first gives $L = R - \tau/2 + \mathcal{O}(R^{-1})$. Insert this once into the bounded correction. The exact phase differs from ρ_0 , on the periodic circle, by $\mathcal{O}(R^{-1})$; smoothness of Ψ_q changes the R^{-1} correction by only $\mathcal{O}(R^{-2})$. Since $R^{-2} = \mathcal{O}((\log Y)^{-1})$, this is [equation \(61\)](#). $\square$

21 Inversion in the base at $q = 0$ and on the real interval

Theorem 21.1 (Real base atlas and the stable small-base jet). *Fix real a .*

1. *If $0 < a < 1$, $q \mapsto (a; q)_\infty$ decreases strictly on $[0, 1)$ from $1 - a$ to 0.*
2. *If $a < 0$, it increases strictly from $1 - a$ to $+\infty$.*
3. *If $a = 0$ it is identically 1, and if $a = 1$ it is identically 0.*
4. *If $a > 1$, it has the distinct simple base zeros $q_j = a^{-1/j}$, $j \geq 1$, accumulating at 1; hence only interval-by-interval base inverses are possible.*

For $a \notin \{0, 1\}$, set $\eta = ((1 - a) - y)/(a(1 - a))$. The branch through $q = 0$ is

$$q = \eta - \eta^2 + (1 + a)\eta^3 - (1 + 4a)\eta^4 + (1 + 11a + 3a^2)\eta^5 - (1 + 26a + 22a^2)\eta^6 + \mathcal{O}(\eta^7). \quad (62)$$

Every coefficient is obtained by finite exact arithmetic: through order η^M , it agrees with every finite product of order $n > M$.

Proof. On $0 < q < 1$, [equation \(48\)](#) has sign $-\text{sgn}(a)$ whenever $a < 1$ and $a \neq 0$. If $0 < a < 1$, then for any M , and q sufficiently close to one, the first M nonconstant factors are at most $1 - a/2 < 1$; the remaining factors lie in $(0, 1)$. Thus the product is at most $(1 - a/2)^M$, and its limit is zero. If $a < 0$, the first M factors are at least $1 + |a|/2$, while all remaining factors exceed one, so the limit is $+\infty$. The values at $q = 0$ and the two constant cases are immediate. For $a > 1$, exactly the j -th factor vanishes at q_j , proving simplicity and the asserted accumulation.

Near $q = 0$, normal convergence makes the product analytic and $\partial_q P(a, 0) = -a(1 - a) \neq 0$, so the analytic inverse exists. The coefficient of q^M uses only factors with index at most M ; hence it equals the stable finite-product coefficient. Direct subset expansion gives

$$\frac{1 - (aq; q)_\infty}{a} = q + q^2 + (1 - a)q^3 + (1 - a)q^4 + (1 - 2a)q^5 + (1 - 2a + a^2)q^6 + \mathcal{O}(q^7).$$

Substitution of $q = \eta + \sum_{j=2}^6 c_j \eta^j$, followed by comparison of successive powers, gives

$$(c_2, c_3, c_4, c_5, c_6) = (-1, 1 + a, -1 - 4a, 1 + 11a + 3a^2, -1 - 26a - 22a^2),$$

which is [equation \(62\)](#). Because the subset calculation through degree M uses no factor beyond index M , it also proves the stated finite exactness mechanism. $\square$

22 Fixed-argument inversion as $q \rightarrow 1$

Theorem 22.1 (Uniform all-order fixed-argument expansion). *Let $q = e^{-t}$, where $t \rightarrow 0$ in a closed subsector of $\Re t > 0$. Uniformly for $|a| \leq \rho < 1$,*

$$\log(a; e^{-t})_\infty \sim -\frac{\text{Li}_2(a)}{t} + \frac{1}{2} \log(1 - a) - \sum_{r \geq 1} \frac{B_{2r}}{(2r)!} \text{Li}_{2-2r}(a) t^{2r-1}. \quad (63)$$

More precisely, for every $M \geq 1$, retain the leading and constant terms and the Bernoulli terms with $1 \leq r \leq M - 1$ (an empty sum when $M = 1$). The remainder is then $\mathcal{O}(|t|^{2M-1})$, uniformly on the stated sets.

Proof. Insert, in [equation \(49\)](#),

$$\frac{1}{1 - e^{-x}} = \frac{1}{x} + \frac{1}{2} + \sum_{r=1}^{M-1} \frac{B_{2r}}{(2r)!} x^{2r-1} + R_M(x).$$

The displayed terms sum respectively to the dilogarithm, logarithm, and polylogarithms in [equation \(63\)](#). To control the remainder, choose a small $\delta > 0$ and split the Lambert index at $m \leq \delta/|t|$. Taylor's theorem is uniform on this first range and gives a contribution bounded by

$$C|t|^{2M-1} \sum_{m \geq 1} \rho^m m^{2M-2} = \mathcal{O}(|t|^{2M-1}).$$

On the complementary range, $\rho^m \leq \exp(-c/|t|)$. Because t stays in a closed subsector of the right half-plane, $1 - e^{-mt}$ is uniformly separated from zero there; the tail is exponentially small and hence smaller than every power of $|t|$. $\square$

Theorem 22.2 (Rigorous large-log base inverse). *Fix a with $|a| < 1$ and $A = \text{Li}_2(a) \neq 0$, and choose compatible branches of all logarithms. Put*

$$D = \frac{\text{Li}_0(a)}{12}, \quad E = \frac{\text{Li}_{-2}(a)}{720}, \quad F = \frac{\text{Li}_{-4}(a)}{30240}, \quad X = -\log y + \frac{1}{2} \log(1-a).$$

Let $X \rightarrow \infty$ so that A/X remains in a closed subsector of $\Re t > 0$. On the inverse sheet with $t \sim A/X$,

$$t = \frac{A}{X} + \frac{DA^2}{X^3} + \frac{A^3(2D^2 - AE)}{X^5} + \frac{A^4(A^2F - 6ADE + 5D^3)}{X^7} + \mathcal{O}(X^{-9}), \quad q = e^{-t}. \quad (64)$$

For real $0 < a < 1$, this is the unique real base inverse as $y \downarrow 0$.

Proof. The preceding theorem gives

$$X = \frac{A}{t} + Dt - Et^3 + Ft^5 + \mathcal{O}(t^7).$$

Insert an ansatz $t = A/X + \alpha X^{-3} + \beta X^{-5} + \gamma X^{-7}$. Comparison of X^{-1}, X^{-3}, X^{-5} gives

$$\alpha = DA^2, \quad \beta = A^3(2D^2 - AE), \quad \gamma = A^4(A^2F - 6ADE + 5D^3).$$

This proves the formal coefficients, but the error also needs transfer. For the displayed approximation t_0 , substitution in the forward equation leaves a residual $\mathcal{O}(X^{-7})$, while

$$\frac{d}{dt} \left(\frac{A}{t} + Dt - Et^3 + Ft^5 \right) = -\frac{A}{t^2} + \mathcal{O}(1)$$

has modulus comparable with $|X|^2$. The remainder is holomorphic and uniformly $\mathcal{O}(t^7)$ on a slightly wider closed subsector, so Cauchy's estimate makes its derivative $\mathcal{O}(t^6)$. Taylor expansion of the exact target equation on $|t - t_0| = K|X|^{-9}$ therefore consists of a constant $\mathcal{O}(X^{-7})$, a linear term of modulus comparable with $K|X|^{-7}$, and a quadratic error $\mathcal{O}(|X|^3|X|^{-18})$. For fixed sufficiently large K , the linear term dominates the other two. Rouché's theorem gives one zero inside that circle, proving the $\mathcal{O}(X^{-9})$ remainder. The real uniqueness follows from [theorem 21.1](#). $\square$

23 The coalescing parameter $a = q^x$

Theorem 23.1 (All-order coalescing expansion). *Let $\Re x > 0$, $q = e^{-t}$, and let $t \rightarrow 0$ in a closed subsector of $|\arg t| < \pi/2$. Use the logarithm obtained by analytic continuation from positive t . Then, locally uniformly for x in compact subsets of $\Re x > 0$,*

$$\begin{aligned} \log(q^x; q)_\infty \sim & -\frac{\pi^2}{6t} + \left(\frac{1}{2} - x\right) \log t + \frac{1}{2} \log(2\pi) - \log \Gamma(x) \\ & + \frac{B_2(x)}{4} t - \sum_{r \geq 1} \frac{B_{2r} B_{2r+1}(x)}{2r(2r+1)!} t^{2r}. \end{aligned} \quad (65)$$

Proof. For $c > 1$, initially with $t > 0$, Mellin inversion of the exponential in the Lambert sum gives

$$\log(e^{-xt}; e^{-t})_\infty = -\frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \Gamma(s) \zeta(s+1) \zeta(s, x) t^{-s} ds. \quad (66)$$

Absolute convergence for $c > 1$ justifies the interchange with the Lambert sum. Both sides then continue analytically to the stated sector, on the logarithm sheet fixed in the theorem. Indeed, uniformly for x in a fixed compact subset of $\Re x > 0$, sufficiently small t in a smaller closed subsector gives no factor zero: for bounded j the exponent $(x+j)t$ has modulus below 2π , while for large j its argument stays strictly between $-\pi/2$ and $\pi/2$, so it cannot be a nonzero multiple of $2\pi i$. We may therefore shift the continued contour left. The pole at $s = 1$ contributes $-\pi^2/(6t)$. At $s = 0$ the double pole, together with

$$\zeta(0, x) = \frac{1}{2} - x, \quad \zeta'(0, x) = \log \Gamma(x) - \frac{1}{2} \log(2\pi),$$

gives the logarithmic and constant terms. At $s = -1$, the residue of Γ , $\zeta(0) = -1/2$, and $\zeta(-1, x) = -B_2(x)/2$ give $B_2(x)t/4$. At $s = -2r$,

$$\operatorname{Res}_{s=-2r} \Gamma(s) = \frac{1}{(2r)!}, \quad \zeta(1-2r) = -\frac{B_{2r}}{2r}, \quad \zeta(-2r, x) = -\frac{B_{2r+1}(x)}{2r+1},$$

which yields the remaining coefficients, including the outer minus sign in [equation \(66\)](#). At negative odd integers below -1 , a trivial zero of $\zeta(s+1)$ cancels the gamma pole.

Here is the required uniform remainder estimate. Fix a compact set $K \subseteq \{\Re x > 0\}$, a closed subsector $|\arg t| \leq \pi/2 - \delta$, and a vertical line $\Re s = \sigma$ avoiding the poles just listed. Uniformly for $x \in K$, Stirling's estimate and the standard polynomial vertical bounds for the two zeta functions give constants C, N such that

$$|\Gamma(\sigma + iu) \zeta(\sigma + 1 + iu) \zeta(\sigma + iu, x)| \leq C(1 + |u|)^N e^{-\pi|u|/2}.$$

For the chosen branch of t^{-s} ,

$$|t^{-(\sigma+iu)}| = |t|^{-\sigma} e^{u \arg t} \leq |t|^{-\sigma} e^{(\pi/2-\delta)|u|}.$$

Thus the integrand on the shifted vertical line is bounded by

$$C|t|^{-\sigma} (1 + |u|)^N e^{-\delta|u|},$$

which is integrable uniformly for $x \in K$. The same exponential factor makes the horizontal sides of the truncating rectangles tend to zero. The shifted contour is therefore $\mathcal{O}_K(|t|^{-\sigma})$. Shifting successively past the displayed poles, and choosing σ strictly between the last retained pole and the next uncanceled pole, proves the asserted sectorial Poincaré remainder locally uniformly in x . $\square$

The specialization $x = 1$ has more structure than its power asymptotic: Dedekind modularity restores the exponentially small remainder exactly, both at the positive cusp and on the radial path toward -1 .

Theorem 23.2 (Exact eta completions at $q = 1$ and $q = -1$). *For $t > 0$,*

$$(e^{-t}; e^{-t})_\infty = \sqrt{\frac{2\pi}{t}} \exp\left(-\frac{\pi^2}{6t} + \frac{t}{24}\right) (e^{-4\pi^2/t}; e^{-4\pi^2/t})_\infty. \quad (67)$$

If $Q_c = \exp(-c\pi^2/t)$, then

$$(-e^{-t}; -e^{-t})_\infty = \sqrt{\frac{\pi}{t}} \exp\left(-\frac{\pi^2}{24t} + \frac{t}{24}\right) \frac{(Q_2; Q_2)_\infty^3}{(Q_4; Q_4)_\infty (Q_1; Q_1)_\infty}. \quad (68)$$

Consequently the relative errors after deleting the final product factor in [equation \(67\)](#) and the final product quotient in [equation \(68\)](#) are, respectively, $\mathcal{O}(e^{-4\pi^2/t})$ and $\mathcal{O}(e^{-\pi^2/t})$.

Proof. Recall

$$\eta(\tau) = e^{\pi i \tau / 12} \prod_{n \geq 1} (1 - e^{2\pi i n \tau})$$

and the modular identity $\eta(-1/\tau) = \sqrt{-i\tau} \eta(\tau)$, with the positive square root when τ is positive imaginary. Set $\tau = it/(2\pi)$. Then

$$\begin{aligned} \eta(\tau) &= e^{-t/24} (e^{-t}; e^{-t})_{\infty}, \\ \eta(-1/\tau) &= e^{-\pi^2/(6t)} (e^{-4\pi^2/t}; e^{-4\pi^2/t})_{\infty}, \\ \sqrt{-i\tau} &= \sqrt{t/(2\pi)}. \end{aligned}$$

Solving the modular identity for the first product proves [equation \(67\)](#).

For the negative radial path put $x = e^{-t}$. Separating the even and odd powers in the defining product, and then separating the odd powers in the ordinary positive product, gives

$$\begin{aligned} (-x; -x)_{\infty} &= (x^2; x^2)_{\infty} (-x; x^2)_{\infty}, \\ (-x; x^2)_{\infty} &= \frac{(x^2; x^4)_{\infty}}{(x; x^2)_{\infty}}, \\ (x; x^2)_{\infty} &= \frac{(x; x)_{\infty}}{(x^2; x^2)_{\infty}}, \quad (x^2; x^4)_{\infty} = \frac{(x^2; x^2)_{\infty}}{(x^4; x^4)_{\infty}}. \end{aligned}$$

Thus

$$(-x; -x)_{\infty} = \frac{(x^2; x^2)_{\infty}^3}{(x; x)_{\infty} (x^4; x^4)_{\infty}}. \quad (69)$$

Apply [equation \(67\)](#) with $t, 2t, 4t$. The square-root factors simplify to $\sqrt{\pi/t}$, the elementary exponent to $-\pi^2/(24t) + t/24$, and the three complementary nomes to Q_4, Q_2, Q_1 , respectively. Substitution in [equation \(69\)](#) proves [equation \(68\)](#).

Finally, as $Q \downarrow 0$, expansion of the first factor and the product tail estimate [equation \(44\)](#) give $(Q; Q)_{\infty} = 1 + \mathcal{O}(Q)$; the same estimate keeps its reciprocal bounded. Applying this observation to the exact final factors proves both relative error assertions. $\square$

For real $x > 0$, the product decreases from 1 to 0 as q runs from 0 to 1, since every factor $1 - q^{x+j}$ does. It therefore has a unique real base inverse. Put

$$\Lambda = -\log y, \quad A = \frac{\pi^2}{6}, \quad B = x - \frac{1}{2}, \quad C = \log \Gamma(x) - \frac{1}{2} \log(2\pi).$$

The leading equation is $\Lambda = A/t + B \log t + C$. If $B \neq 0$, its exact real Lambert- W solution for sufficiently large Λ is

$$t_0 = -\frac{A}{B W_k(-\frac{A}{B} \exp(-\frac{\Lambda-C}{B}))}, \quad (70)$$

where $k = -1$ when $B > 0$, and $k = 0$ when $B < 0$. When $B = 0$, $t_0 = A/(\Lambda - C)$. Exponentiating the leading equation and putting $w = -A/(Bt)$ proves [equation \(70\)](#); all quantities are real, so this exponentiation loses no logarithm-sheet information. The omitted part of [equation \(65\)](#) is $\mathcal{O}(t)$, while the derivative of $A/t + B \log t + C$ has magnitude comparable with t^{-2} . The real mean-value theorem, applied between the exact inverse and t_0 , therefore gives $t = t_0 + \mathcal{O}(t_0^3)$.

24 Fixed-target double scaling at $q = 1$

Theorem 24.1 (All-order fixed-target inverse and recurrence). *Fix $L > 0$. The unique real solution*

$$(a; e^{-\varepsilon})_{\infty} = e^{-L}, \quad 0 < a < 1,$$

has, for every M , an asymptotic expansion $a(\varepsilon) = \sum_{k=1}^M A_k(L)\varepsilon^k + \mathcal{O}(\varepsilon^{M+1})$. Define

$$C_{r,n} = [\varepsilon^n] \left(\sum_{k \geq 1} A_k \varepsilon^k \right)^r.$$

Then $A_1 = L$, and for $n \geq 1$,

$$\begin{aligned} A_{n+1} = & - \sum_{r=2}^{n+1} \frac{C_{r,n+1}}{r^2} - \frac{1}{2} \sum_{r=1}^n \frac{C_{r,n}}{r} \\ & - \sum_{j=1}^{\lfloor n/2 \rfloor} \frac{B_{2j}}{(2j)!} \sum_{r=1}^{n-2j+1} r^{2j-2} C_{r,n-2j+1}. \end{aligned} \quad (71)$$

In particular,

$$\begin{aligned} a = L\varepsilon - \frac{L(L+2)}{4}\varepsilon^2 + \frac{L(L^2+9L+12)}{72}\varepsilon^3 \\ - \frac{L(L^3+4L^2+12L+24)}{576}\varepsilon^4 \\ - \frac{L(31L^4-75L^3-800L^2-720)}{86400}\varepsilon^5 + \mathcal{O}(\varepsilon^6). \end{aligned} \quad (72)$$

Every A_k is a rational polynomial in L divisible by L .

Proof. Taking the negative logarithm gives the exact positive Lambert equation

$$\Phi_\varepsilon(a) := \sum_{r \geq 1} \frac{a^r}{r(1 - e^{-r\varepsilon})} = L. \quad (73)$$

Since the first summand is at least a/ε , every solution obeys $0 < a \leq L\varepsilon$. Moreover

$$\partial_a \Phi_\varepsilon(a) = \sum_{r \geq 1} \frac{a^{r-1}}{1 - e^{-r\varepsilon}} \geq \frac{1}{1 - e^{-\varepsilon}} \geq \frac{1}{\varepsilon}.$$

Thus Φ_ε is strictly increasing, and existence and uniqueness also follow from the principal real argument atlas.

Expand

$$\frac{1}{1 - e^{-r\varepsilon}} = \frac{1}{r\varepsilon} + \frac{1}{2} + \sum_{j \geq 1} \frac{B_{2j}}{(2j)!} (r\varepsilon)^{2j-1}.$$

The coefficient of ε^n in [equation \(73\)](#) contains the new unknown A_{n+1} only in a/ε , with coefficient one. The remaining contributions are, respectively, the three sums on the right of [equation \(71\)](#); setting the coefficient to zero proves the recurrence. It gives the five displayed polynomials by successive substitution. Inductively its operations preserve rational polynomiality, and every term vanishes at $L = 0$, proving the factor L .

It remains to justify the asymptotic remainder. For a truncation $a_M = \sum_{k=1}^M A_k \varepsilon^k = \mathcal{O}(\varepsilon)$, use

$$1 - e^{-r\varepsilon} \geq c \min(r\varepsilon, 1)$$

for real $\varepsilon > 0$. Splitting at $r = \varepsilon^{-1}$ then bounds the Lambert tail $r \geq M + 1$ by

$$\frac{C}{\varepsilon} \sum_{M+1 \leq r \leq \varepsilon^{-1}} \frac{|a_M|^r}{r^2} + C \sum_{r > \varepsilon^{-1}} \frac{|a_M|^r}{r} = \mathcal{O}(\varepsilon^M).$$

For the finitely many retained indices, Taylor's theorem for the elementary kernel, together with the recurrence, gives $\Phi_\varepsilon(a_M) - L = \mathcal{O}(\varepsilon^M)$. The derivative lower bound and the real mean-value theorem then imply

$$|a - a_M| \leq \varepsilon |\Phi_\varepsilon(a) - \Phi_\varepsilon(a_M)| = \mathcal{O}(\varepsilon^{M+1}).$$

This completes both the formal and analytic parts of the proof. $\square$

Warning 24.2 (Complex double scaling). For complex $L = -\log y$, the recurrence still defines a formal branch, but the poles $2\pi i k/r$ of the Lambert kernels accumulate at $\varepsilon = 0$. The real theorem above does not imply a convergent complex Taylor series. A complex version requires a sector, a logarithm branch, and uniform remainder estimates; absent those data it remains a formal asymptotic construction.

25 Raw root-of-unity endpoints and radial inverses

Proposition 25.1 (Raw boundary partial products). *Let ζ be primitive of order d , and write $N = kd + s$, $0 \leq s < d$. Then*

$$(a; \zeta)_N = (1 - a^d)^k \prod_{j=0}^{s-1} (1 - a\zeta^j). \quad (74)$$

Hence the partial products tend to zero when $|1 - a^d| < 1$, are unbounded when $|1 - a^d| > 1$, and do not converge when $|1 - a^d| = 1$ and $a \neq 0$. For $a = 0$, they are identically one.

Proof. Every complete residue block contributes $\prod_{j=0}^{d-1} (1 - a\zeta^j) = 1 - a^d$, proving [equation \(74\)](#). The first two conclusions follow from the geometric factor. In the unit-modulus case, the subsequence $s = 0$ is $(1 - a^d)^k$; it could converge only if $1 - a^d = 1$, which forces $a = 0$. $\square$

This proposition concerns raw partial products on the boundary. The next theorem instead approaches the boundary from $|q| < 1$, where the analytic product is defined.

Theorem 25.2 (Uniform radial cyclotomic expansion and inverse). *Let ζ be primitive of order d , let $|a| \leq \rho < 1$, and put $q = \zeta e^{-t}$, with $t \rightarrow 0$ in a closed subsector of $\Re t > 0$. All logarithms below are the holomorphic determinations obtained by continuation from $a = 0$; in particular,*

$$\log(1 - a^d) = - \sum_{\ell \geq 1} \frac{a^{d\ell}}{\ell} \quad (|a| < 1).$$

Define

$$A_\zeta(a) = \frac{\text{Li}_2(a^d)}{d^2}, \quad (75)$$

$$C_\zeta(a) = \frac{1}{2d} \log(1 - a^d) - \sum_{\substack{m \geq 1 \\ d \nmid m}} \frac{a^m}{m(1 - \zeta^m)}, \quad (76)$$

$$D_\zeta(a) = \frac{a^d}{12(1 - a^d)} - \sum_{\substack{m \geq 1 \\ d \nmid m}} \frac{a^m \zeta^m}{(1 - \zeta^m)^2}. \quad (77)$$

The coefficient series converge absolutely, and

$$\log(a; \zeta e^{-t})_\infty = -\frac{A_\zeta(a)}{t} + C_\zeta(a) - D_\zeta(a)t + \mathcal{O}(t^2) \quad (78)$$

uniformly on the stated compact sets. If $A_\zeta(a) \neq 0$, choose a branch of $\log y$, put $X = -\log y + C_\zeta(a)$, and suppose that $X \rightarrow \infty$ with $A_\zeta(a)/X$ in the chosen closed subsector. The corresponding inverse satisfies

$$t = \frac{A_\zeta(a)}{X} + \frac{D_\zeta(a)A_\zeta(a)^2}{X^3} + \mathcal{O}(X^{-4}), \quad q = \zeta e^{-t}. \quad (79)$$

Proof. Split [equation \(49\)](#) into indices $m = d\ell$ and $d \nmid m$. For the multiples,

$$\frac{1}{1 - e^{-d\ell t}} = \frac{1}{d\ell t} + \frac{1}{2} + \frac{d\ell t}{12} + \mathcal{O}((\ell t)^3).$$

Their singular, constant, and linear sums are respectively

$$-\frac{\text{Li}_2(a^d)}{d^2 t}, \quad \frac{1}{2d} \log(1 - a^d), \quad -\frac{a^d}{12(1 - a^d)} t.$$

For nonmultiples,

$$\frac{1}{1 - \zeta^m e^{-mt}} = \frac{1}{1 - \zeta^m} - \frac{m\zeta^m}{(1 - \zeta^m)^2} t + \mathcal{O}(m^2 t^2).$$

These are exactly the remaining terms in [equations \(76\)](#) and [\(77\)](#).

Absolute convergence of the coefficient sums is not by itself a remainder proof, so split again at $m \leq \delta/|t|$. Choose δ below the nearest singularity of each of the finitely many residue-class kernels. On the first range Taylor's theorem gives a uniform remainder, and summation of $\rho^m m|t|^2$ gives $\mathcal{O}(t^2)$. On the tail, $\rho^m = \mathcal{O}(\exp(-c/|t|))$; the sector condition also separates every denominator from zero. This proves [equation \(78\)](#).

The target equation is

$$X = \frac{A_\zeta}{t} + D_\zeta t + \mathcal{O}(t^2).$$

Substitution of $t_0 = A_\zeta/X + D_\zeta A_\zeta^2/X^3$ leaves residual $\mathcal{O}(X^{-2})$, while the derivative has size comparable with X^2 . Uniform holomorphy on a slightly wider subsector and Cauchy's estimate control the derivatives of the $\mathcal{O}(t^2)$ remainder. Thus, on a circle of radius $K|X|^{-4}$ around t_0 , the linear term has size comparable with $K|X|^{-2}$, the constant residual is $\mathcal{O}(X^{-2})$, and all higher variations are smaller. For fixed sufficiently large K , Rouché's theorem supplies a unique nearby solution and proves [equation \(79\)](#). $\square$

Corollary 25.3 (The radial points 1 and -1). *For real $0 < a < 1$, [equation \(64\)](#) is the unique inverse toward $q = 1^-$. For real $0 < |a| < 1$, put*

$$A_- = \frac{1}{4} \text{Li}_2(a^2), \quad C_- = \frac{1}{2} \log(1 - a), \quad D_- = \frac{a(a+3)}{12(1-a^2)},$$

$$E_- = \frac{a(15a^4 + 4a^3 + 90a^2 + 4a + 15)}{720(1-a^2)^3}, \quad X = -\log y + C_-.$$

On the real radial branch $q = -e^{-t}$, $t \downarrow 0$,

$$\log(a; -e^{-t})_\infty = -\frac{A_-}{t} + C_- - D_- t + E_- t^3 + \mathcal{O}(t^5), \quad (80)$$

$$t = \frac{A_-}{X} + \frac{D_- A_-^2}{X^3} + \frac{2D_-^2 A_-^3 - E_- A_-^4}{X^5} + \mathcal{O}(X^{-7}). \quad (81)$$

Here the inverse formula is taken as $y = (a; -e^{-t})_\infty \downarrow 0$, equivalently $X \rightarrow +\infty$.

Proof. Split the canonical Lambert logarithm into even and odd indices:

$$\log(a; -e^{-t})_{\infty} = - \sum_{m \geq 1} \frac{a^m}{m(1 - (-1)^m e^{-mt})}.$$

For even indices use

$$\frac{1}{1 - e^{-x}} = \frac{1}{x} + \frac{1}{2} + \frac{x}{12} - \frac{x^3}{720} + \mathcal{O}(x^5),$$

and for odd indices use

$$\frac{1}{1 + e^{-x}} = \frac{1}{2} + \frac{x}{4} - \frac{x^3}{48} + \mathcal{O}(x^5).$$

Only the even indices $m = 2r$ contribute a singular term, namely

$$- \sum_{r \geq 1} \frac{a^{2r}}{2r} \frac{1}{2rt} = - \frac{\text{Li}_2(a^2)}{4t}.$$

The constant term is

$$- \frac{1}{2} \sum_{m \geq 1} \frac{a^m}{m} = \frac{1}{2} \log(1 - a).$$

The coefficient of t is

$$- \frac{1}{12} \sum_{r \geq 1} a^{2r} - \frac{1}{4} \sum_{\substack{m \geq 1 \\ m \text{ odd}}} a^m = - \frac{a(a+3)}{12(1-a^2)}.$$

Writing $z = a^2$, the coefficient of t^3 is

$$\frac{1}{180} \sum_{r \geq 1} r^2 z^r + \frac{1}{48} \sum_{\substack{m \geq 1 \\ m \text{ odd}}} m^2 a^m.$$

The elementary identities

$$\sum_{r \geq 1} r^2 z^r = \frac{z(1+z)}{(1-z)^3}, \quad \sum_{\substack{m \geq 1 \\ m \text{ odd}}} m^2 a^m = \frac{a(1+a)}{(1-a)^3} - \frac{4a^2(1+a^2)}{(1-a^2)^3}$$

simplify this coefficient to

$$\frac{a(15a^4 + 4a^3 + 90a^2 + 4a + 15)}{720(1-a^2)^3} = E_-.$$

For fixed $|a| < 1$, the same finite-index/tail split used in the proof of [theorem 25.2](#) justifies the expansions under the sum; the summed fifth-order remainders are $\mathcal{O}(t^5)$. This proves [equation \(80\)](#) with every displayed coefficient.

Now

$$X = \frac{A_-}{t} + D_- t - E_- t^3 + \mathcal{O}(t^5).$$

Substituting $t = A_-/X + \alpha X^{-3} + \beta X^{-5}$ and comparing the coefficients of X^{-1} and X^{-3} gives

$$\alpha = D_- A_-^2, \quad \beta = 2D_-^2 A_-^3 - E_- A_-^4.$$

The resulting residual is $\mathcal{O}(X^{-5})$, whereas the derivative of the target equation is $-A_-/t^2 + \mathcal{O}(1) \asymp -X^2$. The real mean-value theorem therefore improves the inverse error to $\mathcal{O}(X^{-7})$, proving [equation \(81\)](#). The exclusion $a = 0$ is necessary: there the product is identically one and $A_- = 0$. $\square$

Proposition 25.4 (Fixed-level cyclotomic coalescence with winding). *Let ζ be primitive of order d , let $q = \zeta e^{-t}$, and let $t \rightarrow 0$ in a closed subsector of $\Re t > 0$. Fix $y \in \mathbb{C}^\times$, choose a logarithm ℓ of y , and, for $\nu \in \mathbb{Z}$, put*

$$L_\nu := -\ell - 2\pi i\nu.$$

Suppose $L_\nu \neq 0$. For the Lambert logarithm that is zero at $a = 0$, the lifted equation

$$\log(a; q)_\infty = \ell + 2\pi i\nu$$

has exactly d solution germs tending to $a = 0$. They satisfy

$$a^d \sim d^2 L_\nu t, \quad a \sim \omega(d^2 L_\nu t)^{1/d}, \quad \omega^d = 1. \quad (82)$$

Consequently the raw equation $(a; q)_\infty = y$ has one such d -sheeted cluster for every fixed winding ν with $L_\nu \neq 0$. Every continuous raw solution branch tending to zero belongs to exactly one of these clusters. The exceptional lift $L_\nu = 0$, when it occurs, is a different degenerate problem and is not covered by the displayed scale.

Proof. Write $\Lambda(a, t)$ for the Lambert logarithm in [equation \(78\)](#). The lifted equation is $-\Lambda(a, t) = L_\nu$. Suppose first that a solution germ has $a \rightarrow 0$. In the uniform forward formula, $C_\zeta(a) = o(1)$ and $D_\zeta(a)t = o(1)$; its remainder is uniform for $|a| \leq \rho$, so

$$\frac{\text{Li}_2(a^d)}{d^2 t} = L_\nu + o(1).$$

Since $\text{Li}_2(z)/z \rightarrow 1$, the last identity first gives $a^d = \mathcal{O}(t)$ and then $a^d/(d^2 t) \rightarrow L_\nu$, proving the first relation and the scale $a = \mathcal{O}(t^{1/d})$.

Choose a determination $\tau = t^{1/d}$ on the subsector and write $a = \tau b$. Uniformly for b in a fixed compact set,

$$-\Lambda(\tau b, t) - L_\nu = \frac{b^d}{d^2} - L_\nu + o(1),$$

because $\text{Li}_2(tb^d)/(d^2 t) = b^d/d^2 + \mathcal{O}(t)$, while $C_\zeta(\tau b) = o(1)$ and $tD_\zeta(\tau b) = o(1)$. The limiting polynomial has the d distinct roots of $b^d = d^2 L_\nu$. Rouché's theorem on disjoint small circles about those roots supplies one zero on each circle for all sufficiently small t . The preceding converse estimate shows that every solution tending to zero lies on one of these circles. Each circle contains one zero counted with multiplicity, so that zero is simple. For every nonzero t in the subsector, the holomorphic implicit-function theorem therefore makes it a locally holomorphic function of the chosen coordinate $\tau = t^{1/d}$. The circles are fixed and pairwise disjoint, and each contains exactly one zero; uniqueness makes these local functions agree on overlaps and patch to d sectorial solution germs. Conversely, any solution with $a \rightarrow 0$ has

$$b^d = d^2 L_\nu + o(1), \quad b = a/\tau,$$

and hence eventually lies in exactly one of these circles. This proves the exact germ count, its exhaustivity, and both asymptotic relations.

Finally, exponentiating a lifted solution gives the raw equation. Conversely, on a continuous raw solution branch the quantity $(\Lambda(a, t) - \ell)/(2\pi i)$ is integer-valued and continuous, hence is one fixed integer ν . Thus the winding clusters are disjoint and exhaust all continuous raw branches tending to zero. $\square$

Warning 25.5 (Exterior normalization). The defining product does not converge for $|q| > 1$. There is therefore no canonical unnormalized value of $(a; q)_\infty$, and no canonical inverse at $q = \infty$. One may instead specify, for example, $(a; q^{-1})_\infty$ for $|q| > 1$, or a q -gamma normalization with its explicit powers removed. Different choices can have different inverse asymptotics and must not be conflated.

Part IV

Gaussian and multinomial q -analogs

26 Polynomial structure and the positive real inverse

Fix integers $0 < k < n$, put $b = n - k$, $d = kb$, and write

$$G_{n,k}(q) := \begin{bmatrix} n \\ k \end{bmatrix}_q.$$

Although the quotient in [section 3](#) is useful away from roots of unity, the polynomial itself is the primary object.

Proposition 26.1 (Positive palindromic Gaussian polynomial). *The function $G_{n,k}$ is a monic polynomial of degree d , with nonnegative integer coefficients, constant coefficient one, and*

$$G_{n,k}(q) = q^d G_{n,k}(q^{-1}) \quad (q \neq 0).$$

Its coefficient of q is one.

Proof. The finite q -binomial recurrence is

$$G_{n,k}(q) = G_{n-1,k}(q) + q^{n-k} G_{n-1,k-1}(q).$$

It follows directly from the product quotient: after putting the two terms over the denominator $(q; q)_k (q; q)_{n-k}$, the numerator becomes

$$(q; q)_{n-1} ((1 - q^{n-k}) + q^{n-k}(1 - q^k)) = (q; q)_{n-1} (1 - q^n).$$

Starting from $G_{m,0} = G_{m,m} = 1$, the recurrence proves inductively that all coefficients are non-negative integers. The same induction gives degree $k(n - k)$, monicity, and constant coefficient one.

For reciprocity, replace q by q^{-1} in

$$G_{n,k}(q) = \prod_{j=1}^k \frac{1 - q^{b+j}}{1 - q^j}.$$

Since $1 - q^{-m} = -q^{-m}(1 - q^m)$, the signs cancel and the accumulated power is

$$-\sum_{j=1}^k (b + j) + \sum_{j=1}^k j = -kb = -d.$$

Thus $G_{n,k}(q^{-1}) = q^{-d} G_{n,k}(q)$. Finally, the recurrence or the partition interpretation shows that the coefficient of q is one: there is exactly one partition of 1 in a nonempty $k \times b$ rectangle. $\square$

Theorem 26.2 (Global positive-base inverse). *The restriction of $G_{n,k}$ to $[0, \infty)$ is a continuous strictly increasing bijection onto $[1, \infty)$. Consequently, for every real $y > 1$ there is exactly one $q > 0$ satisfying $G_{n,k}(q) = y$.*

Proof. Write $G_{n,k}(q) = 1 + \sum_{r=1}^d p_r q^r$. By [theorem 26.1](#), $p_r \geq 0$ and $p_1 = 1$. Hence

$$G'_{n,k}(q) = \sum_{r=1}^d r p_r q^{r-1} \geq 1 \quad (q \geq 0).$$

The function is therefore strictly increasing, starts at $G_{n,k}(0) = 1$, and tends to infinity because it is monic of positive degree. Continuity and the intermediate value theorem give existence; strict increase gives uniqueness. $\square$

27 The inverse at $q = 0$

Let $p_{k,b}(r)$ denote the number of partitions of r whose Ferrers diagram fits in a $k \times b$ rectangle. The recurrence in the preceding proof gives the standard generating identity

$$G_{n,k}(q) = \sum_{r=0}^d p_{k,b}(r) q^r.$$

In particular, $p_{k,b}(0) = p_{k,b}(1) = 1$. Put

$$c_2 = \mathbf{1}_{k \geq 2} + \mathbf{1}_{b \geq 2}, \quad c_3 = \mathbf{1}_{b \geq 3} + \mathbf{1}_{k \geq 2, b \geq 2} + \mathbf{1}_{k \geq 3}.$$

These are exactly $p_{k,b}(2)$ and $p_{k,b}(3)$.

Theorem 27.1 (Gaussian reversion at zero). *Let $u = y - 1$. The positive inverse from [theorem 26.2](#) has the convergent local expansion*

$$q = u - c_2 u^2 + (2c_2^2 - c_3) u^3 + \mathcal{O}(u^4).$$

More generally,

$$[u^m]q(u) = \frac{1}{m} [z^{m-1}] \left(\frac{z}{G_{n,k}(z) - 1} \right)^m \quad (m \geq 1).$$

Proof. Set $F(z) = G_{n,k}(z) - 1$. Then $F(0) = 0$ and $F'(0) = 1$, so the analytic inverse function theorem gives a convergent inverse near zero. The Lagrange–Bürmann formula from [theorem 8.3](#) gives the coefficient identity. For the first terms, substitute

$$u = q + c_2 q^2 + c_3 q^3 + \mathcal{O}(q^4), \quad q = u + Au^2 + Bu^3 + \mathcal{O}(u^4).$$

Comparison of u^2 and u^3 gives $A = -c_2$ and $B = 2c_2^2 - c_3$. □

28 Complete expansion and inversion at $q = 1$

Let

$$S_m(N) := \sum_{j=1}^N j^m, \quad \Delta_m(n, k) := S_m(n) - S_m(k) - S_m(n - k).$$

Lemma 28.1 (The elementary Bernoulli factor). *On $|x| < 2\pi$, let $\ell(x)$ be the unique holomorphic logarithm of $(1 - e^{-x})/x$ normalized by $\ell(0) = 0$. Then*

$$\ell(x) = -\frac{x}{2} + \sum_{r \geq 1} \frac{B_{2r}}{2r(2r)!} x^{2r}.$$

Proof. The generating function

$$\frac{x}{e^x - 1} = \sum_{m \geq 0} B_m \frac{x^m}{m!}$$

has nearest nonzero singularities at $\pm 2\pi i$. Therefore, for $|x| < 2\pi$,

$$\frac{1}{e^x - 1} - \frac{1}{x} = -\frac{1}{2} + \sum_{r \geq 1} B_{2r} \frac{x^{2r-1}}{(2r)!}.$$

The quotient $(1 - e^{-x})/x$, extended by the value one at zero, is zero-free in the open disk because its first nonzero zeros are $\pm 2\pi i$. Hence its normalized holomorphic logarithm exists there. The left side above is its derivative. Termwise integration from 0 to x gives the formula, with zero integration constant by the normalization. □

Theorem 28.2 (Complete Gaussian logarithmic expansion). *Let $q = e^{-t}$. For $|t| < 2\pi/n$, take the holomorphic logarithm of the normalized quotient whose value at $t = 0$ is zero. Then*

$$\log \frac{G_{n,k}(e^{-t})}{\binom{n}{k}} = -\frac{d}{2}t + \sum_{r \geq 1} \frac{B_{2r} \Delta_{2r}(n, k)}{2r(2r)!} t^{2r}.$$

If

$$\lambda_1 = -\frac{d}{2}, \quad \lambda_{2r} = \frac{B_{2r}}{2r} \Delta_{2r}(n, k), \quad \lambda_{2r+1} = 0 \quad (r \geq 1),$$

and the complete exponential Bell polynomials $\mathcal{B}_m$ are defined by

$$\exp \left(\sum_{j \geq 1} x_j \frac{z^j}{j!} \right) = \sum_{m \geq 0} \mathcal{B}_m(x_1, \dots, x_m) \frac{z^m}{m!},$$

then

$$G_{n,k}(e^{-t}) = \binom{n}{k} \sum_{m \geq 0} \mathcal{B}_m(\lambda_1, \dots, \lambda_m) \frac{t^m}{m!}.$$

Proof. Use

$$G_{n,k}(e^{-t}) = \prod_{j=1}^k \frac{1 - e^{-(b+j)t}}{1 - e^{-jt}}.$$

Apply [theorem 28.1](#) to every numerator and denominator. The logarithms of the scale factors $(b+j)/j$ sum to $\log \binom{n}{k}$. The linear difference is

$$\sum_{j=1}^k (b+j-j) = kb = d,$$

and the even differences are precisely $\Delta_{2r}(n, k)$. Every factor expansion converges when $|(b+j)t| < 2\pi$; the sufficient common disk is $|t| < 2\pi/n$. Exponentiating the resulting series and invoking the defining generating identity for $\mathcal{B}_m$ proves the second formula. $\square$

Corollary 28.3 (Literal Taylor coordinate $h = q - 1$). *As $h \rightarrow 0$,*

$$\frac{G_{n,k}(1+h)}{\binom{n}{k}} = 1 + \frac{d}{2}h + \frac{d(3d+n-5)}{24}h^2 + \frac{d(d-2)(d+n-3)}{48}h^3 + \mathcal{O}(h^4).$$

Proof. In [theorem 28.2](#), substitute

$$t = -\log(1+h) = -h + \frac{h^2}{2} - \frac{h^3}{3} + \mathcal{O}(h^4),$$

use $\Delta_2 = d(n+1)$, and exponentiate through degree three. All discarded products have degree at least four. Straight collection gives the displayed coefficients. $\square$

Put $v = \log(y/\binom{n}{k})$. The linear term in [theorem 28.2](#) is nonzero.

Corollary 28.4 (Ordinary Gaussian inverse at $q = 1$). *On the local branch through $q = 1$,*

$$t = -\frac{2v}{d} + \frac{\Delta_2}{3d^3}v^2 - \frac{\Delta_2^2}{9d^5}v^3 + \mathcal{O}(v^4), \quad q = e^{-t}.$$

The coefficient of v^4 is the first one involving Δ_4 .

Proof. Write

$$v = c_1 t + c_2 t^2 + c_4 t^4 + \dots, \quad c_1 = -d/2, \quad c_2 = \Delta_2/24.$$

There is no cubic term. Substituting $t = Av + Bv^2 + Cv^3 + \dots$ and comparing coefficients gives

$$A = c_1^{-1}, \quad B = -c_2 c_1^{-3}, \quad C = 2c_2^2 c_1^{-5},$$

which are the displayed values. Since c_4 first contributes in degree four, Δ_4 cannot occur earlier. $\square$

29 Centered palindromic inversion

The ordinary inverse above retains the deterministic reciprocity drift. Once that drift is removed, $q = 1$ becomes a fold.

Theorem 29.1 (Centered palindromic inverse). *Let*

$$P(q) = \sum_{j=0}^D c_j q^j$$

be nonconstant, with $c_j \geq 0$, $c_j = c_{D-j}$, and $c_0 = c_D > 0$. Define

$$H_P(t) := e^{Dt/2} \frac{P(e^{-t})}{P(1)}.$$

Then H_P is even, $H_P(0) = 1$, strictly increases on $(0, \infty)$, and tends to infinity there. For every $h > 1$, the equation $H_P(t) = h$ therefore has exactly two real solutions $t = \pm E_P(h)$. At $h = 1$ the two branches meet in a square-root fold.

Proof. Let X take value j with probability $c_j/P(1)$, and set $Z = X - D/2$. Palindromy says Z and $-Z$ have the same law. Hence

$$H_P(t) = \mathbb{E}(e^{-tZ}) = \mathbb{E}(\cosh(tZ)),$$

which is even and equals one at zero. For $t > 0$,

$$H'_P(t) = \mathbb{E}(Z \sinh(tZ)) > 0.$$

Indeed, the integrand is nonnegative and is positive whenever $Z \neq 0$; the endpoint masses $c_0 = c_D > 0$ ensure that this happens with positive probability. Those same endpoints yield

$$H_P(t) \geq \frac{c_0}{P(1)} e^{Dt/2} \rightarrow \infty.$$

This proves the global branch count.

Finally,

$$H_P(t) = 1 + \frac{\mathbb{E}(Z^2)}{2} t^2 + \mathcal{O}(t^4),$$

and $\mathbb{E}(Z^2) > 0$ by nondegeneracy. Thus

$$t = \pm \sqrt{\frac{2(h-1)}{\mathbb{E}(Z^2)}} (1 + \mathcal{O}(h-1)),$$

which is exactly a square-root fold. □

Let κ_{2r} be the even cumulants of Z , so that

$$u := \log H_P(t) = \frac{\kappa_2}{2!} t^2 + \frac{\kappa_4}{4!} t^4 + \frac{\kappa_6}{6!} t^6 + \dots$$

Proposition 29.2 (Centered inverse through $u^{5/2}$). *As $u \downarrow 0$,*

$$t = \pm \sqrt{\frac{2u}{\kappa_2}} \left[1 - \frac{\kappa_4}{12\kappa_2^2} u + \left(\frac{7\kappa_4^2}{288\kappa_2^4} - \frac{\kappa_6}{180\kappa_2^3} \right) u^2 + \mathcal{O}(u^3) \right].$$

Proof. Put $s = t^2$. Then

$$u = As + Bs^2 + Cs^3 + \mathcal{O}(s^4), \quad A = \kappa_2/2, \quad B = \kappa_4/24, \quad C = \kappa_6/720.$$

Substitute

$$t = (u/A)^{1/2}(1 + \alpha u + \beta u^2 + \mathcal{O}(u^3)).$$

Comparison in $u = At^2 + Bt^4 + Ct^6 + \dots$ gives

$$\alpha = -\frac{B}{2A^2}, \quad \beta = \frac{7B^2 - 4AC}{8A^4}.$$

Inserting A, B, C yields the stated coefficients. The two signs are the two branches supplied by [theorem 29.1](#). $\square$

For $P = G_{n,k}$, [theorem 28.2](#) gives

$$\kappa_{2r} = \frac{B_{2r}}{2r} \Delta_{2r}(n, k), \quad \kappa_2 = \frac{d(n+1)}{12}.$$

Thus the centered Gaussian inverse is an explicit Bernoulli–Faulhaber Puiseux series.

30 Cyclotomic roots and the alternating base

Theorem 30.1 (Square-free Gaussian covering). *The cyclotomic factorization is*

$$G_{n,k}(q) = \prod_{m=2}^n \Phi_m(q)^{e_m}, \quad e_m = \left\lfloor \frac{n}{m} \right\rfloor - \left\lfloor \frac{k}{m} \right\rfloor - \left\lfloor \frac{n-k}{m} \right\rfloor.$$

Every e_m is either zero or one. Consequently every complex zero of $G_{n,k}$ is a simple root of unity, and the local inverse at a zero is an ordinary Taylor branch rather than a Puiseux branch of higher order.

Proof. The factorization $1 - q^j = \prod_{m|j} \Phi_m(q)$ shows that the multiplicity of Φ_m in $(q; q)_N$ is $\lfloor N/m \rfloor$. Subtracting the denominator multiplicities proves the formula for e_m . Write $n = k + b$. The elementary carry identity

$$\left\lfloor \frac{k+b}{m} \right\rfloor - \left\lfloor \frac{k}{m} \right\rfloor - \left\lfloor \frac{b}{m} \right\rfloor \in \{0, 1\}$$

proves square-freeness. Distinct cyclotomic polynomials are coprime and each is separable over characteristic zero. Thus every zero is simple. The analytic inverse function theorem then supplies an ordinary local inverse. $\square$

If ζ is primitive of order m and $e_m = 1$, put

$$R_m(q) := \prod_{\substack{2 \leq r \leq n \\ r \neq m}} \Phi_r(q)^{e_r}.$$

Then

$$G'_{n,k}(\zeta) = \Phi'_m(\zeta) R_m(\zeta) \neq 0, \quad q(y) = \zeta + \frac{y}{\Phi'_m(\zeta) R_m(\zeta)} + \mathcal{O}(y^2).$$

Proposition 30.2 (Value and zero derivative at $q = -1$). *One has*

$$G_{n,k}(-1) = \begin{cases} 0, & n \text{ even and } k \text{ odd,} \\ \binom{\lfloor n/2 \rfloor}{\lfloor k/2 \rfloor}, & \text{otherwise.} \end{cases}$$

In the zero case, writing $n = 2m$ and $k = 2r + 1$,

$$G'_{2m,2r+1}(-1) = (m-r) \binom{m}{r}.$$

Proof. Separate even and odd exponents in the product quotient and pair every vanishing factor $1 - q^{2s}$ with a factor $1 - q^2$. Since

$$\lim_{q \rightarrow -1} \frac{1 - q^{2s}}{1 - q^2} = s, \quad \lim_{q \rightarrow -1} (1 - q^{2s+1}) = 2,$$

the surviving factorial ratios reduce to the displayed binomial coefficient. When n is even and k odd, there is exactly one unpaired even numerator factor, so the value is zero.

In that case the quotient after removing the unique factor $1 + q$ has limit

$$\frac{m!}{r!(m-r-1)!} = (m-r) \binom{m}{r}.$$

Because $1 + q$ has derivative one at -1 , this limit is exactly $G'_{2m, 2r+1}(-1)$. □

31 The reciprocal and large-target inverse

Write the small- Q expansion

$$G_{n,k}(Q) = 1 + Q + c_2 Q^2 + c_3 Q^3 + \mathcal{O}(Q^4)$$

with the c_2, c_3 of [section 27](#), and choose a branch $Y = y^{1/d}$.

Theorem 31.1 (Correct large-target Gaussian inverse). *On every branch for which $Y \rightarrow \infty$,*

$$\begin{aligned} q = Y - \frac{1}{d} + \left(\frac{d-1}{2d^2} - \frac{c_2}{d} \right) Y^{-1} \\ + \left(-\frac{(d-1)(d-2)}{3d^3} + \frac{(d-2)c_2}{d^2} - \frac{c_3}{d} \right) Y^{-2} + \mathcal{O}(Y^{-3}). \end{aligned}$$

Proof. Reciprocity gives

$$y = q^d G_{n,k}(q^{-1}).$$

Set $x = Y^{-1}$ and

$$q = Y(1 + Ax + Bx^2 + Cx^3 + \mathcal{O}(x^4)).$$

Then

$$q^{-1} = x - Ax^2 + (A^2 - B)x^3 + \mathcal{O}(x^4),$$

and therefore

$$G_{n,k}(q^{-1}) = 1 + x + (c_2 - A)x^2 + (A^2 - B - 2Ac_2 + c_3)x^3 + \mathcal{O}(x^4).$$

Divide the equation by Y^d and take logarithms. Comparing the coefficients of x, x^2, x^3 in

$$0 = d \log(1 + Ax + Bx^2 + Cx^3) + \log G_{n,k}(q^{-1})$$

gives successively

$$A = -\frac{1}{d}, \quad B = \frac{d-1}{2d^2} - \frac{c_2}{d},$$

and

$$C = -\frac{(d-1)(d-2)}{3d^3} + \frac{(d-2)c_2}{d^2} - \frac{c_3}{d}.$$

Multiplication by Y produces the displayed expansion and leaves a remainder of order Y^{-3} . □

Remark 31.2. Several source reports printed $-(d-1)(2d-1)/(3d^3) + 2c_2(d-1)/d^2 - c_3/d$ at order Y^{-2} . The coefficient comparison above shows that expression is incorrect. The correct coefficient vanishes in the expected low-degree special cases and is the one retained here.

32 Multinomial extension

Throughout this section let $s \geq 1$, let $n_1, \dots, n_s \in \mathbb{Z}_{\geq 0}$, and put $n = n_1 + \dots + n_s > 0$. Set

$$M_{\mathbf{n}}(q) := \frac{(q; q)_n}{(q; q)_{n_1} \cdots (q; q)_{n_s}}, \quad D(\mathbf{n}) := \sum_{i < j} n_i n_j,$$

$$\Delta_m(\mathbf{n}) := S_m(n) - \sum_{i=1}^s S_m(n_i).$$

Theorem 32.1 (Complete multinomial expansion at one). *For $q = e^{-t}$ and $|t| < 2\pi/n$, take the holomorphic logarithm of the normalized quotient whose value at $t = 0$ is zero. Then*

$$\log \frac{M_{\mathbf{n}}(e^{-t})}{\binom{n}{n_1, \dots, n_s}} = -\frac{D(\mathbf{n})}{2}t + \sum_{r \geq 1} \frac{B_{2r} \Delta_{2r}(\mathbf{n})}{2r(2r)!} t^{2r}.$$

If $D(\mathbf{n}) > 0$, equivalently at least two of the n_i are positive, then

$$e^{D(\mathbf{n})t/2} \frac{M_{\mathbf{n}}(e^{-t})}{\binom{n}{n_1, \dots, n_s}}$$

is a nonconstant even function with a nondegenerate minimum at $t = 0$, and its inverse has the two Puiseux branches of [theorem 29.2](#). If $D(\mathbf{n}) = 0$, then $M_{\mathbf{n}} \equiv 1$ and no local inverse exists.

Proof. Apply [theorem 28.1](#) to the numerator $(q; q)_n$ and every denominator $(q; q)_{n_i}$. Since $|jt| < 2\pi$ for every factor, the normalized factor logarithms sum to the holomorphic branch specified in the statement. Constants give the ordinary multinomial coefficient and the power sums give $\Delta_{2r}(\mathbf{n})$. The linear difference is

$$\frac{1}{2} \left(n^2 - \sum_i n_i^2 \right) = \sum_{i < j} n_i n_j = D(\mathbf{n}).$$

It remains to justify the hypotheses for centered inversion. Put $N_i = n_1 + \dots + n_i$. The telescoping factorization

$$M_{\mathbf{n}}(q) = \prod_{i=2}^s \left[\frac{N_i}{n_i} \right]_q$$

proves polynomiality, with the empty product for $s = 1$ understood as one. Boundary factors with $n_i = 0$ or $N_{i-1} = 0$ are one. By [theorem 26.1](#), every nontrivial Gaussian factor has nonnegative coefficients, equal positive endpoints, and is palindromic; hence so does their product. Its degree is

$$\sum_{i=2}^s n_i N_{i-1} = \sum_{j < i} n_i n_j = D(\mathbf{n}).$$

Thus it is nonconstant exactly when $D(\mathbf{n}) > 0$, in which case reciprocity makes the centered normalization even and [theorems 29.1](#) and [29.2](#) applies. If $D(\mathbf{n}) = 0$, at most one part is positive and the same factorization gives $M_{\mathbf{n}} = 1$. $\square$

Theorem 32.2 (Cyclotomic carry and exact ramification). *Under the standing hypotheses above, let $m \geq 1$, let ζ be a primitive m -th root of unity, and write $n_i = m\ell_i + r_i$ with $0 \leq r_i < m$. The multiplicity of ζ as a zero of $M_{\mathbf{n}}$ is*

$$e_m = \left\lfloor \frac{n}{m} \right\rfloor - \sum_{i=1}^s \left\lfloor \frac{n_i}{m} \right\rfloor = \left\lfloor \frac{r_1 + \dots + r_s}{m} \right\rfloor.$$

Thus $0 \leq e_m \leq s - 1$. If $e_m > 0$, the inverse at target zero has exactly e_m local Puiseux determinations, counted with their choices of the e_m -th root.

Proof. As in [theorem 30.1](#), cyclotomic factorization of each factorial gives the first expression. Since

$$n = m \sum_i \ell_i + \sum_i r_i,$$

substitution gives the residue-carry formula. The residue sum lies between zero and $s(m-1)$, proving $e_m \leq s-1$.

Factor locally

$$M_n(q) = (q - \zeta)^{e_m} U(q), \quad U(\zeta) \neq 0.$$

After choosing an analytic e_m -th root of U in a sufficiently small disk, the equation $M_n(q) = y$ is equivalent to

$$(q - \zeta) U(q)^{1/e_m} = \omega y^{1/e_m},$$

where ω runs through the e_m -th roots of unity. The analytic inverse function theorem applies to the left side, whose derivative at ζ is $U(\zeta)^{1/e_m} \neq 0$. This produces precisely the stated Puiseux determinations. $\square$

For $s = 2$, the carry count is at most one, recovering the square-free Gaussian case. For many parts, the exact same cyclotomic mechanism produces higher ramification. Centered inversion at $q = 1$ and cyclotomic inversion therefore represent genuinely different singular mechanisms even when they occur in the same polynomial family.

Part V

Other principal q -analogs

33 q -numbers

Choose a logarithm value $\text{Log } q$ for $q \in \mathbb{C} \setminus \{0, 1\}$, put $q^x = e^{x \text{Log } q}$, and define

$$[x]_q := \frac{1 - q^x}{1 - q}.$$

Proposition 33.1 (Exact inverse in the exponent). *Let $1 - (1 - q)y \neq 0$, and choose a value of $\text{Log}(1 - (1 - q)y)$. Then*

$$x = \frac{\text{Log}(1 - (1 - q)y)}{\text{Log } q}$$

solves $[x]_q = y$. Every solution is obtained by allowing the logarithm value in the numerator to vary by $2\pi i\mathbb{Z}$.

Proof. The equation $[x]_q = y$ is equivalent to

$$e^{x \text{Log } q} = 1 - (1 - q)y.$$

Taking a logarithm value gives the displayed expression. Conversely, exponentiating that expression recovers the equation. The complete preimage of a nonzero complex number under the exponential consists of one logarithm value plus $2\pi i\mathbb{Z}$, which proves the last assertion. $\square$

For an integer $m \geq 2$,

$$[m]_q = 1 + q + \cdots + q^{m-1},$$

so inversion in the base is an algebraic covering.

Proposition 33.2 (Distinguished base jets of the q -number). *Let $u = y - 1$. The inverse branch through $q = 0$ satisfies*

$$\begin{aligned} m = 2 : \quad q &= u, & \text{exactly,} \\ m = 3 : \quad q &= u - u^2 + 2u^3 + \mathcal{O}(u^4), \\ m \geq 4 : \quad q &= u - u^2 + u^3 + \mathcal{O}(u^4). \end{aligned}$$

For a fixed complex x and $q = e^{-t} \rightarrow 1$,

$$[x]_{e^{-t}} = x + \frac{x(1-x)}{2}t + \frac{x(2x^2-3x+1)}{12}t^2 - \frac{x^2(x-1)^2}{24}t^3 + \mathcal{O}(t^4).$$

If $x \notin \{0, 1\}$, its base inverse is

$$t = \frac{2(y-x)}{x(1-x)} + \frac{2(2x-1)}{3x^2(x-1)^2}(y-x)^2 + \mathcal{O}((y-x)^3).$$

Proof. At zero, the forward series begins $u = q$ for $m = 2$, $u = q + q^2$ for $m = 3$, and $u = q + q^2 + q^3 + \mathcal{O}(q^4)$ for $m \geq 4$. Direct substitution of $q = u + Au^2 + Bu^3 + \dots$ gives the three formulas.

For $q = e^{-t}$, divide the Taylor series of $1 - e^{-xt}$ by that of $1 - e^{-t}$, or equivalently use

$$[x]_{e^{-t}} = x \exp \left(-\frac{x-1}{2}t + \sum_{r \geq 1} \frac{B_{2r}(x^{2r}-1)}{2r(2r)!}t^{2r} \right).$$

Expansion through degree three gives the stated forward jet. Reversion of $\delta := y - x = at + bt^2 + \dots$, where

$$a = \frac{x(1-x)}{2}, \quad b = \frac{x(x-1)(2x-1)}{12},$$

gives $t = \delta/a - b\delta^2/a^3 + \dots$, which is the displayed inverse. For $x = 0$ or 1 , the forward function is identically 0 or 1 , so base recovery is impossible rather than singular of finite order. $\square$

Proposition 33.3 (Alternating and reciprocal q -number branches). *For $r \geq 1$,*

$$[2r]_{-1} = 0, \quad \frac{\partial}{\partial q}[2r]_q \Big|_{q=-1} = r,$$

and

$$[2r+1]_{-1} = 1, \quad \frac{\partial}{\partial q}[2r+1]_q \Big|_{q=-1} = -r.$$

At infinity,

$$[m]_q = q^{m-1}(1 + q^{-1} + \dots + q^{-(m-1)}),$$

so the inverse at a large target has $m - 1$ Puiseux determinations, indexed by the choices of $y^{1/(m-1)}$.

Proof. Evaluate $\sum_{j=0}^{m-1} q^j$ and its derivative $\sum_{j=1}^{m-1} j q^{j-1}$ at -1 , pairing consecutive terms. The reciprocal identity is obtained by factoring out q^{m-1} . Its parenthesized factor tends to one, so the finite-ramification normal form of [theorem 11.1](#) gives exactly $m - 1$ inverse determinations. $\square$

34 q -factorials and their exact ramification

For $n \in \mathbb{N}_0$, put

$$[n]_q! := \prod_{j=1}^n [j]_q.$$

Proposition 34.1 (Zero-base factorial inverse). *For $n \geq 2$,*

$$[n]_q! = 1 + (n-1)q + \frac{(n-2)(n+1)}{2}q^2 + \mathcal{O}(q^3).$$

Consequently, with $u = y - 1$,

$$q = \frac{u}{n-1} - \frac{(n-2)(n+1)}{2(n-1)^3}u^2 + \mathcal{O}(u^3).$$

Proof. In the product

$$[n]_q! = \prod_{j=2}^n (1 + q + \cdots + q^{j-1}),$$

the coefficient of q is $n-1$. A term of degree two is obtained either by choosing q from two distinct factors, in $\binom{n-1}{2}$ ways, or by choosing q^2 from one of the $n-2$ factors with $j \geq 3$. Their sum is $(n-2)(n+1)/2$. Reversion of $u = (n-1)q + Aq^2 + \cdots$ proves the inverse formula. $\square$

Theorem 34.2 (Exact root-of-unity ramification of q -factorials). *For every $n \geq 0$, let $\zeta \neq 1$ be a primitive d -th root of unity. Then*

$$\text{ord}_{q=\zeta}[n]_q! = \left\lfloor \frac{n}{d} \right\rfloor.$$

If $n \geq 2$, put $M = \lfloor n/2 \rfloor \geq 1$. Then

$$[n]_q! = M!(q+1)^M (1 + \mathcal{O}(q+1)) \quad (q \rightarrow -1),$$

and the inverse at target zero has the M branches

$$q = -1 + \omega_M^j \left(\frac{y}{M!} \right)^{1/M} + \mathcal{O}(y^{2/M}), \quad 0 \leq j < M.$$

For $n = 0, 1$, one has $[n]_q! \equiv 1$, so there is no inverse at target zero.

Proof. Because $\zeta \neq 1$, the denominator in $[j]_q = (1 - q^j)/(1 - q)$ is nonzero at ζ . The numerator has a simple zero there exactly when $d \mid j$. Exactly $\lfloor n/d \rfloor$ factors therefore vanish, each simply.

For $d = 2$ and $n \geq 2$, the even factors satisfy

$$[2r]_q = r(q+1) + \mathcal{O}((q+1)^2),$$

while the odd factors tend to one. Multiplying $r = 1, \dots, M$ gives the coefficient $M!$. Applying [theorem 11.1](#) to the resulting exact zero order gives the Puiseux branches and their remainder. $\square$

For $n \geq 2$, write $N = \binom{n}{2} > 0$. One also has

$$[n]_q! = q^N \left(1 + \frac{n-1}{q} + \mathcal{O}(q^{-2}) \right),$$

and the same logarithmic coefficient comparison as in [theorem 31.1](#) yields, on each choice $Y = y^{1/N} \rightarrow \infty$,

$$q = Y - \frac{n-1}{N} + \mathcal{O}(Y^{-1}) = Y - \frac{2}{n} + \mathcal{O}(Y^{-1}).$$

For $n = 0, 1$, the forward function is constant and has no large-target inverse.

35 The q -gamma function

For $0 < q < 1$ and $x > 0$, define

$$\Gamma_q(x) = (1 - q)^{1-x} \frac{(q; q)_\infty}{(q^x; q)_\infty}.$$

It satisfies

$$\Gamma_q(x + 1) = [x]_q \Gamma_q(x), \quad \Gamma_q(1) = \Gamma_q(2) = 1.$$

Proposition 35.1 (Exact generalized expansion at $q = 0$). *For $0 < q < 1$ and $x > 0$,*

$$\log \Gamma_q(x) = \sum_{m \geq 1} \frac{q^{mx}}{m(1 - q^m)} - \sum_{m \geq 1} \frac{q^m}{m(1 - q^m)} - (1 - x) \sum_{m \geq 1} \frac{q^m}{m}.$$

All three series converge absolutely. Expanding $(1 - q^m)^{-1} = \sum_{r \geq 0} q^{mr}$ gives a convergent generalized power series with exponents in

$$\{m(x + r) : m \geq 1, r \geq 0\} \cup \mathbb{N}_0.$$

For rational x it is a Puiseux series; for irrational x , its natural ordering is a generalized-power ordering rather than an ordinary Taylor ordering. Before coincident exponents are collected, the $(m, r) = (1, 0)$ term of the first double series contributes q^x , while the two integer-power series contribute $-(2 - x)q$ at exponent one. These are bookkeeping contributions, not necessarily the first two terms in the ordered generalized-power expansion. Whenever exponents coincide, all contributions at that exponent must be collected.

Proof. Take logarithms in the product definition and use the absolutely convergent Lambert identity

$$\log(a; q)_\infty = - \sum_{m \geq 1} \frac{a^m}{m(1 - q^m)}$$

from [theorem 18.2](#), together with $\log(1 - q) = - \sum_{m \geq 1} q^m/m$. Absolute convergence permits expansion of each geometric denominator and rearrangement. The displayed exponent set is then immediate. The $(m, r) = (1, 0)$ term and the integer exponent-one terms give the two stated uncollected contributions; collecting all terms with a common exponent is essential when collisions occur. $\square$

The logarithmic derivative and its derivative are

$$\psi_q(x) = -\log(1 - q) + (\log q) \sum_{j \geq 0} \frac{q^{x+j}}{1 - q^{x+j}},$$

$$\psi'_q(x) = (\log q)^2 \sum_{j \geq 0} \frac{q^{x+j}}{(1 - q^{x+j})^2} > 0.$$

Termwise differentiation is justified locally uniformly for $x > 0$.

Theorem 35.2 (Unique q -gamma minimum and two real inverse branches). *For every $0 < q < 1$, there is a unique $x_q > 0$ with $\psi_q(x_q) = 0$. The function Γ_q strictly decreases on $(0, x_q)$, strictly increases on (x_q, ∞) , and tends to infinity at both endpoints. Thus each target strictly above $\Gamma_q(x_q)$ has exactly two positive argument inverses, the minimum has one double inverse point, and smaller targets have none. The two branches meet in a square-root fold.*

Proof. The formula above makes ψ_q strictly increasing. As $x \downarrow 0$, the $j = 0$ summand is asymptotic to

$$(\log q) \frac{1}{-x \log q} = -\frac{1}{x},$$

so $\psi_q(x) \rightarrow -\infty$. As $x \rightarrow \infty$, the series tends to zero and $\psi_q(x) \rightarrow -\log(1 - q) > 0$. Hence it has one zero.

Since $\Gamma'_q/\Gamma_q = \psi_q$, the derivative has the asserted signs. The factor $(q^x; q)_\infty$ has a simple zero at $x = 0$, so $\Gamma_q(x) \rightarrow \infty$ there. As $x \rightarrow \infty$, the denominator tends to one while $(1 - q)^{1-x} \rightarrow \infty$. The branch count follows by continuity and strict monotonicity. At the minimum,

$$(\log \Gamma_q)''(x_q) = \psi'_q(x_q) > 0,$$

so Taylor's theorem and [theorem 11.1](#) give the square-root fold. $\square$

Theorem 35.3 (All-order $q \rightarrow 1$ expansion of Γ_q). *Let $q = e^{-t}$, with $t \rightarrow 0$ in a closed subsector of $|\arg t| < \pi/2$. Uniformly for x in compact subsets of $\Re x > 0$,*

$$\log \Gamma_{e^{-t}}(x) \sim \log \Gamma(x) + c_1(x)t + \sum_{r \geq 1} c_{2r}(x)t^{2r},$$

where

$$c_1(x) = \frac{(x-1)(2-x)}{4}$$

and

$$c_{2r}(x) = \frac{B_{2r}}{2r(2r)!} \left(1 - x + \frac{B_{2r+1}(x)}{2r+1} \right).$$

No odd algebraic power beyond t^1 occurs.

Proof. Insert the all-order coalescing expansion of $(q^x; q)_\infty$ from [theorem 23.1](#) into

$$\log \Gamma_q(x) = (1-x) \log(1-q) + \log(q; q)_\infty - \log(q^x; q)_\infty.$$

Use the $x = 1$ specialization for $(q; q)_\infty$ and

$$\log \frac{1 - e^{-t}}{t} = -\frac{t}{2} + \sum_{r \geq 1} \frac{B_{2r}}{2r(2r)!} t^{2r}.$$

The t^{-1} , logarithmic, and constant normalization terms cancel, leaving $\log \Gamma(x)$. The linear coefficient simplifies to

$$\frac{1}{24} - \frac{B_2(x)}{4} - \frac{1-x}{2} = \frac{(x-1)(2-x)}{4}.$$

At every even order, the $x = 1$ Bernoulli-polynomial term vanishes and the remaining two terms combine to $c_{2r}(x)$. The expansions invoked above are locally uniform with uniform differentiated remainders, so subtraction preserves the asserted uniform asymptotic meaning. $\square$

In particular,

$$c_2(x) = \frac{(x-2)(x-1)(2x+3)}{144},$$

and, for fixed $x \notin \{1, 2\}$, if

$$v = \log \frac{y}{\Gamma(x)},$$

ordinary reversion gives

$$t = \frac{v}{c_1(x)} - \frac{c_2(x)}{c_1(x)^3} v^2 + \frac{2c_2(x)^2}{c_1(x)^5} v^3 + \mathcal{O}(v^4), \quad q = e^{-t}.$$

At $x = 1, 2$, $\Gamma_q(x) \equiv 1$, so there is no base inverse.

36 q -digamma inversion

Proposition 36.1 (Global argument inverse and the exceptional reciprocal fold). *For fixed $0 < q < 1$, the function ψ_q is a strictly increasing bijection*

$$(0, \infty) \longrightarrow (-\infty, -\log(1 - q)).$$

For $q = e^{-t}$,

$$\psi_{e^{-t}}(x) \sim \psi(x) + \frac{3-2x}{4}t + \sum_{r \geq 1} c'_{2r}(x)t^{2r}.$$

At fixed $x \neq 3/2$, base inversion at $q = 1$ is ordinary. At $x = 3/2$,

$$\psi_{e^{-t}}(3/2) - \psi(3/2) = -\frac{t^2}{288} + \mathcal{O}(t^4),$$

so, within the original domain $0 < q < 1$, $q = 1$ is a one-sided quadratic endpoint. More precisely, if

$$\Delta = \psi_{e^{-t}}(3/2) - \psi(3/2),$$

then the unique local inverse with $t \downarrow 0$ is

$$t = \sqrt{-288\Delta} (1 + \mathcal{O}(\Delta)), \quad \Delta \uparrow 0, \quad \Delta < 0.$$

There is also a natural genuine two-sided statement. Extend the standard normalization to $q > 1$ by

$$\Gamma_q(x) := q^{(x-1)(x-2)/2} \Gamma_{1/q}(x), \quad \psi_q(x) := \partial_x \log \Gamma_q(x),$$

and use the classical functions at $q = 1$. Then

$$\psi_q(x) = \psi_{1/q}(x) + \left(x - \frac{3}{2}\right) \log q.$$

In particular, $t \mapsto \psi_{e^{-t}}(3/2)$ is reciprocal-even, is C^2 at $t = 0$, and has a genuine two-sided quadratic fold there. Its branches are $t = \pm \sqrt{-288\Delta} (1 + \mathcal{O}(\Delta))$, equivalently $q = e^{\pm \sqrt{-288\Delta} (1 + \mathcal{O}(\Delta))}$.

Proof. Strict increase and the two endpoint limits were proved in [theorem 35.2](#); these give the bijection. Differentiate [theorem 35.3](#) termwise in x , which is permitted by its compact local uniformity. The linear coefficient $(3 - 2x)/4$ vanishes only at $3/2$. Direct differentiation of

$$c_2(x) = \frac{(x-2)(x-1)(2x+3)}{144}$$

gives $c'_2(3/2) = -1/288$. The inverse function theorem handles the nonexceptional points. The differentiated remainder in [theorem 35.3](#) gives, at the exceptional point,

$$\partial_t \psi_{e^{-t}}(3/2) = -\frac{t}{144} + \mathcal{O}(t^3) < 0$$

for all sufficiently small $t > 0$. Hence the physical one-sided inverse exists and is locally unique. Factoring the forward relation as $\Delta = -t^2(1 + \mathcal{O}(t^2))/288$ and taking the positive square root gives its displayed expansion.

The twice-differentiated remainder likewise gives

$$\partial_t^2 \psi_{e^{-t}}(3/2) = -\frac{1}{144} + \mathcal{O}(t^2).$$

For the reciprocal extension, logarithmic differentiation in x gives the stated identity for ψ_q . At $x = 3/2$ its correction term vanishes, so replacing q by $1/q$, or t by $-t$, changes nothing. The one-sided expansion and its differentiated remainder patch with this exact reflection to a C^2 function at zero, with second derivative $-1/144$. The reflected positive and negative solutions are therefore the asserted two branches. This real argument requires no holomorphic continuation across the boundary $|q| = 1$. $\square$

37 q -beta coordinate and base inverses

For $x, y > 0$ and initially $0 < q < 1$, define

$$B_q(x, y) := \frac{\Gamma_q(x)\Gamma_q(y)}{\Gamma_q(x+y)}.$$

Theorem 37.1 (Coordinate monotonicity and base type change). *For every $0 < q < 1$, $B_q(x, y)$ is strictly decreasing in either positive coordinate while the other is fixed. Hence each coordinate inverse is unique on its range.*

As $q = e^{-t} \rightarrow 1$,

$$\log \frac{B_{e^{-t}}(x, y)}{B(x, y)} = b_1(x, y)t + b_2(x, y)t^2 + \mathcal{O}(t^4),$$

where

$$b_1(x, y) = \frac{xy - 1}{2}, \quad b_2(x, y) = -\frac{x^2y + xy^2 - xy - 1}{24}.$$

The base inverse is ordinary when $xy \neq 1$. On $xy = 1$,

$$b_2(x, 1/x) = -\frac{(x-1)^2}{24x};$$

For $x \neq 1$, this is a one-sided quadratic endpoint in the original domain. If

$$\Delta = \log \frac{B_{e^{-t}}(x, 1/x)}{B(x, 1/x)},$$

then its unique local inverse with $t \downarrow 0$ is

$$t = \sqrt{-\frac{24x}{(x-1)^2}\Delta} (1 + \mathcal{O}(\Delta)), \quad \Delta \uparrow 0, \quad \Delta < 0.$$

Under the reciprocal extension of Γ_q from [theorem 36.1](#),

$$B_q(x, y) = q^{1-xy} B_{1/q}(x, y).$$

Consequently, on $xy = 1$ the extended beta function is exactly reciprocal-even and the one-sided endpoint becomes a genuine two-sided quadratic fold. At $(x, y) = (1, 1)$, the identity $B_q(1, 1) \equiv 1$ makes the base completely unidentifiable.

Proof. The logarithmic coordinate derivatives are

$$\partial_x \log B_q(x, y) = \psi_q(x) - \psi_q(x+y) < 0,$$

$$\partial_y \log B_q(x, y) = \psi_q(y) - \psi_q(x+y) < 0,$$

because ψ_q is strictly increasing. This proves strict coordinate monotonicity and uniqueness of each coordinate inverse.

Subtract the $x + y$ instance of [theorem 35.3](#) from the sum of its x and y instances. Elementary expansion of c_1 and c_2 gives the displayed b_1, b_2 ; no cubic term exists because the q -gamma expansion has no odd term beyond the linear one. The inverse function theorem applies when $b_1 \neq 0$. On $xy = 1$, substitution gives the stated negative b_2 , nonzero unless $x = 1$. Uniform differentiated remainders give

$$\partial_t \Delta = -\frac{(x-1)^2}{12x}t + \mathcal{O}(t^3) < 0$$

for sufficiently small $t > 0$. Thus the one-sided inverse exists and is unique, and taking the positive square root in $\Delta = -(x-1)^2 t^2 / (24x) + \mathcal{O}(t^4)$ gives its displayed expansion. The twice-differentiated remainder is

$$\partial_t^2 \Delta = -\frac{(x-1)^2}{12x} + \mathcal{O}(t^2),$$

which will control the regularity at the reflected endpoint.

For $q > 1$, substitution of the reciprocal q -gamma normalization into the definition of B_q gives an exponent

$$\frac{(x-1)(x-2) + (y-1)(y-2) - (x+y-1)(x+y-2)}{2} = 1 - xy,$$

which proves the reciprocal identity. On $xy = 1$ it gives exact evenness under $t \mapsto -t$; the differentiated one-sided remainder patches to a C^2 function with nonzero second derivative, and reflection supplies the second branch. At $(1, 1)$, $\Gamma_q(1) = \Gamma_q(2) = 1$ proves the exact constant identity. $\square$

Warning 37.2. The claim that q -beta is always strictly decreasing in its base is false: $B_q(1, 1) = 1$ for every base. The theorem above is the corrected statement. It proves strictness in the two shape coordinates and gives the exact base-inverse type change.

38 Euler q -exponentials

It is useful to separate the product variable from the classical-limit scaling:

$$e_q(z) := \frac{1}{(z; q)_\infty}, \quad E_q(z) := (-z; q)_\infty,$$

$$e_q(x) := e_q((1-q)x), \quad E_q(x) := E_q((1-q)x).$$

Proposition 38.1 (Exact inverse reduction and $q \rightarrow 1$ generator). *Let $|q| < 1$. If A_q is a selected argument-inverse branch of $a \mapsto (a; q)_\infty$, then*

$$e_q^{-1}(y) = A_q(y^{-1}), \quad E_q^{-1}(y) = -A_q(y).$$

Writing $h = 1 - q$, suppose additionally that $|hx| < 1$, and choose the logarithms determined by their power series at $x = 0$. Then the scaled functions satisfy the absolutely convergent identities

$$\log e_q(x) = \sum_{m \geq 1} \frac{h^m x^m}{m(1 - (1-h)^m)},$$

$$\log E_q(x) = \sum_{m \geq 1} \frac{(-1)^{m+1} h^m x^m}{m(1 - (1-h)^m)}.$$

For every fixed x , the condition $|hx| < 1$ holds for all sufficiently small positive h . In the resulting $h \downarrow 0$ asymptotic expansion, only finitely many indices m contribute at each fixed order.

Proof. The inverse formulas are immediate after rewriting the two target equations as $(z; q)_\infty = y^{-1}$ and $(-z; q)_\infty = y$, with the selected branch data retained. Apply [theorem 18.2](#) with $a = \pm hx$ and $q = 1 - h$, remembering the minus sign introduced by the reciprocal in e_q . Since the m -th summand begins at order h^{m-1} after the denominator is expanded, only finitely many m can affect any prescribed coefficient. For completeness, fix $K \geq 0$, take real $0 < h < 1$, and shrink h until $h|x| \leq 1/2$. Since $1 - (1-h)^m \geq h$, the tail satisfies

$$\sum_{m \geq K+2} \left| \frac{h^m x^m}{m(1 - (1-h)^m)} \right| \leq \frac{1}{h} \sum_{m \geq K+2} \frac{(h|x|)^m}{m} = \mathcal{O}(h^{K+1}).$$

The alternating numerator obeys the same bound. The finitely many terms with $m \leq K + 1$ may be expanded individually, proving the asserted coefficientwise asymptotic statement with a genuine remainder estimate. $\square$

For example,

$$\begin{aligned}\log e_q(x) &= x + \frac{x^2}{4}h + \left(\frac{x^2}{8} + \frac{x^3}{9}\right)h^2 + \mathcal{O}(h^3), \\ \log E_q(x) &= x - \frac{x^2}{4}h + \left(-\frac{x^2}{8} + \frac{x^3}{9}\right)h^2 + \mathcal{O}(h^3).\end{aligned}$$

Bell exponentiation gives the functions themselves, while $e_q(z)E_q(-z) = 1$ is exact.

39 Basic hypergeometric argument inversion

Theorem 39.1 (Universal local series-variable inverse). *Let a basic hypergeometric function, after all removable parameter cancellations, have a convergent local series*

$$F(z; \mathbf{a}, \mathbf{b}, q) = 1 + c_1 z + c_2 z^2 + c_3 z^3 + \cdots, \quad c_1 \neq 0.$$

Then its argument inverse at $y = 1$ is

$$z = \frac{y-1}{c_1} - \frac{c_2}{c_1^3}(y-1)^2 + \left(\frac{2c_2^2}{c_1^5} - \frac{c_3}{c_1^4}\right)(y-1)^3 + \mathcal{O}((y-1)^4).$$

All higher coefficients are given by the analytic Lagrange–Bürmann formula.

Proof. The condition $c_1 \neq 0$ gives a local analytic inverse. Put $u = y - 1$ and

$$z = Au + Bu^2 + Cu^3 + \mathcal{O}(u^4).$$

Substitution in $u = c_1 z + c_2 z^2 + c_3 z^3 + \cdots$ yields

$$c_1 A = 1, \quad c_1 B + c_2 A^2 = 0, \quad c_1 C + 2c_2 AB + c_3 A^3 = 0.$$

Solving gives the displayed coefficients. The all-order statement is [theorem 8.3](#). $\square$

If a numerator parameter reaches q^{-N} , the series truncates and the argument inverse becomes an algebraic covering. If a denominator parameter approaches the same truncation locus, common finite q -Pochhammer factors must be cancelled before applying the theorem; otherwise a removable singularity is misclassified as a branch point.

40 Multiplicative theta products

For $0 < |q| < 1$, let

$$\vartheta(z; q) := (z; q)_\infty (q/z; q)_\infty (q; q)_\infty, \quad z \in \mathbb{C}^\times.$$

Proposition 40.1 (Simple theta zero atlas). *The zeros of $z \mapsto \vartheta(z; q)$ are exactly $z = q^m$, $m \in \mathbb{Z}$, and every zero is simple. Hence the inverse at target zero is ordinary on every sheet:*

$$z = q^m + \frac{y}{\partial_z \vartheta(q^m; q)} + \mathcal{O}(y^2).$$

Moreover,

$$\vartheta(qz; q) = -z^{-1} \vartheta(z; q),$$

so one zero derivative determines all the others.

Proof. The factor $(z; q)_\infty$ has one simple vanishing factor precisely at $z = q^{-j}$, $j \geq 0$. The factor $(q/z; q)_\infty$ has one simple vanishing factor precisely at $z = q^{j+1}$, $j \geq 0$. Because $|q| < 1$, the two lists do not overlap, and every other factor is nonzero. Their union is $\{q^m : m \in \mathbb{Z}\}$, proving the zero and multiplicity claims. The local inverse formula is the analytic inverse function theorem.

Finally,

$$(qz; q)_\infty = \frac{(z; q)_\infty}{1 - z}, \quad (1/z; q)_\infty = (1 - z^{-1})(q/z; q)_\infty.$$

Multiplying and using $(1 - z^{-1})/(1 - z) = -z^{-1}$ proves quasi-periodicity. Differentiating it at a zero transports the derivative normalization. $\square$

Part VI

Cyclotomic limits and the Fabius–Rvachev deformation

41 The boundary and probability regimes are different

Two uses of a geometric parameter occur in this volume. In [section 25](#), the base q approaches the unit circle and the Lambert series develops resonant residue classes. In the present part, $0 < q < 1$ is the contraction ratio of a random series. The Rvachev member is the perfectly regular interior value $q = 1/2$. Thus the radial inverse [theorem 25.2](#) and the finite cyclotomic analysis of [section 30](#) remain available when a product formula reaches the boundary, but they should not be conflated with statistical recovery of the geometric contraction parameter.

The law, support, CDF refinement, smooth density, and half-base identification are represented in Lean by the modules `FabiusFunction.GeometricUniformLaw`, `FabiusFunction.GeometricUniformCDF`, and `FabiusFunction.ProbabilityRepresentation`. In particular, `geometricUniformCDF_one_half_eq_fabiusReal` is the exact formal bridge at $q = 1/2$. The phase-bearing sinc-product bridge is formalized separately in `FabiusFunction.GeometricSincCharacteristicFunction`, as `charFun_geometricUniformDistribution_eq_phase_mu_1_geometricSincProduct`. The arbitrary-power Fourier decay in [equation \(86\)](#), its parameter-derivative strengthening in [equation \(110\)](#), and the cumulant inversions later in this chapter are complete human proofs at the frontier level; they are not claimed here as existing Lean declarations.

42 The geometric uniform law and its sinc product

Let $V_0, V_1, \dots$ be independent random variables uniformly distributed on $[0, 1]$. We choose the usual bounded versions, so $0 \leq V_j(\omega) \leq 1$ for every j and every outcome ω ; replacing arbitrary representatives on their null exceptional sets does not change their joint law or independence. Let $U_j = 2V_j - 1$, so that U_j is uniform on $[-1, 1]$ and satisfies $|U_j(\omega)| \leq 1$ pointwise. For $0 < q < 1$, define

$$Y_q = (1 - q) \sum_{j=0}^{\infty} q^j V_j, \tag{83}$$

$$X_q = 2Y_q - 1 = (1 - q) \sum_{j=0}^{\infty} q^j U_j. \tag{84}$$

The first variable takes values in $[0, 1]$, while the centered variable takes values in $[-1, 1]$. Write $F_q(x) = \mathbb{P}(Y_q \leq x)$.

Theorem 42.1 (Geometric law, refinement, and smooth density). *The series in equations (83) and (84) converge absolutely for every outcome. The characteristic function of X_q is*

$$\phi_q(t) = \mathbb{E} e^{itX_q} = \prod_{j=0}^{\infty} \text{sinc}_{\text{rad}}((1-q)q^j t), \quad \text{sinc}_{\text{rad}} z = \frac{\sin z}{z}, \quad (85)$$

with the removable value $\text{sinc}_{\text{rad}} 0 = 1$; equivalently, $\text{sinc}_{\pi} x = \text{sinc}_{\text{rad}}(\pi x)$. For every $A > 0$ there is a constant $C_{A,q}$ such that

$$|\phi_q(t)| \leq C_{A,q}(1 + |t|)^{-A}. \quad (86)$$

Consequently X_q has a compactly supported C^∞ density u_q , and F_q is C^∞ . The distributional decomposition

$$X_q \stackrel{\text{law}}{=} (1-q)U_0 + qX'_q,$$

where X'_q is an independent copy, gives

$$u_q(x) = \frac{1}{2(1-q)} \int_{(x-(1-q))/q}^{(x+(1-q))/q} u_q(s) \, ds, \quad (87)$$

$$u'_q(x) = \frac{u_q((x+(1-q))/q) - u_q((x-(1-q))/q)}{2q(1-q)}. \quad (88)$$

Equivalently, the CDF of Y_q satisfies

$$F_q(x) = \int_0^1 F_q\left(\frac{x - (1-q)v}{q}\right) \, dv, \quad (89)$$

$$F'_q(x) = \frac{F_q(x/q) - F_q((x - (1-q))/q)}{1-q}. \quad (90)$$

Proof. Since the nonnegative weights $(1-q)q^j$ sum to one and $|U_j| \leq 1$, both random series converge absolutely and have the asserted supports. For the N -th partial sum, independence and $\mathbb{E} e^{isU_j} = \text{sinc}_{\text{rad}} s$ give the finite version of equation (85). The partial sums converge pointwise and are bounded in modulus by one, so dominated convergence gives the infinite product.

For real s , $|\text{sinc}_{\text{rad}} s| \leq \min(1, |s|^{-1})$. Fix an integer $r > A$. Once $|t| \geq ((1-q)q^{r-1})^{-1}$, the first r factors therefore give

$$|\phi_q(t)| \leq \prod_{j=0}^{r-1} \frac{1}{(1-q)q^j |t|} = \frac{q^{-r(r-1)/2}}{(1-q)^r} |t|^{-r}.$$

The remaining factors have modulus at most one. Enlarging the constant on a bounded interval proves equation (86). Hence $t^m \phi_q(t)$ is integrable for every m . Fourier inversion and differentiation under the integral produce a C^∞ density for the law of X_q ; its support is already contained in $[-1, 1]$. Since $Y_q = (X_q + 1)/2$, its density is $2u_q(2x - 1)$, and hence $F'_q(x) = 2u_q(2x - 1)$. Thus F_q is C^∞ as claimed.

Removing the first summand in equation (84) leaves q times an independent copy. Conditioning on either summand gives equation (87); the fundamental theorem of calculus gives equation (88). The same conditioning applied to equation (83) gives equation (89). Changing variables in that integral yields

$$F_q(x) = \frac{q}{1-q} \int_{(x-(1-q))/q}^{x/q} F_q(s) \, ds,$$

whose derivative is equation (90). □

43 The exact Fabius and Rvachev member

Definition 43.1 (Normalized bounded Fabius function). A normalized bounded Fabius function is a continuously differentiable map $G : \mathbb{R} \rightarrow [0, 1]$ such that

$$G(x) = 0 \quad (x \leq 0), \quad G(x) = 1 \quad (x \geq 1), \quad G(1-x) = 1 - G(x),$$

and

$$G'(x) = 2G(2x) \quad (0 \leq x \leq \tfrac{1}{2}).$$

This exterior-tail normalization removes the additive and scaling ambiguities from the differential equation.

Theorem 43.2 (The half-base CDF is the Fabius function). *There is a unique normalized bounded Fabius function in the sense of [theorem 43.1](#); denote it by F_{bd} . Then*

$$F_{1/2}(x) = F_{\text{bd}}(x) \quad (x \in \mathbb{R}). \quad (91)$$

Moreover $u_{1/2}$ is Rvachev's compactly supported function:

$$u_{1/2}(s) = \text{up}(s) = \begin{cases} F_{\text{bd}}(s+1), & -1 \leq s \leq 0, \\ F_{\text{bd}}(1-s), & 0 \leq s \leq 1, \\ 0, & |s| \geq 1. \end{cases} \quad (92)$$

With the Fourier convention $\widehat{u}_{2\pi}(z) = \int_{\mathbb{R}} u(s) e^{-2\pi i s z} ds$, for every $z \in \mathbb{R}$,

$$\widehat{\text{up}}_{2\pi}(z) = \prod_{j=0}^{\infty} \text{sinc}_{\pi} \left(\frac{z}{2^j} \right). \quad (93)$$

Thus the CDF statement and the translated-density statement are compatible; they are not competing normalizations.

Proof. The law of $Y_{1/2}$ is contained in $[0, 1]$, and [theorem 42.1](#) gives it a smooth density; in particular it has no atoms. Replacing every V_j by $1 - V_j$ shows that $Y_{1/2}$ and $1 - Y_{1/2}$ have the same law. Therefore

$$F_{1/2}(1-x) = \mathbb{P}(Y_{1/2} \geq x) = 1 - F_{1/2}(x),$$

where atomlessness justifies the second equality. When $0 \leq x \leq 1/2$, [equation \(90\)](#) becomes

$$F'_{1/2}(x) = 2(F_{1/2}(2x) - F_{1/2}(2x-1)) = 2F_{1/2}(2x),$$

because $2x-1 \leq 0$. Together with the two constant exterior tails and the reflection law, these are precisely the normalized bounded Fabius axioms.

For completeness, uniqueness follows from a contraction argument. Any candidate G in [theorem 43.1](#) is nondecreasing: its derivative is nonnegative on $[0, 1/2]$, and differentiating reflection gives $G'(1-x) = G'(x)$. Thus G is the CDF of a probability measure μ on $[0, 1]$. For such a CDF define

$$(\mathcal{T}G)(x) = \int_0^1 G(2x-v) dv = \int_{2x-1}^{2x} G(s) ds.$$

This is the CDF of $(V+Z)/2$, where V is uniform on $[0, 1]$, Z has CDF G , and they are independent. On $(0, 1/2)$,

$$(\mathcal{T}G)'(x) = 2G(2x) = G'(x),$$

while on $(1/2, 1)$, reflection and its differentiated form give

$$(\mathcal{T}G)'(x) = 2(1 - G(2x - 1)) = 2G(2 - 2x) = G'(1 - x) = G'(x).$$

Both functions vanish at zero and have the same constant exterior tails, so $\mathcal{T}G = G$: every normalized bounded Fabius law is a fixed point.

On probability measures on $[0, 1]$, let $\mathcal{T}\mu$ denote the law just constructed. Given any coupling (Z_1, Z_2) of μ_1, μ_2 , use the same independent V in both images. Then

$$\mathbb{E} \left| \frac{V + Z_1}{2} - \frac{V + Z_2}{2} \right| = \frac{1}{2} \mathbb{E} |Z_1 - Z_2|,$$

and taking the infimum over couplings yields $W_1(\mathcal{T}\mu_1, \mathcal{T}\mu_2) \leq \frac{1}{2} W_1(\mu_1, \mu_2)$. Two fixed points therefore have zero Wasserstein distance. Finally, separating the first summand shows $Y_{1/2} \stackrel{\text{law}}{=} (V_0 + Y'_{1/2})/2$, so its law is a fixed point. It is the unique one, proving [equation \(91\)](#). This is the human-readable argument whose conclusion is formalized by `geometricUniformCDF_one_half_eq_fabiusReal`.

Since $X_{1/2} = 2Y_{1/2} - 1$, change of variables gives

$$u_{1/2}(s) = \frac{1}{2} F'_{1/2} \left(\frac{s+1}{2} \right).$$

For $-1 \leq s \leq 0$, the Fabius derivative equation makes the right side $F_{\text{bd}}(s+1)$. Reflection gives the branch $F_{\text{bd}}(1-s)$ for $0 \leq s \leq 1$, and the support gives zero outside. Finally, for real z , the Fourier-sign convention gives $\widehat{\text{up}}_{2\pi}(z) = \phi_{1/2}(-2\pi z) = \phi_{1/2}(2\pi z)$; the second equality uses the evenness of the centered law. Substituting $q = 1/2$ in [equation \(85\)](#) and using $\text{sinc}_{\pi} x = \text{sinc}_{\text{rad}}(\pi x)$ gives [equation \(93\)](#). $\square$

44 Thue–Morse spline approximants

Let $\text{wt}_2(k)$ be the number of ones in the binary expansion of k , and put $\varepsilon_k = (-1)^{\text{wt}_2(k)}$. The occurrence of these signs is ordinary inclusion–exclusion on the dyadic geometric comb.

Theorem 44.1 (Exact Thue–Morse partial CDF). *For $N \geq 1$, put*

$$Y^{(N)} = \sum_{j=0}^{N-1} 2^{-j-1} V_j, \quad F_N(x) = \mathbb{P}(Y^{(N)} \leq x), \quad r_+ = \max(r, 0).$$

Then

$$F_N(x) = \frac{2^{N(N+1)/2}}{N!} \sum_{k=0}^{2^N-1} \varepsilon_k \left(x - \frac{k}{2^N} \right)_+^N. \quad (94)$$

For the bounded Fabius function F_{bd} and every real x ,

$$F_N(x - 2^{-N}) \leq F_{\text{bd}}(x) \leq F_N(x). \quad (95)$$

In particular the explicit splines converge pointwise to F_{bd} , and uniformly with the quantitative bound

$$0 \leq F_N(x) - F_{\text{bd}}(x) \leq F_{\text{bd}}(x + 2^{-N}) - F_{\text{bd}}(x) \leq \omega_{F_{\text{bd}}}(2^{-N}), \quad (96)$$

where $\omega_{F_{\text{bd}}}(\delta) = \sup_{|a-b| \leq \delta} |F_{\text{bd}}(a) - F_{\text{bd}}(b)|$. Consequently $\|F_N - F_{\text{bd}}\|_{\infty} \leq \omega_{F_{\text{bd}}}(2^{-N}) \rightarrow 0$.

Proof. For positive s , the Laplace transform of a uniform variable on $[0, w]$ is $(1 - e^{-sw})/(sw)$. Therefore the Laplace transform, in the space variable x , of the CDF of a sum of N such variables is

$$\frac{1}{s^{N+1} \prod_j w_j} \prod_{j=0}^{N-1} (1 - e^{-sw_j}).$$

Expanding the last product over subsets and using

$$\int_0^\infty e^{-sx} \frac{(x-a)_+^N}{N!} dx = \frac{e^{-sa}}{s^{N+1}}$$

gives the usual inclusion–exclusion formula. Here $\prod_j w_j = 2^{-N(N+1)/2}$. The subset sums $\sum_j \varepsilon_j 2^{-j-1}$ are $k/2^N$ in bit-reversed order, and bit reversal preserves the number of selected indices. The subset sign is thus ε_k , proving [equation \(94\)](#).

The omitted tail of $Y_{1/2}$ lies pointwise, for every outcome, between zero and $\sum_{j \geq N} 2^{-j-1} = 2^{-N}$. Hence

$$\{Y^{(N)} \leq x - 2^{-N}\} \subseteq \{Y_{1/2} \leq x\} \subseteq \{Y^{(N)} \leq x\}.$$

Taking probabilities and using [equation \(91\)](#) proves the sandwich. Its right inequality gives $F_{\text{bd}}(x) \leq F_N(x)$. Applying its left inequality at $x + 2^{-N}$ gives $F_N(x) \leq F_{\text{bd}}(x + 2^{-N})$, and hence [equation \(96\)](#). The continuous function F_{bd} is constant off $[0, 1]$, so it is uniformly continuous on $\mathbb{R}$ and $\omega_{F_{\text{bd}}}(2^{-N}) \rightarrow 0$. This proves the claimed uniform convergence. $\square$

The finite sign identity behind the theorem is

$$\prod_{j=0}^{N-1} (1 - z^{2^j}) = \sum_{k=0}^{2^N-1} \varepsilon_k z^k. \quad (97)$$

It follows immediately by choosing either 1 or $-z^{2^j}$ from each factor; uniqueness of binary expansion identifies the chosen subset with k , and its cardinality with $\text{wt}_2(k)$.

45 Cumulants as exact inverse observables

We use the Bernoulli-number convention $B_2 = 1/6$, $B_4 = -1/30$.

Theorem 45.1 (All cumulants and monotone base recovery). *For every $m \geq 1$,*

$$\kappa_{2m}(X_q) = \frac{2^{2m} B_{2m}}{2m} \frac{(1-q)^{2m}}{1-q^{2m}}, \quad (98)$$

and every odd cumulant vanishes:

$$\kappa_{2m+1}(X_q) = 0 \quad (m \geq 0). \quad (99)$$

After division by the nonzero uniform cumulant, set

$$R_m(q) = \frac{\kappa_{2m}(X_q)}{\kappa_{2m}(U_0)} = \frac{(1-q)^{2m}}{1-q^{2m}}.$$

Then R_m is a strictly decreasing bijection from $(0, 1)$ to $(0, 1)$. Thus, given an observed normalized cumulant $\rho_m \in (0, 1)$, the unique recovered base is equivalently the unique $q \in (0, 1)$ satisfying

$$\rho_m(1 - q^{2m}) = (1 - q)^{2m}. \quad (100)$$

Its endpoint expansions are

$$q = \frac{1 - \rho_m}{2m} + \mathcal{O}((1 - \rho_m)^2), \quad \rho_m \uparrow 1, \quad (101)$$

$$1 - q = (2m\rho_m)^{1/(2m-1)} \left(1 + \mathcal{O}(\rho_m^{1/(2m-1)})\right), \quad \rho_m \downarrow 0. \quad (102)$$

Proof. The moment generating function of U_0 is $\sinh z/z$. Near the origin, differentiating its logarithm gives

$$\frac{d}{dz} \log \frac{\sinh z}{z} = \coth z - \frac{1}{z} = \sum_{m \geq 1} \frac{2^{2m} B_{2m}}{(2m)!} z^{2m-1}.$$

After integration, comparison with $\log \mathbb{E} e^{zU_0} = \sum_r \kappa_r(U_0) z^r / r!$ gives $\kappa_{2m}(U_0) = 2^{2m} B_{2m} / (2m)$ and zero odd cumulants.

Here is the limit argument that justifies adding the countably many cumulants. Put

$$X_{q,N} = (1-q) \sum_{j=0}^N q^j U_j.$$

The chosen bounded versions give the deterministic estimate $|X_q - X_{q,N}| \leq q^{N+1}$, while both variables take values in $[-1, 1]$. Hence their moment generating functions converge locally uniformly for complex z . On a sufficiently small fixed disk about zero,

$$|\mathbb{E} e^{zX_{q,N}} - 1| \leq e^{|z|} - 1 < 1$$

uniformly in N , and the same estimate holds for X_q . The analytic logarithms normalized to vanish at zero therefore converge locally uniformly; by the Cauchy integral formula, all their derivatives at zero converge.

For each finite N and $r \geq 1$, independence and homogeneity of cumulants give

$$\kappa_r(X_{q,N}) = \kappa_r(U_0)(1-q)^r \sum_{j=0}^N q^{jr}.$$

Passing to the limit and summing the geometric series proves [equation \(98\)](#). Symmetry of every finite partial sum, or directly of the full series, proves [equation \(99\)](#), including $\kappa_1(X_q) = \mathbb{E} X_q = 0$.

Logarithmic differentiation gives

$$\frac{R'_m(q)}{R_m(q)} = -\frac{2m}{1-q} + \frac{2mq^{2m-1}}{1-q^{2m}} < 0,$$

because $q^{2m-1}(1-q) < 1 - q^{2m} = (1-q)(1+q+\dots+q^{2m-1})$. The endpoint limits are one and zero, so strict monotonicity proves the bijection. Substituting the observed value $\rho_m = R_m(q)$ gives [equation \(100\)](#) and its unique root in $(0, 1)$. At $q = 0$, $R_m(q) = 1 - 2mq + \mathcal{O}(q^2)$. At $q = 1 - \varepsilon$,

$$R_m(q) = \frac{\varepsilon^{2m-1}}{2m} (1 + \mathcal{O}(\varepsilon)).$$

Ordinary and Puiseux reversion give [equations \(101\) and \(102\)](#). □

The case $m = 1$ is especially simple.

Corollary 45.2 (Exact variance inverse). *The centered variance and its inverse are*

$$V(q) = \text{Var}(X_q) = \frac{1-q}{3(1+q)}, \quad q = \frac{1-3V}{1+3V}. \quad (103)$$

The Rvachev value $q = 1/2$ is characterized by $V = 1/9$.

Proof. Either put $m = 1$ in [equation \(98\)](#), or sum $\frac{1}{3}(1-q)^2 \sum_{j \geq 0} q^{2j}$. Solving the resulting linear-fractional equation gives the inverse and the specialization. □

46 Scale-free recovery and an upgraded conjecture

Theorem 46.1 (Kurtosis inverse and total location–scale identifiability). *The excess kurtosis of X_q is*

$$\gamma_2(q) = \frac{\kappa_4(X_q)}{\kappa_2(X_q)^2} = -\frac{6}{5} \frac{1 - q^2}{1 + q^2}, \quad (104)$$

and is a strictly increasing bijection from $(0, 1)$ to $(-6/5, 0)$, with inverse

$$q = \sqrt{\frac{1 + 5\gamma_2/6}{1 - 5\gamma_2/6}}. \quad (105)$$

Consequently the mean, positive variance, and excess kurtosis uniquely identify every member

$$Z = \mu + cX_q, \quad \mu \in \mathbb{R}, \quad c > 0, \quad 0 < q < 1.$$

Explicitly, $\mu = \mathbb{E}Z$, q is given by [equation \(105\)](#), and

$$c = \sqrt{\text{Var}(Z) \frac{3(1+q)}{1-q}}. \quad (106)$$

Thus the total cumulant identifiability statement left conjectural in one source report is a theorem for the stated three-parameter family.

Proof. Insert $B_2 = 1/6$ and $B_4 = -1/30$ into [equation \(98\)](#). Cancellation gives

$$\frac{-2}{15} \frac{(1-q)^4}{1-q^4} \left(\frac{3(1-q^2)}{(1-q)^2} \right)^2 = -\frac{6}{5} \frac{1-q^2}{1+q^2}.$$

The last rational function is strictly increasing, with the stated endpoint limits. Solving it for q^2 gives [equation \(105\)](#). Translation changes only the mean, positive scaling multiplies variance by c^2 , and excess kurtosis is unchanged by both operations. Hence the three displayed reconstruction formulas exist and are unique. Formula [equation \(106\)](#) is just $\text{Var}(Z) = c^2 V(q)$ with [equation \(103\)](#) substituted. $\square$

For redundancy checks, the next standardized cumulant is

$$\frac{\kappa_6(X_q)}{\kappa_2(X_q)^3} = \frac{48}{7} \frac{(1-q^2)^2}{1+q^2+q^4}. \quad (107)$$

At $q = 1/2$, the exact diagnostic triple begins

$$\text{Var}(X_{1/2}) = \frac{1}{9}, \quad \gamma_2(X_{1/2}) = -\frac{18}{25}, \quad \frac{\kappa_6(X_{1/2})}{\kappa_2(X_{1/2})^3} = \frac{144}{49}.$$

Equation [equation \(107\)](#) follows from $\kappa_6(U_0) = 16/63$ and the same geometric-series cancellation as above.

47 Endpoint regimes and conditioning

Proposition 47.1 (Uniform and Gaussian endpoint laws). *As $q \downarrow 0$, for the chosen pointwise bounded versions, $X_q \rightarrow U_0$ uniformly in the sample point (and hence almost surely). As $q \uparrow 1$, $X_q \rightarrow 0$ in L^2 . After standardization,*

$$\frac{X_q}{\sqrt{V(q)}} \xrightarrow{\text{law}} \mathcal{N}(0, 1). \quad (108)$$

Proof. The first limit has the deterministic bound

$$|X_q - U_0| \leq q|U_0| + (1 - q) \sum_{j \geq 1} q^j |U_j| \leq 2q.$$

The L^2 assertion follows from $\mathbb{E}X_q^2 = V(q) \rightarrow 0$. For the Gaussian limit, put

$$a_{q,j} = \frac{(1 - q)q^j}{\sqrt{V(q)}}.$$

Then

$$\sum_{j \geq 0} a_{q,j}^2 = 3, \quad \max_{j \geq 0} a_{q,j} = a_{q,0} = \sqrt{3(1 - q^2)} \rightarrow 0.$$

The sinc product gives the characteristic function of the standardized law as $\prod_{j \geq 0} \text{sinc}_{\text{rad}}(ta_{q,j})$. For fixed real t and q sufficiently close to one, every argument lies in a fixed neighborhood of zero on which

$$\log \text{sinc}_{\text{rad}} z = -\frac{z^2}{6} + \mathcal{O}(z^4),$$

with a uniform remainder constant. The resulting logarithmic series is absolutely convergent because $\sum_j a_{q,j}^2 = 3$. Consequently

$$\log \prod_{j \geq 0} \text{sinc}_{\text{rad}}(ta_{q,j}) = -\frac{t^2}{6} \sum_{j \geq 0} a_{q,j}^2 + \mathcal{O}\left(t^4 \sum_{j \geq 0} a_{q,j}^4\right) = -\frac{t^2}{2} + o(1),$$

because $\sum_j a_{q,j}^4 \leq (\max_j a_{q,j}^2) \sum_j a_{q,j}^2 \rightarrow 0$. Thus the characteristic functions converge pointwise to $e^{-t^2/2}$, and Lévy's continuity theorem proves [equation \(108\)](#). This direct Fourier argument also avoids applying a finite-row triangular-array theorem to the countably infinite series. $\square$

Variance recovery is absolutely well-conditioned on every closed subinterval of $0 < V < 1/3$, since

$$\frac{dq}{dV} = -\frac{6}{(1 + 3V)^2}.$$

Its relative condition number diverges as $q \downarrow 0$, but that is the inevitable normalization by a recovered parameter tending to zero, not an absolute singularity. Near $q = 1$, the Puiseux behavior in [equation \(102\)](#) shows why high-order cumulant recovery is progressively flatter than variance recovery.

48 Parameter scores, local CDF inversion, and the symmetry obstruction

Put $w_j(q) = (1 - q)q^j$. Then

$$w'_j(q) = q^{j-1}(j - (j + 1)q),$$

where the formula at $j = 0$ is interpreted by cancellation as -1 . Write

$$H_q(x) = \mathbb{P}(X_q \leq x) = F_q\left(\frac{x + 1}{2}\right)$$

for the centered CDF.

Proposition 48.1 (Fourier score and quantile sensitivity). *Away from the zeros of the sinc factors,*

$$\partial_q \log \phi_q(t) = \sum_{j \geq 0} w'_j(q) \left(t \cot(w_j(q)t) - \frac{1}{w_j(q)} \right). \quad (109)$$

The bracket is interpreted by analytic continuation at $t = 0$, where its value is zero. The series converges locally uniformly in $q \in (0, 1)$ and real t on the open set where none of those factors vanishes. More quantitatively, for every compact $K \subseteq (0, 1)$, every $A > 0$, and all integers $k \geq 0$, there is a constant $C_{A,k,K}$ such that

$$\sup_{q \in K} \left| \partial_q^k \phi_q(t) \right| \leq C_{A,k,K} (1 + |t|)^{-A}. \quad (110)$$

Consequently the characteristic function, density, centered CDF H_q , and uncentered CDF $F_q(x) = H_q(2x - 1)$ are C^∞ in q on compact subintervals of $(0, 1)$, and parameter differentiation may be passed through Fourier inversion. If a centered quantile branch satisfies $H_q(x_p(q)) = p$ and $u_q(x_p(q)) > 0$, then it is locally smooth and

$$x'_p(q) = - \frac{\partial_q \mathbb{P}(X_q \leq x)|_{x=x_p(q)}}{u_q(x_p(q))}. \quad (111)$$

Proof. For a nonzero sinc factor,

$$\frac{d}{dw} \log \text{sinc}_{\text{rad}}(wt) = t \cot(wt) - \frac{1}{w}.$$

At $t = 0$ the apparent singularity is removable because

$$t \cot(wt) - \frac{1}{w} = -\frac{wt^2}{3} - \frac{w^3 t^4}{45} + \mathcal{O}(w^5 t^6).$$

The chain rule gives each summand in [equation \(109\)](#). For large j , the bracket equals $-w_j t^2/3 + \mathcal{O}(w_j^3 t^4)$, while $|w'_j| \leq C_K(j+1)q_+^{j-1}$ on any compact $K \subset (0, 1)$, where $q_+ = \max K < 1$. The resulting majorant is a convergent multiple of $\sum (j+1)q_+^{2j}$, proving local uniform convergence away from the zeros.

For completeness, we quantify the product differentiation, including at its zeros. For every $r \geq 0$, direct differentiation of $w_j = (1-q)q^j$ gives constants depending only on K, r such that

$$\sup_{q \in K} |\partial_q^r w_j(q)| \leq C_{K,r} (j+1)^r q_+^j, \quad \sum_{j \geq 0} \sup_{q \in K} |\partial_q^r w_j(q)| < \infty.$$

The integral representation $\text{sinc}_{\text{rad}} y = \frac{1}{2} \int_{-1}^1 e^{iyu} du$ shows that every real derivative of sinc_{rad} is bounded. These two estimates, together with the chain rule, give normally convergent differentiated products on compact (q, t) -sets.

Now fix k and choose an integer $R > A + 2k$. On K , each of the first R weights is bounded away from zero, so an undifferentiated leading factor is $\mathcal{O}_K(|t|^{-1})$. In a term of the k -fold Leibniz rule, suppose that $s \leq k$ of those leading factors are differentiated. Their total derivative order is at most k , so boundedness of the sinc derivatives makes the term $\mathcal{O}_{K,k,R}(|t|^{k+s-R})$. Since $s \leq k$ and $R > A + 2k$, this is $\mathcal{O}_{K,k,R}(|t|^{-A})$. Derivatives falling on later factors are summable by the preceding weight estimates and do not change this worst power. Summing the normally convergent Leibniz expansion proves [equation \(110\)](#); enlarging the constant handles bounded t .

Given any numbers of spatial and parameter derivatives, choose A large enough that the corresponding power of t is integrable. Dominated Fourier inversion then proves smooth dependence of u_q . Since all centered laws are supported in the common interval $[-1, 1]$, $H_q(x) = \int_{-1}^x u_q(s) ds$ on that interval, with constant exterior tails; hence the same conclusion holds for H_q and F_q . Finally, the implicit-function theorem applies at a quantile where $u_q(x_p) > 0$. Differentiating its defining identity and solving $\partial_q H_q(x_p) + u_q(x_p)x'_p = 0$ gives [equation \(111\)](#). $\square$

For fixed x , the C^2 inverse sensitivity theorem [theorem 10.1](#) applies to $q \mapsto F_q(x)$ at any point where its q -derivative is nonzero. If $A = \partial_q F_q(x)|_{q=q_0}$, $B = \partial_q^2 F_q(x)|_{q=q_0}$, and $\Delta = p - F_{q_0}(x)$, the first terms are

$$q - q_0 = \frac{\Delta}{A} - \frac{B}{2A^3} \Delta^2 + \mathcal{O}(\Delta^3).$$

At $x = 1/2$, however, reflection forces $F_q(1/2) = 1/2$ for every q . The entire parameter score vanishes there, so the midpoint contains no information about q . This is an exact identifiability obstruction, not a numerical conditioning accident.

49 The geometric comb as a q -Pochhammer interpolation basis

Proposition 49.1 (Newton remainder on geometric nodes). *Let $n \geq 1$, let $0 < q < 1$, let $I \subset \mathbb{R}$ be an interval containing $x, 1, q, \dots, q^{n-1}$, and let $g : I \rightarrow \mathbb{R}$ be n -times continuously differentiable. Let $N_{n-1}g$ be its interpolation polynomial at those n distinct geometric nodes. Then some point ξ in the convex hull of $x, 1, q, \dots, q^{n-1}$ satisfies*

$$g(x) - N_{n-1}g(x) = \frac{g^{(n)}(\xi)}{n!} \prod_{j=0}^{n-1} (x - q^j) = \frac{g^{(n)}(\xi)}{n!} x^n (x^{-1}; q)_n, \quad (112)$$

where the final expression is understood by polynomial continuation at $x = 0$.

Proof. If $x = q^j$ for one of the interpolation nodes, both sides of the first equality vanish, and any ξ in the convex hull may be chosen. Otherwise the $n + 1$ points are distinct. Put

$$c = \frac{g(x) - N_{n-1}g(x)}{\prod_{j=0}^{n-1} (x - q^j)}, \quad h(t) = g(t) - N_{n-1}g(t) - c \prod_{j=0}^{n-1} (t - q^j).$$

The real-valued function h has those $n + 1$ distinct zeros. Applying Rolle's theorem n times gives a point ξ in their convex hull with $h^{(n)}(\xi) = 0$. Since $N_{n-1}g$ has degree at most $n - 1$ and the node polynomial is monic of degree n , this identity reads $g^{(n)}(\xi) - n!c = 0$, proving the first equality. For $x \neq 0$, the second equality is the factor-by-factor identity

$$x^n (x^{-1}; q)_n = x^n \prod_{j=0}^{n-1} (1 - x^{-1}q^j) = \prod_{j=0}^{n-1} (x - q^j).$$

The final product is a polynomial in x , so this identity defines the polynomial continuation of the middle expression at $x = 0$. $\square$

This is the precise interpolation link behind the informal phrase *geometric comb*. The same nodes q^j govern the random-series weights, the cumulant sums, Newton divided differences, and the finite q -Pochhammer basis. Parameter recovery from Fabius–Rvachev observables is therefore connected to inverse q -products structurally, rather than by a mere resemblance of notation.

Part VII

Certification and research frontiers

50 Branch-aware numerical certification

An inverse approximation is not a certificate. A certificate must identify the forward map, the active parameter, the target, the domain containing the selected branch, and the arithmetic enclosure used

to control every discarded tail and rounding error. In the complex setting it must additionally record the continuation path and every logarithm or fractional-power determination. Changing any one of these data can change the inverse being computed.

The basic workflow is therefore:

1. choose a coordinate in which the selected sheet is regular;
2. evaluate the forward map and its derivatives with outward-rounded enclosures, including explicit infinite tails;
3. obtain a candidate from an asymptotic expansion, continuation, or a floating-point solve;
4. prove existence and uniqueness in a real interval or complex disk;
5. if all sheets are required, certify the remaining zero count by the argument principle.

The candidate-generation step may be heuristic. The last two steps are not.

50.1 Certified tails for infinite products

The next estimate controls the value and the first two argument derivatives with one truncation index.

Theorem 50.1 (Complex q -Pochhammer tail). *Let $a, q \in \mathbb{C}$, put $\rho = |q| < 1$, and choose $N \in \mathbb{N}_0$ so that*

$$\eta := |a|\rho^N < 1.$$

The tail

$$T_N(a, q) := \prod_{j=N}^{\infty} (1 - aq^j)$$

is nonzero and has the logarithm

$$L_N(a, q) := - \sum_{j=N}^{\infty} \sum_{k=1}^{\infty} \frac{(aq^j)^k}{k}.$$

With

$$B_N := \frac{|a|\rho^N}{(1 - \eta)(1 - \rho)},$$

one has

$$|L_N(a, q)| \leq B_N, \quad |T_N(a, q) - 1| \leq e^{B_N} - 1, \quad (113)$$

$$|\partial_a L_N(a, q)| \leq \frac{\rho^N}{(1 - \eta)(1 - \rho)}, \quad |\partial_a^2 L_N(a, q)| \leq \frac{\rho^{2N}}{(1 - \eta)^2(1 - \rho^2)}. \quad (114)$$

More explicitly, suppose interval arithmetic certifies scalar bounds

$$\rho \leq \bar{\rho} < 1, \quad \eta \leq \bar{\eta} < 1.$$

Then the value bounds hold with

$$\bar{B}_N := \frac{\bar{\eta}}{(1 - \bar{\eta})(1 - \bar{\rho})}$$

in place of B_N , and the two derivative bounds hold after replacing their right-hand sides respectively by

$$\frac{\bar{\rho}^N}{(1 - \bar{\eta})(1 - \bar{\rho})}, \quad \frac{\bar{\rho}^{2N}}{(1 - \bar{\eta})^2(1 - \bar{\rho}^2)}.$$

For interval balls $\mathcal{A}$ and $\mathcal{Q}$ of possible a and q , it suffices to certify

$$\bar{\rho} \geq \sup_{q \in \mathcal{Q}} |q|, \quad \bar{\eta} \geq \left(\sup_{a \in \mathcal{A}} |a| \right) \bar{\rho}^N,$$

with both upper endpoints still strictly below one.

Proof. For $|z| < 1$, the power-series logarithm gives

$$|-\log(1-z)| \leq \sum_{k \geq 1} |z|^k = \frac{|z|}{1-|z|}.$$

For $j \geq N$, $|aq^j| \leq \eta$. Hence

$$|L_N(a, q)| \leq \sum_{j \geq N} \frac{|a|\rho^j}{1-\eta} = B_N.$$

The majorant is summable, so the logarithmic series converges absolutely and the product converges to $T_N = e^{L_N}$. No tail factor vanishes because $|aq^j| < 1$. The elementary estimate $|e^z - 1| \leq e^{|z|} - 1$ proves the second bound in (113).

On a slightly smaller neighbourhood of the given a , the differentiated series have the summable majorants used below; termwise differentiation is therefore legitimate. Explicitly,

$$\partial_a L_N = - \sum_{j \geq N} \frac{q^j}{1-aq^j}, \quad \partial_a^2 L_N = - \sum_{j \geq N} \frac{q^{2j}}{(1-aq^j)^2}.$$

Using $|1-aq^j| \geq 1-\eta$ and summing the two geometric series gives (114). Each factor in the three displayed majorants is monotone in ρ and η on $[0, 1)$, so substituting $\bar{\rho}, \bar{\eta}$ proves the interval versions. $\square$

Near a zero of a finite prefix, that factor should be removed symbolically. The theorem then controls the nonvanishing tail without hiding the local zero inside a logarithmic sum.

The Rvachev Fourier product has an equally simple real-axis certificate. We use $\text{sinc}_{\text{rad}} x = \sin(x)/x$ for $x \neq 0$ and $\text{sinc}_{\text{rad}}(0) = 1$.

Theorem 50.2 (Geometric sinc-product tail). *Let $0 < q < 1$, $t \in \mathbb{R}$, and $w_j = (1-q)q^j$. Then*

$$S_N(t) := \prod_{j=N}^{\infty} \text{sinc}_{\text{rad}}(w_j t)$$

converges, and

$$|S_N(t) - 1| \leq \exp\left(\frac{t^2(1-q)^2 q^{2N}}{6(1-q^2)}\right) - 1. \quad (115)$$

Proof. Taylor's formula with integral remainder gives, for real x ,

$$|\sin x - x| \leq \int_0^{|x|} (|x| - u) u \, du = \frac{|x|^3}{6}.$$

Consequently $|\text{sinc}_{\text{rad}} x - 1| \leq x^2/6$, with the value at zero understood by continuity. Put $u_j = \text{sinc}_{\text{rad}}(w_j t) - 1$. For a finite product,

$$\left| \prod_{j=N}^M (1 + u_j) - 1 \right| \leq \prod_{j=N}^M (1 + |u_j|) - 1 \leq \exp\left(\sum_{j=N}^M |u_j|\right) - 1.$$

Moreover,

$$\sum_{j \geq N} |u_j| \leq \frac{t^2}{6} \sum_{j \geq N} w_j^2 = \frac{t^2(1-q)^2 q^{2N}}{6(1-q^2)}.$$

The summability proves convergence of the product, and passage to the limit in the finite-product inequality proves (115). $\square$

50.2 Real branches: brackets, residuals, and interval Newton

On a real branch, monotonicity is the cleanest existence and uniqueness certificate. Newton acceleration may be used inside the bracket, but it is the bracket that carries the proof.

Theorem 50.3 (Monotone branch certificate). *Let $f : [\alpha, \beta] \rightarrow \mathbb{R}$ be continuous and strictly monotone, and let y lie between $f(\alpha)$ and $f(\beta)$. Then there is a unique $x_* \in [\alpha, \beta]$ with $f(x_*) = y$. Sign-preserving bisection produces nested brackets X_r containing x_* with*

$$\text{length}(X_r) = 2^{-r}(\beta - \alpha).$$

If f is differentiable and $|f'(x)| \geq m > 0$ throughout the original bracket, then every $x_0 \in [\alpha, \beta]$ satisfies

$$|x_0 - x_*| \leq \frac{|f(x_0) - y|}{m}. \quad (116)$$

Proof. The intermediate value theorem gives existence, and strict monotonicity gives uniqueness. At each bisection step, one of the two half-intervals still has endpoint values bracketing y ; choosing it preserves x_* and halves the length. Finally, the mean-value theorem gives

$$f(x_0) - f(x_*) = f'(\xi)(x_0 - x_*)$$

for some ξ between x_0 and x_* . Taking absolute values and using the derivative lower bound proves (116). $\square$

The residual estimate assumes that the root already exists. Interval Newton can certify existence as well, provided its derivative enclosure is separated from zero.

Theorem 50.4 (One-dimensional interval Newton). *Let $X = [\alpha, \beta]$, let f be continuously differentiable on X , and let D be a nonempty closed bounded real interval containing $f'(X)$ with $0 \notin D$. For $x_0 \in X$, define*

$$N(X) := x_0 - \frac{f(x_0) - y}{D} = \left\{ x_0 - \frac{f(x_0) - y}{d} : d \in D \right\}.$$

If $N(X) \subseteq X$, then $f(x) = y$ has exactly one solution in X . If $N(X) \subseteq \text{int } X$, the solution lies in the interior of X .

Proof. First suppose $D = [m, M]$ with $0 < m \leq M$. Then $f' \geq m$, so f is strictly increasing. Put $r = f(x_0) - y$. If $r = 0$, x_0 is the unique root. If $r > 0$, the left endpoint of $N(X)$ is $x_0 - r/m$. Inclusion in X gives $x_0 - \alpha \geq r/m$. The mean-value theorem therefore yields

$$f(x_0) - f(\alpha) \geq m(x_0 - \alpha) \geq r,$$

so $f(\alpha) \leq y < f(x_0)$, and the intermediate value theorem gives a root. If $r < 0$, the right endpoint of $N(X)$ is $x_0 - r/m$; inclusion gives $\beta - x_0 \geq -r/m$, whence $f(\beta) - f(x_0) \geq -r$. Thus $f(x_0) < y \leq f(\beta)$, and again a root exists. Strict increase makes it unique.

If D is negative, apply the preceding argument to $-f$ and target $-y$; its interval Newton set is the same. The stronger inclusion $N(X) \subseteq \text{int } X$ makes the relevant endpoint inequality strict, and therefore excludes an endpoint root when $r \neq 0$. When $r = 0$, it says directly that the root $x_0 \in N(X)$ is interior. $\square$

Directed rounding is essential here. A derivative sampled at many points is not an interval extension of $f'(X)$, and an ordinary floating-point inclusion is not a mathematical inclusion.

50.3 Complex disks: contraction, Rouché, and total zero counts

The complex analogue of interval Newton is particularly transparent in one variable. It certifies a selected sheet and gives a constructive refinement map.

Theorem 50.5 (Complex Krawczyk disk). *Let f be holomorphic on a neighbourhood of the closed disk $D = \{z : |z - z_0| \leq r\}$, where $r > 0$, and let $A \in \mathbb{C} \setminus \{0\}$. If*

$$\kappa := \sup_{z \in D} |1 - Af'(z)| < 1, \quad |Af(z_0)| \leq (1 - \kappa)r,$$

then f has exactly one zero in D . If the second inequality is strict, the zero lies in the open disk.

Proof. Set $T(z) = z - Af(z)$. The disk is convex, and integration along the line segment from w to z gives

$$|T(z) - T(w)| \leq \kappa|z - w|.$$

Also, for $z \in D$,

$$|T(z) - z_0| \leq |T(z) - T(z_0)| + |Af(z_0)| \leq \kappa r + (1 - \kappa)r = r.$$

Thus T is a contraction of the complete metric space D into itself. The contraction theorem supplies a unique fixed point z_* . Since $A \neq 0$, $T(z_*) = z_*$ is equivalent to $f(z_*) = 0$. Every zero is a fixed point, so there are no others. Under the strict second inequality the displayed estimate is strict on the boundary, hence z_* is interior. $\square$

When a second-derivative bound is sharper than an enclosure of the Newton map, the following Rouché certificate is convenient.

Theorem 50.6 (One-zero Rouché disk). *Let f be holomorphic on a neighbourhood of $D = \{z : |z - z_0| \leq r\}$, where $r > 0$, and suppose $M \geq 0$ and*

$$\sup_{z \in D} |f''(z)| \leq M, \quad |f(z_0)| + \frac{1}{2}Mr^2 < |f'(z_0)|r.$$

Then f has exactly one zero, counted with multiplicity, in the open disk.

Proof. For $z \in D$, Taylor's formula along the segment from z_0 to z is

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + (z - z_0)^2 \int_0^1 (1 - s)f''(z_0 + s(z - z_0)) \, ds.$$

The last term has modulus at most $Mr^2/2$ on the boundary. Therefore, on $|z - z_0| = r$, the difference between $f(z)$ and the linear function $f'(z_0)(z - z_0)$ has modulus strictly smaller than $|f'(z_0)(z - z_0)|$. Rouché's theorem says that the two functions have the same number of zeros in the disk. The linear function has one, counted with multiplicity. $\square$

Disjoint one-zero disks do not by themselves prove that every inverse sheet has been found. A validated argument-principle count closes that gap.

Theorem 50.7 (Validated argument-principle count). *Let Ω be a bounded Jordan domain, let f be holomorphic on a neighbourhood of $\bar{\Omega}$, and suppose interval evaluation proves $0 \notin f(\partial\Omega)$. Write the positively oriented boundary as a finite concatenation of C^3 panels $\gamma_k : [0, 1] \rightarrow \partial\Omega$, and put*

$$g_k(s) := \frac{f'(\gamma_k(s))\gamma_k'(s)}{f(\gamma_k(s))}.$$

Suppose validated bounds $M_k \geq \sup_{0 \leq s \leq 1} |g_k''(s)|$ are available. Let

$$Q := \sum_k \frac{g_k(0) + g_k(1)}{2}, \quad E := \frac{1}{12} \sum_k M_k.$$

Suppose outward-rounded arithmetic provides $\hat{Q} \in \mathbb{C}$ and $\varepsilon_Q \geq 0$ with $|Q - \hat{Q}| \leq \varepsilon_Q$. If the closed disk

$$\left\{ z : \left| z - \frac{\hat{Q}}{2\pi i} \right| \leq \frac{E + \varepsilon_Q}{2\pi} \right\}$$

contains exactly one integer n , then f has exactly n zeros in Ω , counted with multiplicity.

Proof. For a twice continuously differentiable complex-valued function g on $[0, 1]$, two integrations by parts give the trapezoidal remainder formula

$$\int_0^1 g(s) \, ds - \frac{g(0) + g(1)}{2} = -\frac{1}{2} \int_0^1 s(1-s)g''(s) \, ds.$$

Since $\int_0^1 s(1-s) \, ds = 1/6$, its modulus is at most $\sup |g''|/12$. Summing over the panels gives

$$\left| \int_{\partial\Omega} \frac{f'(z)}{f(z)} \, dz - Q \right| \leq E.$$

The boundary nonvanishing hypothesis makes the argument principle applicable, so

$$\frac{1}{2\pi i} \int_{\partial\Omega} \frac{f'(z)}{f(z)} \, dz$$

is the integer equal to the number of zeros of f in Ω . The quadrature and rounding bounds, followed by the triangle inequality, place this integer in the displayed disk. If that disk contains only n , the zero count is n . $\square$

In practice, interval balls for $f(\gamma_k([0, 1]))$ certify boundary nonvanishing, automatic differentiation bounds g_k'' , and panel subdivision reduces E . Counting one zero in each selected disk and then the same total number in a containing contour certifies both uniqueness and completeness. For a meromorphic forward map, the same integral counts zeros minus poles; the poles and their multiplicities must be enclosed and added back explicitly.

50.4 Asymptotic truncation is part of the certificate

An $O(t^{N+1})$ printed in a manuscript has no numerical content until its constant, domain, and branch are known. For a convergent Taylor expansion, Cauchy's estimate supplies all three.

Theorem 50.8 (Cauchy tail bound). *Let H be holomorphic on a neighbourhood of $\{z : |z| \leq R\}$, let $M \geq \max_{|z|=R} |H(z)|$, and write $H(z) = \sum_{j \geq 0} c_j z^j$. For $N \in \mathbb{N}_0$, $0 \leq r < R$, and $|z| \leq r$,*

$$\left| H(z) - \sum_{j=0}^N c_j z^j \right| \leq M \frac{(r/R)^{N+1}}{1 - r/R}.$$

Proof. Cauchy's coefficient formula gives $|c_j| \leq M/R^j$. Hence

$$\left| \sum_{j>N} c_j z^j \right| \leq M \sum_{j>N} (r/R)^j = M \frac{(r/R)^{N+1}}{1 - r/R}.$$

$\square$

For an inverse expansion, it is usually sharper to certify its forward residual than to bound inverse coefficients separately.

Theorem 50.9 (Branchwise inverse residual transfer). *For each parameter s , let $p(s)$ be the selected real solution of $F_s(p) = y_s$ in an interval I_s , and suppose $p \mapsto F_s(p)$ is continuously differentiable there. Let $p_N(s) \in I_s$ be any truncated inverse approximation. If*

$$\inf_{p \in I_s} |\partial_p F_s(p)| \geq m_s > 0, \quad |F_s(p_N(s)) - y_s| \leq E_N(s),$$

then

$$|p_N(s) - p(s)| \leq \frac{E_N(s)}{m_s}.$$

Proof. Apply the mean-value theorem to $p_N(s)$ and $p(s)$ and use the derivative lower bound. The fact that both points lie on the certified interval is what makes the bound branch-specific. $\square$

At a fold, m_s tends to zero in the original coordinate. One must first pass to the square-root coordinate in which the sheet is regular, or use a Rouché or Krawczyk disk. Reporting a large residual quotient in the singular coordinate does not certify the branch.

Divergent asymptotic series require a different discipline. The elementary calculation below explains optimal truncation for a Gevrey-one envelope.

Proposition 50.10 (Optimal index for a Gevrey-one remainder envelope). *Let $A, C > 0$ and $t \neq 0$. For every $N \in \mathbb{N}_0$, define the candidate remainder envelope*

$$E_N(t) = C(A|t|)^{N+1}(N+1)!.$$

Suppose the actual remainder after truncation at N has been proved to have modulus at most $E_N(t)$ for every N . Then

$$\frac{E_{N+1}(t)}{E_N(t)} = A|t|(N+2).$$

Put $x = (A|t|)^{-1} - 2$. The certified envelope decreases exactly when $N < x$, is unchanged when $N = x$, and increases when $N > x$. If $A|t| > 1/2$, its unique minimizing index is $N = 0$. If $A|t| = 1/2$, the minimizing indices are 0 and 1. If $A|t| < 1/2$ and $x \notin \mathbb{N}_0$, the unique minimizing index is $\lceil x \rceil$; if $x \in \mathbb{N}_0$, the minimizing indices are x and $x + 1$.

Proof. Cancel the common factors in E_{N+1}/E_N . The monotonicity and location of the discrete minimum follow immediately from whether this ratio is below or above one. $\square$

This proposition is not permission to assume a Gevrey bound. Such a bound must first be proved uniformly in the stated sector. A defensible asymptotic certificate therefore records the coordinate and sheet, the allowed domain, the remainder constants, the chosen truncation order, the outward-rounded forward residual, and the final real or complex root enclosure. Merely observing that successive terms initially decrease is evidence, not proof.

51 Two archived conjectures that are theorems

Two claims repeatedly presented as conjectural in the source reports need no numerical evidence. In their natural precise forms they are elementary theorems.

51.1 Total cumulants identify the geometric uniform law

Let $U_0, U_1, \dots$ be independent random variables, each uniform on $[-1, 1]$, and put

$$X_{\mu,c,q} = \mu + c \sum_{j=0}^{\infty} q^j U_j, \quad \mu \in \mathbb{R}, \quad c > 0, \quad 0 < q < 1.$$

The series converges absolutely almost surely: with probability one all the inequalities $|U_j| \leq 1$ hold, and then its absolute value is bounded by $\sum_j q^j$.

Theorem 51.1 (Total-cumulant identifiability). *Let M, V , and K denote respectively the mean, variance, and standardized fourth cumulant of $X_{\mu,c,q}$, so $K = \kappa_4(X_{\mu,c,q})/V^2$. Then*

$$M = \mu, \quad V = \frac{c^2}{3(1-q^2)}, \quad K = -\frac{6}{5} \frac{1-q^2}{1+q^2}.$$

The resulting map

$$(\mu, c, q) \mapsto (M, V, K)$$

is a bijection from $\mathbb{R} \times (0, \infty) \times (0, 1)$ onto $\mathbb{R} \times (0, \infty) \times (-6/5, 0)$. Its inverse is

$$\mu = M, \quad q = \sqrt{\frac{1+5K/6}{1-5K/6}}, \quad c = \sqrt{3V(1-q^2)}.$$

Proof. For a uniform variable U on $[-1, 1]$,

$$\mathbb{E}U = 0, \quad \mathbb{E}U^2 = \frac{1}{3}, \quad \mathbb{E}U^4 = \frac{1}{5}.$$

Consequently its second and fourth cumulants are

$$\kappa_2(U) = \frac{1}{3}, \quad \kappa_4(U) = \mathbb{E}U^4 - 3(\mathbb{E}U^2)^2 = -\frac{2}{15}.$$

The partial sums converge to $X_{\mu,c,q}$ in every L^r , because their tails are bounded deterministically by a geometric series. Their moments through order four therefore converge. Independence, additivity of cumulants, and homogeneity of κ_r give

$$\kappa_2(X_{\mu,c,q}) = \frac{c^2}{3} \sum_{j \geq 0} q^{2j} = \frac{c^2}{3(1-q^2)}$$

and

$$\kappa_4(X_{\mu,c,q}) = -\frac{2c^4}{15} \sum_{j \geq 0} q^{4j} = -\frac{2c^4}{15(1-q^4)}.$$

Dividing the latter by the square of the former yields the displayed formula for K . Solving it gives

$$q^2 = \frac{1+5K/6}{1-5K/6}.$$

As K runs from $-6/5$ to 0 , this expression runs strictly from 0 to 1 . The formulas for μ and c then follow uniquely from M and V , and they plainly reconstruct the prescribed triple. $\square$

Thus total cumulants identify the exact three-parameter model. Stability under noisy cumulants, missing observations, or a perturbed digit law is a different problem; it becomes a theorem or conjecture only after the perturbation model and metric are specified.

51.2 A transcendental value determines the Gaussian rectangle

For integers $a, b \geq 0$, write

$$G_{a,b}(Q) = \left[\begin{matrix} a+b \\ a \end{matrix} \right]_Q.$$

This is the generating polynomial for partitions whose Ferrers diagrams fit inside an a -by- b rectangle.

Theorem 51.2 (Uniqueness at a transcendental base). *Let $q \in \mathbb{C}$ be transcendental over $\mathbb{Q}$. If*

$$0 < k < n, \quad 0 < k' < n', \quad \left[\begin{matrix} n \\ k \end{matrix} \right]_q = \left[\begin{matrix} n' \\ k' \end{matrix} \right]_q,$$

then $n' = n$ and either $k' = k$ or $k' = n - k$.

Proof. Transcendence says that evaluation at q is injective on $\mathbb{Z}[Q]$. The asserted equality of values is therefore an equality of Gaussian polynomials. Put

$$a = k, \quad b = n - k, \quad m = \min(a, b), \quad M = \max(a, b),$$

and define a', b', m', M' similarly.

The coefficient of Q^d in $G_{a,b}$ counts partitions of d fitting inside the rectangle. Every partition of $d \leq m$ has at most m parts and largest part at most m , so these coefficients equal the unrestricted partition numbers $p(d)$. At $d = m + 1$, at least one partition is excluded: if $a = m$, the partition $1 + \dots + 1$ with $m + 1$ parts is too tall; if $b = m$, the one-part partition $(m + 1)$ is too wide. Hence $m + 1$ is the first degree at which the coefficient is smaller than $p(d)$. Equality of the two polynomials therefore gives $m = m'$.

The degree of $G_{a,b}$ is $ab = mM$, so equality of degrees and positivity of m give $M = M'$. Thus the unordered pairs $\{a, b\}$ and $\{a', b'\}$ agree. Since $n = a + b$ and $k = a$, this says exactly that $n' = n$ and $k' = k$ or $k' = n - k$. $\square$

52 Precisely stated open frontiers

The following list keeps only assertions with a definite truth value. Each conjecture states its domain, normalization, and branch, followed by the present evidence and the missing mathematical obligation.

52.1 Generic monodromy of finite q -products

Set

$$P_{n,q}(a) = \prod_{j=0}^{n-1} (1 - aq^j).$$

The group-theoretic implication needed for this frontier is already a theorem.

Theorem 52.1 (Simple, distinct critical values force full monodromy). *Let $P \in \mathbb{C}[a]$ have degree $n \geq 2$. Suppose P' has $n - 1$ simple zeros and their images under P are pairwise distinct. Then the monodromy group of $P(a) - y = 0$ over $\mathbb{C} \setminus \{\text{critical values of } P\}$ is S_n .*

Proof. Near a simple critical point, the holomorphic Morse lemma identifies the map with $z \mapsto z^2$ in local coordinates. A small loop around its critical value therefore exchanges precisely the two colliding sheets and fixes the others: its monodromy is a transposition. Distinct critical values give one such branch cycle for each critical point. Small positively oriented loops around the finite critical values generate the fundamental group of their complement; the loop at infinity is the inverse of their product. Hence these transpositions generate the monodromy group.

The cover is connected. Algebraically, this follows because $P(a) - y$ is irreducible in $\mathbb{C}(y)[a]$: viewed in $\mathbb{C}[a, y]$, it has degree one in y , and any factor independent of y would divide the coefficient -1 ; Gauss's lemma then gives irreducibility over $\mathbb{C}(y)$. Hence the monodromy action is transitive.

Associate a graph to the sheets, with an edge for every generating transposition. Its connected components are exactly the orbits of the group they generate, so transitivity makes the graph connected. The edge transpositions of a connected graph generate all of S_n . Thus the monodromy group is S_n . $\square$

Conjecture 52.2 (Generic finite q -Pochhammer monodromy). *For every $n \geq 3$ there is a nonzero polynomial $E_n \in \mathbb{Z}[q]$ such that, whenever $q \in \mathbb{C}$ and $qE_n(q) \neq 0$, the critical points of $P_{n,q}$ are simple and their critical values are pairwise distinct. In particular, the monodromy group is then S_n .*

Simplicity is controlled by the discriminant of $P'_{n,q}$, and coincidence of two critical values by a second resultant or discriminant, all with integer coefficients in q . The real obligation is to show that their product is not the zero polynomial for every n . Once that is known, every excluded nonzero parameter is algebraic. A single exact specialization with simple critical points and distinct critical values for each n , or a uniform symbolic argument, would suffice. Floating-point plots cannot establish this nonvanishing.

52.2 Continuous Gaussian monotonicity

For $0 < q < 1$ and $u > 0$, use the standard normalization

$$\Gamma_q(u) = (1 - q)^{1-u} \prod_{j=0}^{\infty} \frac{1 - q^{j+1}}{1 - q^{u+j}}.$$

For $x > y > 0$, define the continuous Gaussian coefficient by

$$\mathcal{G}_q(x, y) = \begin{cases} \frac{\Gamma_q(x+1)}{\Gamma_q(y+1)\Gamma_q(x-y+1)}, & 0 < q < 1, \\ \frac{\Gamma(x+1)}{\Gamma(y+1)\Gamma(x-y+1)}, & q = 1, \\ q^{y(x-y)}\mathcal{G}_{1/q}(x, y), & q > 1. \end{cases}$$

The last line fixes the reciprocal-base normalization rather than leaving it implicit.

Conjecture 52.3 (Continuous Gaussian monotonicity). *For every pair $x > y > 0$, the function*

$$q \mapsto \mathcal{G}_q(x, y)$$

is strictly increasing on $(0, 1)$.

For integral $x = n$ and $y = k$, this is immediate from the nonnegative coefficients of the Gaussian polynomial, with strictness because $0 < k < n$. There is also rigorous local evidence at the upper endpoint. Apply [theorem 35.3](#) at $x+1$, $y+1$, and $x-y+1$. Its linear coefficient satisfies

$$c_1(x+1) - c_1(y+1) - c_1(x-y+1) = -\frac{y(x-y)}{2}.$$

Consequently, if $q = e^{-t}$, then

$$\log \frac{\mathcal{G}_{e^{-t}}(x, y)}{\mathcal{G}_1(x, y)} = -\frac{y(x-y)}{2}t + O(t^2),$$

so $\mathcal{G}_{e^{-t}}(x, y) < \mathcal{G}_1(x, y)$ for every sufficiently small $t > 0$. A global proof must establish the sign of the q -derivative of the displayed q -gamma quotient, most likely by reorganizing the resulting Lambert series into a manifestly positive expression. Conversely, one outward-rounded interval evaluation showing a negative derivative would be a rigorous counterexample.

52.3 A complete-Bernstein principal inverse

Fix $0 < q < 1$ and let

$$P_q(a) = (a; q)_\infty = \prod_{j=0}^{\infty} (1 - aq^j), \quad a < 1.$$

Every factor is positive. Locally uniform differentiation of the logarithm gives

$$\frac{P'_q(a)}{P_q(a)} = - \sum_{j=0}^{\infty} \frac{q^j}{1 - aq^j} < 0.$$

Moreover, $P_q(a) \rightarrow 0$ as $a \uparrow 1$, while $P_q(a) \rightarrow \infty$ as $a \rightarrow -\infty$. For the latter limit, retain any fixed number of factors and then let that finite product tend to infinity. Thus P_q has a decreasing inverse

$$A_q : (0, \infty) \longrightarrow (-\infty, 1), \quad X_q(y) = 1 - A_q(y) > 0.$$

The first two Bernstein signs are theorems. Indeed, if $r_j = q^j / (1 - aq^j) > 0$, then

$$\frac{P''_q(a)}{P_q(a)} = \left(\sum_{j \geq 0} r_j \right)^2 - \sum_{j \geq 0} r_j^2 = 2 \sum_{0 \leq i < j} r_i r_j > 0.$$

Implicit differentiation now gives

$$A'_q(y) = \frac{1}{P'_q(A_q(y))} < 0, \quad A''_q(y) = -\frac{P''_q(A_q(y))}{P'_q(A_q(y))^3} > 0,$$

and therefore $X'_q(y) > 0$ and $X''_q(y) < 0$.

Conjecture 52.4 (Complete-Bernstein principal inverse). *For every $0 < q < 1$, X_q is a complete Bernstein function. Equivalently, there exist $\alpha_q, \beta_q \geq 0$ and a positive measure ν_q on $(0, \infty)$, with*

$$\int_{(0, \infty)} \frac{\nu_q(ds)}{1 + s} < \infty,$$

such that

$$X_q(y) = \alpha_q + \beta_q y + \int_{(0, \infty)} \frac{y}{y + s} \nu_q(ds) \quad (y > 0).$$

The conjecture implies all alternating derivative inequalities $(-1)^{r-1} X_q^{(r)}(y) \geq 0$ for $r \geq 1$. The two proved signs above and the higher signs observed in formal series reversion are evidence, not a proof of the infinite family. The decisive obligation is a Pick continuation of X_q to the slit plane, or an explicit positive representing measure. Notice also that the endpoint above forces $\alpha_q = X_q(0+) = 0$ if the conjecture holds.

52.4 The minimum of the q -gamma function

The existence and uniqueness of the minimizer are not conjectural.

Theorem 52.5 (Unique minimum of Γ_q). *For every $0 < q < 1$, the function $\Gamma_q : (0, \infty) \rightarrow (0, \infty)$ has a unique global minimizer x_q^* .*

Proof. The logarithmic derivative of the standard q -gamma product is

$$\psi_q(x) = -\log(1-q) + (\log q) \sum_{j=0}^{\infty} \frac{q^{x+j}}{1-q^{x+j}}.$$

The series and its derivative converge locally uniformly, and

$$\psi'_q(x) = (\log q)^2 \sum_{j=0}^{\infty} \frac{q^{x+j}}{(1-q^{x+j})^2} > 0.$$

Thus ψ_q is strictly increasing. Its $j = 0$ summand shows that $\psi_q(x) \rightarrow -\infty$ as $x \downarrow 0$, whereas dominated geometric convergence gives $\psi_q(x) \rightarrow -\log(1-q) > 0$ as $x \rightarrow \infty$. It therefore has exactly one zero x_q^* . Since $\Gamma'_q(x) = \Gamma_q(x)\psi_q(x)$ and $\Gamma_q(x) > 0$, the function decreases before this zero and increases after it. The zero is the unique global minimizer. $\square$

Write $m_q = \Gamma_q(x_q^*)$.

Conjecture 52.6 (Monotonic motion of the q -gamma minimum). *On $0 < q < 1$, the minimizer x_q^* is strictly increasing and the minimum value m_q is strictly decreasing.*

The archived safeguarded-Newton computations, which are numerical evidence rather than interval certificates, give

q	0.05	0.20	0.60	0.95
x_q^*	1.37998346149	1.42196580342	1.45066979499	1.46060675866
m_q	0.98270463141	0.95376342969	0.91169489999	0.88840573828

and exhibit both proposed signs. They are compatible with the formal $q \downarrow 0$ asymptotics and the classical-gamma limit as $q \uparrow 1$. A proof requires sign control of the implicit derivative

$$\frac{dx_q^*}{dq} = -\frac{\partial_q \psi_q(x_q^*)}{\partial_x \psi_q(x_q^*)}$$

and, by the envelope identity,

$$\frac{d}{dq} \log m_q = \partial_q \log \Gamma_q(x_q^*).$$

The denominators are already positive by the theorem; the two numerators remain the substantive Lambert-series inequalities.

52.5 Cyclotomic resurgence with fixed normalization

Let ζ be a primitive m -th root of unity and, for $\Re t > 0$ and $|a| < 1$, take the logarithm determined by the absolutely convergent series

$$L_a(t) = \text{Log}(a; \zeta e^{-t})_{\infty} = -\sum_{r=1}^{\infty} \frac{a^r}{r(1 - \zeta^r e^{-rt})}.$$

The singular action and constant term forced by this normalization are

$$S_{-1}(a) = -\frac{\text{Li}_2(a^m)}{m^2},$$

$$S_0(a) = \frac{1}{2m} \log(1 - a^m) - \sum_{\substack{r \geq 1 \\ m \nmid r}} \frac{a^r}{r(1 - \zeta^r)}.$$

A formal series $\widehat{R}(t) = \sum_{n \geq 0} c_n t^n$ is *Gevrey one* here if $|c_n| \leq C A^n n!$ for some $C, A > 0$. Its formal Borel transform is

$$\mathcal{B}\widehat{R}(\xi) = \sum_{n \geq 0} \frac{c_{n+1}}{n!} \xi^n;$$

the constant c_0 is carried separately by Borel–Laplace summation.

Conjecture 52.7 (Cyclotomic resurgent inverse). *Let $K \subseteq \{a : |a| < 1\}$ and $0 < \delta < \pi/2$. Uniformly for $a \in K$, as $t \rightarrow 0$ in $|\arg t| \leq \pi/2 - \delta$, the normalized function*

$$R_a(t) = L_a(t) - \frac{S_{-1}(a)}{t} - S_0(a)$$

has a Gevrey-one asymptotic expansion. Its Borel transform admits endless analytic continuation: for each $a \in K$, there is a locally finite set $\Sigma_{a,\zeta} \subset \mathbb{C}$, with $0 \notin \Sigma_{a,\zeta}$, such that the germ at zero continues along every path in $\mathbb{C} \setminus \Sigma_{a,\zeta}$ issuing from zero. In every direction whose open Borel ray avoids $\Sigma_{a,\zeta}$ and for which the Laplace integral converges, its Borel sum equals the exact $R_a(t)$.

Moreover, fix $a_0 \in \text{int } K$, and let $v(t) \rightarrow 0$ be a resurgent germ on a non-Stokes sector. Let $a(t) \in K$, with $a(t) \rightarrow a_0$, be the chosen local solution of

$$L_{a(t)}(t) = L_{a_0}(t) + v(t).$$

If $\partial_a L_{a(t)}(t)$ is bounded away from zero on every closed subsector under consideration, then $a(t)$ is resurgent, and its Stokes automorphisms are obtained from those of L and v by the same local implicit inversion. Exponentiating the displayed equation gives the corresponding statement for the chosen branch of the product inverse.

The exact Lambert-series identity for L_a , its residue-class splitting, and the Bernoulli expansion of the m -divisible terms explain S_{-1} , S_0 , and the observed Gevrey growth. They do not prove endless continuation or locate every Borel singularity. The remaining obligations are uniform factorial remainder bounds on K , analytic continuation and growth estimates in the Borel plane, identification of the Stokes set, equality of Borel sums with the convergent product, and a quantitative implicit-function argument for the inverse clause.

52.6 Editorial disposition of the remaining proposals

Several former conjectures were deliberately not carried forward.

- The proposed finite-CDF inversion statement combined an elementary simple-fold theorem with an unproved analyticity assertion about a specific density. The fold theorem is covered by the branch-aware certificates above; any density claim must name the density and regularity hypothesis.
- The phrase *generic Gaussian monodromy has the permitted group* did not define either the parameter space or the permitted group. The finite-product conjecture above is its falsifiable replacement.
- Broad inverse-universality and positivity-after-normalization claims did not specify a topology, scaling, normalization, or limiting class.
- Total cumulant identifiability and discrete Gaussian uniqueness at a transcendental base were upgraded to Theorems 51.1 and 51.2.

The omitted themes may still motivate research programs, but they are not mathematical assertions until their quantifiers and normalizations are fixed.

53 Lean source-status map

This section is a source-presence crosswalk, not compilation evidence. The declarations below occur in the indicated Lean source modules; no fresh build status is asserted by this table. A source declaration supports exactly the topic stated in the last column; it does not silently formalize the stronger complex certification theorems or conjectures of this chapter. Module names suppress the common `FabiusFunction.` prefix, and declaration names suppress the common `Fabius.` namespace; the two `Bell.` declarations retain their different namespace explicitly. Every displayed suffix otherwise matches the source exactly.

Lean module	Exact source declarations	Legitimate human crosswalk
<code>FabiusInverse</code>	<code>fabiusReal_fabiusInv;</code> <code>fabiusInv_fabiusReal;</code> <code>strictMonoOn_fabiusInv;</code> <code>continuous_fabiusInv;</code> <code>fabiusInv_hasDerivAt</code>	Two-sided inversion on the proved domain, strict monotonicity, continuity, and the regular inverse derivative.
<code>FabiusInverseEffectiveContinuity</code>	<code>abs_fabiusInv_sub_lt_inverse_two_pow_of_lt_factorialDenominator;</code> <code>fabiusInv_effectivelyUniformContinuous</code>	An explicit factorial-denominator inverse modulus and effective uniform continuity.
<code>FabiusInverseLogarithmicModulus</code>	<code>abs_fabiusInv_sub_lt_inv_nat_of_lt_logarithmicFactorialDenominator;</code> <code>fabiusInv_effectivelyUniformContinuous_logarithmic;</code> <code>fabiusInv_effectivelyUniformContinuous_logarithmicDelta</code>	The later logarithmic inverse moduli, including the delta formulation.
<code>LambertWBranchPointGeometry;</code> <code>LambertWBranchPointAsymptotics</code>	<code>tendsto_deriv_principalLambertW_branchPoint_atTop;</code> <code>tendsto_deriv_lowerLambertW_branchPoint_atBot;</code> <code>principalLambertW_not_differentiableAt_branchPoint;</code> <code>lowerLambertW_not_differentiableAt_branchPoint;</code> <code>principalLambertW_add_one_isEquivalent_branchPoint;</code> <code>lowerLambertW_add_one_isEquivalent_branchPoint</code>	The opposite vertical tangents, failure of finite differentiability, and the signed leading square-root laws at the real Lambert- W branch point. They identify the correct signed leading square-root scale for a possible regularization; they establish neither a regularized analytic coordinate chart nor a higher Puiseux remainder bound.
<code>GeometricUniformLaw</code>	<code>hasSum_geometricUniformWeight;</code> <code>geometricUniformSeries_mem_Icc;</code> <code>geometricUniformDistribution_selfSimilar;</code> <code>geometricUniformDistribution_support_eq_Icc</code>	Convergence, support, and self-similarity of Lean's normalized $[0, 1]$ geometric uniform law. If $Z = (1 - q) \sum_{j \geq 0} q^j V_j$ and $U_j = 2V_j - 1$, then $\sum_{j \geq 0} q^j U_j = (2Z - 1)/(1 - q)$. This affine rescaling underlies Theorem 51.1.
<code>GeometricSincCharacteristicFunction</code>	<code>charFun_geometricUniformDistribution_eq_phase_mul_geometricSincProduct;</code> <code>charFun_geometricUniformDistribution_eq_phase_mul_geometricReciprocalGamma</code>	The sinc-product characteristic function and its reciprocal-gamma form.

Continued on the next page

Lean module	Exact source declarations	Legitimate human crosswalk
SincZetaRemainder; FabiusComputability	norm_sincZeta_tail_le; rvachevFourierProduct_eq_prefix_mul_correction_mul_tail; abs_uniformPartialCDF_sub_fabiusReal_le; fabiusEffectiveUniformModulus	Certified sinc-product and CDF tails and an effective forward Fabius modulus. These are nearby formal certificates, not the general-purpose complex Theorems 50.1–50.7.
QPochhammerElementaryIdentities; QBinomialReciprocity	finiteQPochhammerIn_base_reversal; finiteQPochhammerIn_inv_base_reversal; gaussianBinomial_adjacent_mul; gaussianBinomial_reciprocity; gaussianBinomial_neg_one_eq_zero_of_odd_degree; gaussianBinomial_neg_one_even_odd_eq_zero	Finite base reversal, adjacent ratios, Gaussian reciprocity, and the odd-degree vanishing mechanism at $q = -1$.
GaussianBinomialAtNegOne	gaussianBinomial_neg_one_even_even; gaussianBinomial_neg_one_odd_even; gaussianBinomial_neg_one_odd_odd; finiteQPochhammerIn_neg_one_even; finiteQPochhammerIn_neg_one_odd	The remaining parity evaluations for Gaussian coefficients and the finite q -Pochhammer formulas at $q = -1$; together with the preceding row these form the complete parity atlas.
BellPolynomialInversion; QuadraticAsymptoticInversion	Bell.complete_cumulant; Bell.cumulant_complete; quadratic_asymptotic_inversion	Mutual moment–cumulant inversion and the proved quadratic asymptotic inversion schema.

No declaration in this audited source crosswalk is cited as proving the exact statements of the general q -Pochhammer derivative-tail theorem, interval Newton theorem, complex Krawczyk theorem, Rouché disk theorem, validated argument principle, total-cumulant parameter-identification theorem, or transcendental Gaussian uniqueness theorem as stated here. They are complete human-readable proofs in this chapter, not claimed Lean milestones. Likewise, no Lean counterpart is claimed for the five conjectures in Section 52.

Part VIII

Reference appendices

54 Source corpus and editorial provenance

The canonical volume is a semantic consolidation of six independently delivered source packages. They are not numbered editions of one manuscript: overlap records independent rediscovery or reuse, and disagreement must be settled mathematically rather than by choosing the newest-looking file. The historical paths in [table 2](#) were relative to the former sibling layout under exponent s - and q -series; the pinned pre-retirement revision and repository history preserve them. The row counts are the numbers of theorem-like source environments recorded in the machine-readable concordance.

The editorial disposition is governed by five rules.

1. A semantic duplicate is stated once, with the strongest correct hypotheses and one complete proof. Several source rows may therefore point to one canonical label.
2. A useful source theorem may be strengthened, split into logically independent results, or absorbed into a more general lemma. The concordance records the primary destination and

Source package and principal file	Rows	Distinctive contribution and canonical disposition
<code>inverse_q_analog_functions_report/</code> <code>inverse_q_analog_functions.tex</code>	49	The clearest branchwise narrative, passive-parameter viewpoint, real and complex sheet atlases, and continuation algorithms. Its reusable inverse calculus is absorbed into the universal chapter; its product-specific geometry is distributed among the finite and infinite product chapters.
<code>inverse_q_analog_jet_atlas/</code> <code>inverse_q_analog_jet_atlas.tex</code>	38	Formal triangular reversion, Lagrange–Bürmann and Lagrange–Good formulas, Bell-polynomial organization, mixed jets, and coefficient atlases. These are consolidated, with corrected algebraic hypotheses, in the universal inversion calculus.
<code>inverse_q_analogs_extended_report/</code> <code>inverse_q_analogs_extended_report.tex</code>	48	The broadest collection of exceptional finite regimes, radial inverses, q -special functions, certification ideas, and Fabius–Rvachev recovery. Correct special cases are retained at their natural topical destinations; overgeneralized claims are corrected or retired.
<code>inverse_q_analogs_report/</code> <code>inverse_q_analogs_report.tex</code>	59	Discriminants, remote inverse sheets, real branch geometry, numerical certification, and the earliest large conjecture register. Exact results are deduplicated; symbolic collision claims without independently checkable certificates remain conditional targets rather than theorems.
<code>inverse_q_analogs_report-2/</code> <code>inverse_q_analogs_report.tex</code>	29	An independent synthesis emphasizing reciprocal regimes, roots of unity, compact coefficient tables, and algorithmic cross-checks. It supplies corroborating provenance and occasionally sharper normalizations, not a presumption of chronological priority.
<code>q_pochhammer_q_binomial_expansions_report/</code> <code>q_analog_expansions_report.tex</code>	37	The primary forward-expansion engine: finite and infinite q -Pochhammer products, Gaussian and multinomial coefficients, Bernoulli/polylogarithmic expansions, roots of unity, and q -special functions. Its uniform head–tail remainder arguments are retained wherever termwise formal expansion alone would be insufficient.

Table 2: The six source packages and their canonical roles.

explains any one-to-many refinement in its disposition note.

3. A false, ill-posed, redundant, or unfalsifiable assertion is not preserved merely because it carried a theorem or conjecture heading. Interesting statements with genuine missing obligations survive only in explicitly labelled *Conjecture* environments.
4. Numerical and symbolic calculations are evidence only to the extent certified by reproducible exact arithmetic, interval enclosures, or a proved residual bound. Unsupported large factorizations are certification targets, not hidden premises of unconditional theorems.
5. A script, table, captured output, or figure is migrated only when it uniquely supports a retained result or reproducible calculation. Byte-identical duplicates and generated publication byproducts are not separate mathematical sources. Superseded layouts remain recoverable from repository history.

55 The exhaustive result concordance

The file `theorem_concordance.csv` is the machine-readable ledger for the source corpus. It has one header and exactly 260 data rows, one for every environment recognized by the extraction audit. The package totals are

$$49 + 38 + 48 + 59 + 29 + 37 = 260.$$

By kind, the inventory contains 131 theorems, 28 propositions, 16 corollaries, 2 lemmas, 32 conjectures, 28 problems or research problems, 9 definitions, 5 algorithms, 4 examples, 4 principles, and 1 computational result. The extractor found an intervening `proof` or `proofidea` environment after 163 rows and no such environment after 97. This last field is a syntactic audit signal, not a verdict on correctness: conjectures and definitions normally have no proof, while a source proof environment may still contain a gap.

For an assertion, `canonical_status` uses exactly the three-way vocabulary of [section 2](#): *Lean-proved*, *human-proved frontier result*, or *conjecture*. A definition, notation declaration, or purely descriptive algorithm is not forced into an assertion status; its retention or retirement is explained in the disposition field. Likewise, *merged*, *strengthened*, *corrected*, and *retired* are dispositions, not proof statuses.

The extractor initially generated the five editorial columns empty, because text matching cannot decide mathematical equivalence. In the reviewed ledger, all 260 rows now have a canonical status and disposition note; 180 map to canonical labels, while the remaining 80 are explicitly absorbed, retired, or not separately asserted. Exact Lean module and declaration cells remain empty unless an exact formal counterpart is claimed. The retained reviewed CSV is authoritative. The revision-backed extractor `audit/extract_source_results.py` reproduces and verifies its ten immutable source columns from the commit pinned in `audit/SOURCE_REVISION`; it refuses to overwrite the five editorial columns.

56 Notation and mandatory branch data

An inverse symbol denotes a branch only after its data have been fixed. At minimum one must record the forward map F , the active variable, all passive parameters, the target or limiting coordinate, the base point or real domain, the continuation path or sheet, and every logarithm or fractional-power determination. The following table is a compact decoding guide for the notation used throughout the volume.

CSV field group	Meaning
source_key	Stable key formed from the package name and the source label, or from a deterministic kind-and-ordinal key when the source supplied no label.
source_package, source_file, source_line	Exact source location. Line numbers belong to the archived source snapshot and are provenance coordinates, not canonical page references.
source_label, source_kind, source_title, source_chapter, source_section_path	The source manuscript's own classification and surrounding hierarchy. These fields preserve what was claimed without endorsing it.
source_proof_present	Whether a proof or proofidea environment occurs before the next recognized result. It does not assert that the proof is sound, complete, or relevant to the exact statement.
canonical_label, canonical_status	The primary canonical destination and its strict status. Repeated labels across source rows express semantic deduplication. A split or partial retention is described in disposition_notes.
lean_module, lean_declaration	An exact formal counterpart, when one has actually been identified and validated. Related supporting lemmas do not justify filling these fields for a stronger human theorem.
disposition_notes	The reason for strengthening, merging, splitting, correcting, making conditional, retaining as a conjecture, or retiring the source result.

Table 3: Interpretation of the concordance columns.

Table 4: Notation and the branch information suppressed by a bare inverse symbol.

Notation	Meaning and domain	Branch data that must accompany inversion
$P_n(a, q) = (a; q)_n$	Finite product $\prod_{j=0}^{n-1} (1 - aq^j)$, with an empty product equal to one.	State whether a or q is active; fix n and the other parameter; record a base point, real monotonicity interval, or Puiseux sheet.
$P(a, q) = (a; q)_\infty$	Normally convergent product only for $ q < 1$. Boundary radial limits and exterior reciprocal normalizations are different objects.	For argument inversion, give the principal interval or the zero-gap index and side. For base inversion, give the interior path or the radial normalization and target logarithm.
$\mathcal{A}_q(y)$	The principal real inverse of $a \mapsto (a; q)_\infty$ for $0 < q < 1$, $y > 0$, selected by $a < 1$.	No sheet index is needed on this monotone branch; a nonprincipal real or complex inverse must not reuse this symbol without an explicit sheet label.
$\begin{bmatrix} n \\ k \end{bmatrix}_q$	Gaussian polynomial, including its polynomial continuation where the factorial quotient reads $0/0$.	Fix n, k ; distinguish the positive real branch, a reciprocal $q \leftrightarrow q^{-1}$ branch, and a cyclotomic zero or local sheet.
$q = e^{-t}$	Logarithmic coordinate near $q = 1$, with $t = -\text{Log } q$.	Record the logarithm branch and, in a complex asymptotic statement, a closed sector contained in $\Re t > 0$.

Notation	Meaning and domain	Branch data that must accompany inversion
$q = \zeta e^{-t}$	Radial or sectorial approach to a primitive root ζ of order d .	Record ζ , d , the sector or real radial path, and the continuation of every logarithm. This notation never asserts a germ across $ q = 1$.
$s = 1 - y, \eta = ((1 - a) - y)/(a(1 - a))$	Regular target coordinates at the unit level and at the small-base endpoint.	Record the source base point and the nonvanishing denominators. A different normalization changes every printed inverse coefficient.
$X = -\text{Log } y + C$	Reciprocal-log scale used when a forward logarithm begins $-A/t + C + \dots$.	Fix the target logarithm, the constant C , the sheet with $t \sim A/X$, and a sector on which A/X is admissible.
W_k	The k -th Lambert- W determination.	Record k and verify the original chosen-log equation. Solving an exponentiated equation alone can admit candidates on the wrong logarithm sheet.
$(y - y_0)^{1/m}$	Ramified coordinate at a zero of order m .	Choose the root on a simply connected punctured target domain, or work on the ramified cover; all m determinations coalesce at y_0 .
Γ_q, ψ_q, B_q	The q -gamma, q -digamma, and q -beta normalizations specified in the q -special-function chapter.	State the real parameter interval or complex continuation, the active parameter, the base regime, and any reciprocal continuation for $q > 1$.
$\phi_q(t)$, sinc products	Characteristic functions of the normalized geometric-uniform law and the Fabius–Rvachev member.	Fix the probability normalization, phase convention, real base $0 < q < 1$, and whether inversion concerns a distribution parameter, a Fourier value, or a cumulant observable.

57 Source-to-canonical topical crosswalk

The row-level concordance is the authoritative provenance mechanism. For human navigation, [table 5](#) records where the major source themes landed. A source can appear in several canonical chapters, and a canonical theorem can synthesize several sources.

Source package	Principal source themes	Canonical destinations
<code>inverse_q_analog_functions_report</code>	Branch orientation, regular inverse germs, sensitivities, real and complex sheet atlases, continuation.	Sections 6, 10, 16, 19 and 50.
<code>inverse_q_analog_jet_atlas</code>	Triangular coefficient recurrences, Lagrange inversion, Bell polynomials, mixed jets, multivariate and Good inversion.	Sections 7 to 9, 13 and 14.
<code>inverse_q_analogs_extended_report</code>	Exceptional finite bases, $q = 0, \pm 1$, radial cyclotomic inversion, q -special functions, certification, Fabius–Rvachev recovery.	Sections 17, 17.4, 21, 25, 35, 42 and 50.
<code>inverse_q_analogs_report</code>	Discriminants, real gaps and remote sheets, q -special inverses, certification algorithms, conjecture register.	Sections 16, 19, 36, 50 and 52 and theorem 19.2.
<code>inverse_q_analogs_report-2</code>	Reciprocal regimes, roots of unity, compact inverse tables, independent algorithmic derivations.	Sections 17, 22, 25, 31, 38 and 50.
<code>q_pochhammer_q_binomial_expansions_report</code>	Exact forward coefficient engines, uniform remainders, Bernoulli and polylogarithmic expansions, Gaussian/multinomial limits, roots of unity, q -gamma and q -beta.	Sections 16, 22, 23, 28, 32, 35 and 37.

Table 5: Topical source-to-canonical crosswalk.

58 Formalization boundary

The exact declaration-level source map is already given in [section 53](#) and is not duplicated here. Its current formal surface has five principal islands:

1. classical Fabius inverse identities, monotonicity, continuity, differentiability, and explicit effective-continuity moduli;
2. the geometric-uniform probability law, its support and self-similarity, the geometric sinc characteristic-function identity, certified sinc and CDF tails, and an effective forward Fabius modulus;
3. selected finite q -Pochhammer and Gaussian-polynomial identities, including reciprocity and exact values at $q = -1$;
4. Bell/cumulant conversion and a quadratic asymptotic-inversion lemma;
5. real Lambert- W branch-point geometry: the opposite vertical tangents and failure of finite differentiability, together with the signed leading square-root equivalences for the principal and lower branches.

The last island comes from the companion Lambert- W corpus, specifically `LambertWBranchPointGeometry.lean` and `LambertWBranchPointAsymptotics.lean`. It is supporting formalization for the universal fold discussion, not a seventh inverse- q precursor; accordingly it adds no row to the frozen 260-result source ledger. The signed leading equivalences do not include a higher Puiseux coefficient or remainder theorem, still less a formal inversion theorem for a q -product.

These declarations support parts of the canonical narrative; they do not silently formalize stronger neighbors. In particular, no Lean status is claimed here for the general analytic and formal inverse theorems beyond the named Bell and quadratic support, or for the human proofs of infinite-product normal convergence, remote-gap and log-periodic asymptotics, all-order $q \rightarrow 1$ and cyclotomic expansions, Dedekind-eta completion, Gaussian and q -special global inverse atlases, complex certification algorithms, or the conjectures in [section 52](#). A theorem in those families remains a *human-proved frontier result* unless an exact named declaration is listed for that exact statement.

The Lean table in [section 53](#) is explicitly a source inspection, not a build report. No Lean process was run while preparing this appendix, so it contributes no fresh compiler evidence. Likewise, a nonempty `lean_module` or `lean_declaration` field in a future concordance row will be a cross-reference, not by itself proof of a current aggregate build. Compiler validation, source inspection, a focused module build, and a full repository build are distinct evidence levels and must be reported separately.

59 Repository and source note

No citation key occurs in the canonical package at the time of this consolidation, so this volume does not manufacture a bibliography from the longer reading lists of its precursors. Its immediate documentary sources were the six packages in [table 2](#). The retained package-level ledger `PROVENANCE.md`, row-level ledger `theorem_concordance.csv`, editorial contract `README.md`, and revision-backed extractor `audit/extract_source_results.py` are the live canonical records. Repository history preserves superseded layouts and retired duplicates.

Classical named tools used in proofs—the inverse-function theorem, Lagrange–Bürmann and Lagrange–Good inversion, Rouché’s theorem, Dedekind-eta modularity, and standard special-function identities—are invoked with their exact hypotheses at the point of use. If external citations are introduced in a later edition, their actual citation keys and bibliographic entries should be added together; an uncited generic reading list would not improve the present provenance record.